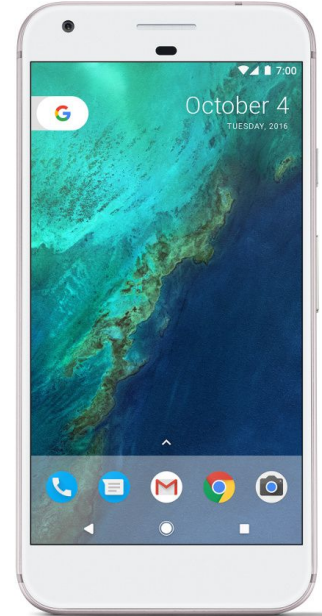


Part I: What is Federated Learning?

Data is born at the edge

Billions of phones & IoT devices constantly generate data

Data enables better products and smarter models

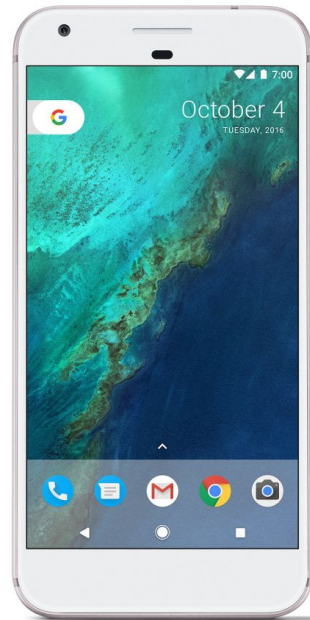


Can data live at the edge?

Data processing is moving on device:

- Improved latency
- Works offline
- Better battery life
- Privacy advantages

E.g., on-device inference for mobile keyboards and cameras.



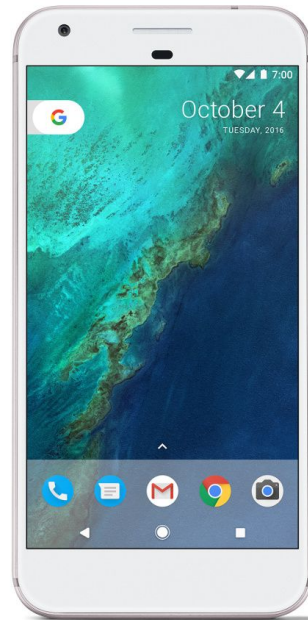
Can data live at the edge?

Data processing is moving on device:

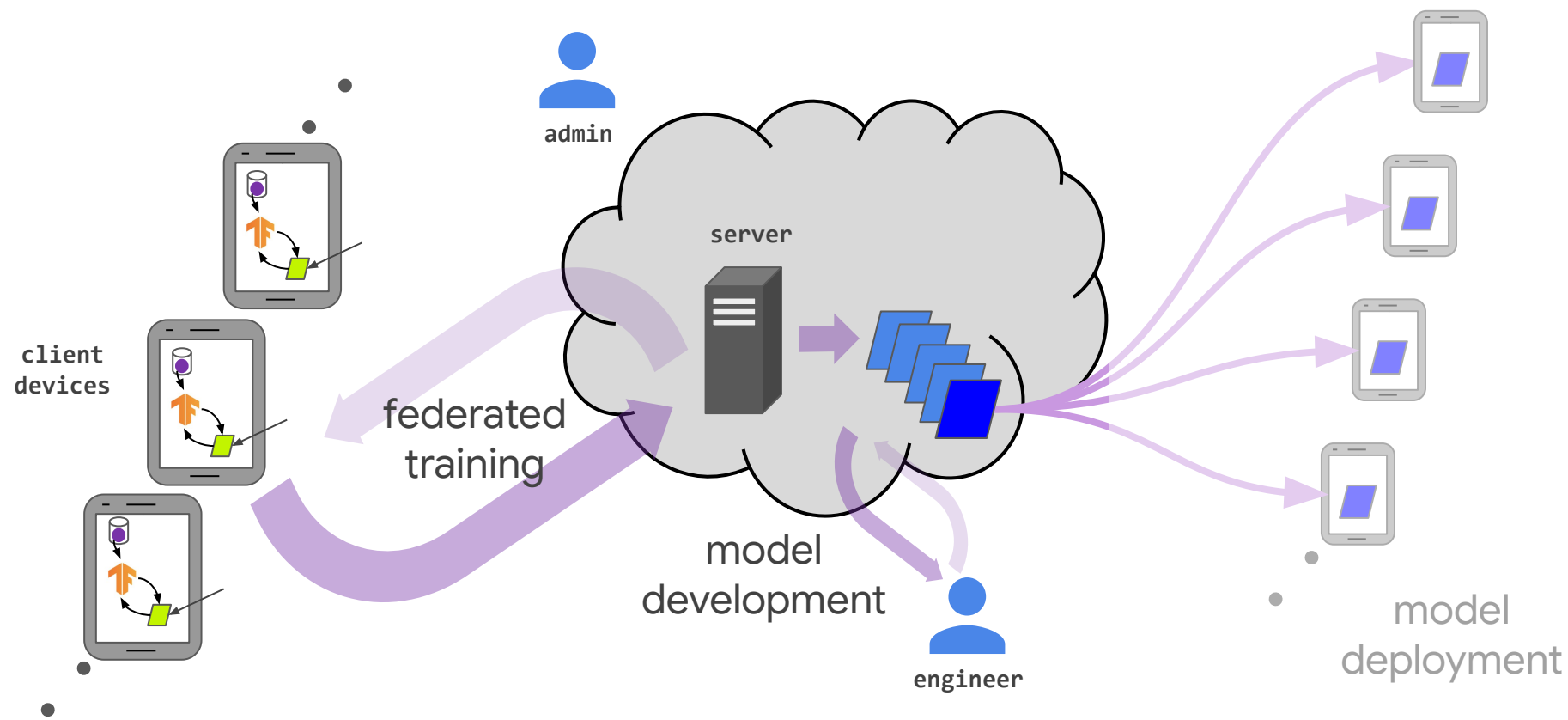
- Improved latency
- Works offline
- Better battery life
- Privacy advantages

E.g., on-device inference for mobile keyboards and cameras.

What about analytics?
What about learning?



Cross-device federated learning



Applications of cross-device federating learning

What makes a good application?

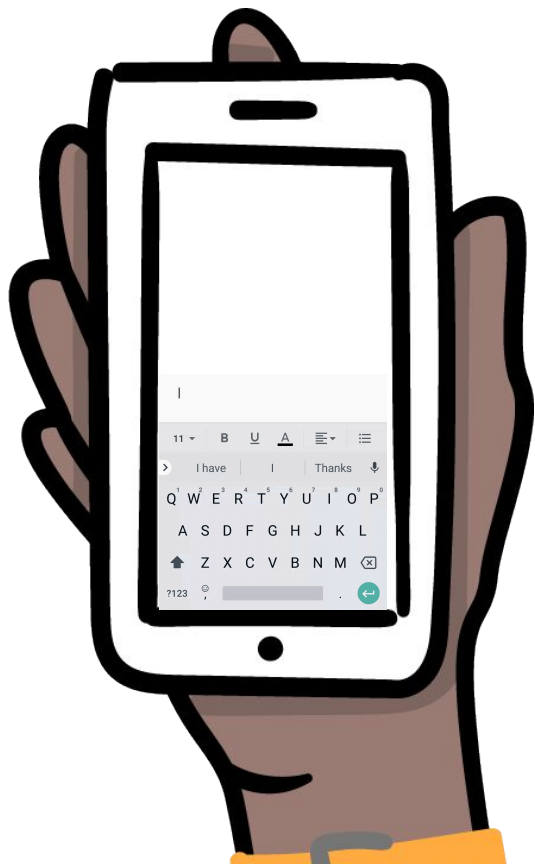
- On-device data is more relevant than server-side proxy data
- On-device data is privacy sensitive or large
- Labels can be inferred naturally from user interaction

Example applications

- Language modeling for mobile keyboards and voice recognition
- Image classification for predicting which photos people will share
- ...

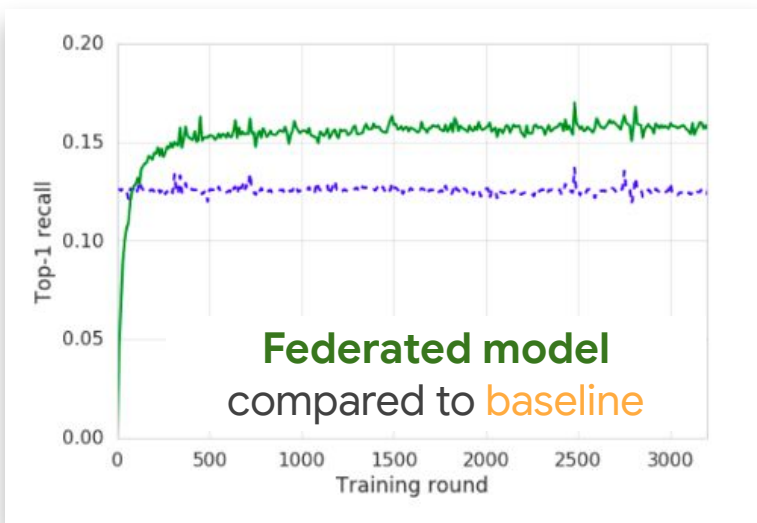


Gboard: next-word prediction



Federated RNN (compared to prior n-gram model):

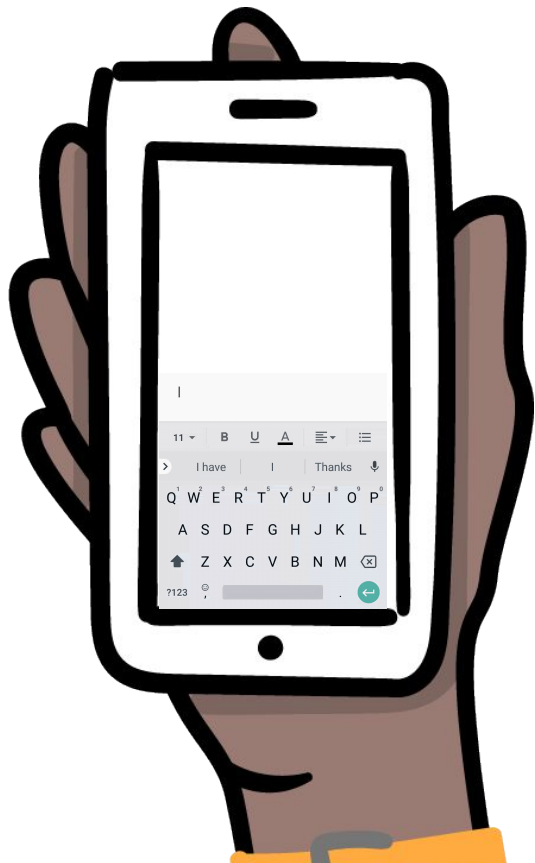
- Better next-word prediction accuracy: +24%
- More useful prediction strip: +10% more clicks



A. Hard, et al. **Federated Learning for Mobile Keyboard Prediction.** *arXiv:1811.03604*



Other federated models in Gboard



Emoji prediction

- 7% more accurate emoji predictions
- prediction strip clicks +4% more
- 11% more users share emojis!

Ramaswamy, *et al.* **Federated Learning for Emoji Prediction in a Mobile Keyboard.** *arXiv:1906.04329.*

Action prediction

When is it useful to suggest a gif, sticker, or search query?

- 47% reduction in unhelpful suggestions
- increasing overall emoji, gif, and sticker shares

T. Yang, *et al.* **Applied Federated Learning: Improving Google Keyboard Query Suggestions.** *arXiv:1812.02903*

Discovering new words

Federated discovery of what words people are typing that Gboard doesn't know.

M. Chen, *et al.* **Federated Learning Of Out-Of-Vocabulary Words.** *arXiv:1903.10635*

Cross-device federated learning at Apple

MIT Technology Review

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Artificial intelligence / Machine learning

How Apple personalizes Siri without hoovering up your data

The tech giant is using privacy-preserving machine learning to improve its voice assistant while keeping your data on your phone.

by **Karen Hao**

December 11, 2019



*"Instead, it relies primarily on a technique called **federated learning**, Apple's head of privacy, Julien Freudiger, told an audience at the Neural Processing Information Systems conference on December 8. Federated learning is a privacy-preserving machine-learning method that was first introduced by Google in 2017. It allows Apple to train different copies of a speaker recognition model across all its users' devices, using only the audio data available locally. It then sends just the updated models back to a central server to be combined into a master model. In this way, raw audio of users' Siri requests never leaves their iPhones and iPads, but the assistant continuously gets better at identifying the right speaker."*

<https://www.technologyreview.com/2019/12/11/131629/apple-ai-personalizes-siri-federated-learning/>

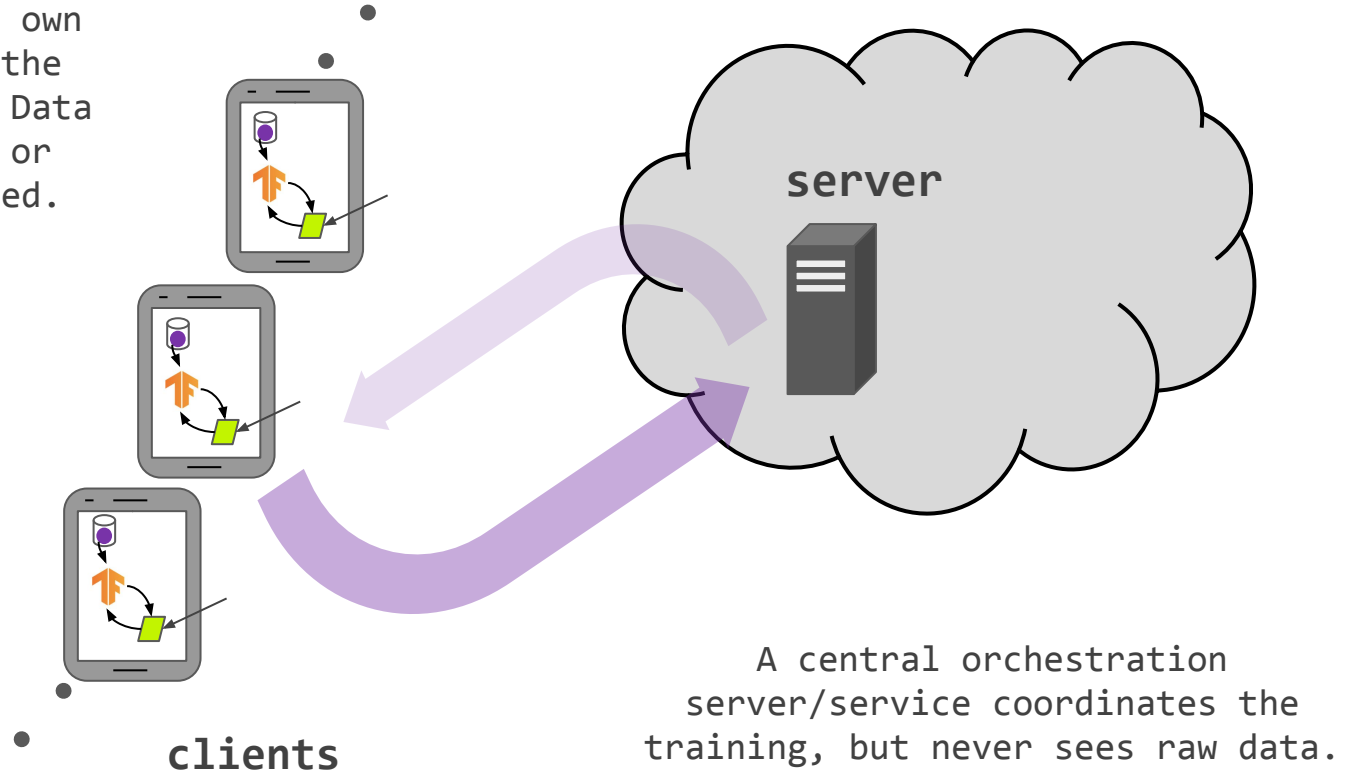
Federated Learning

Federated learning is a machine learning setting where multiple entities (clients) collaborate in solving a machine learning problem, under the coordination of a central server or service provider. Each client's raw data is stored locally and not exchanged or transferred; instead, focused updates intended for immediate aggregation are used to achieve the learning objective.

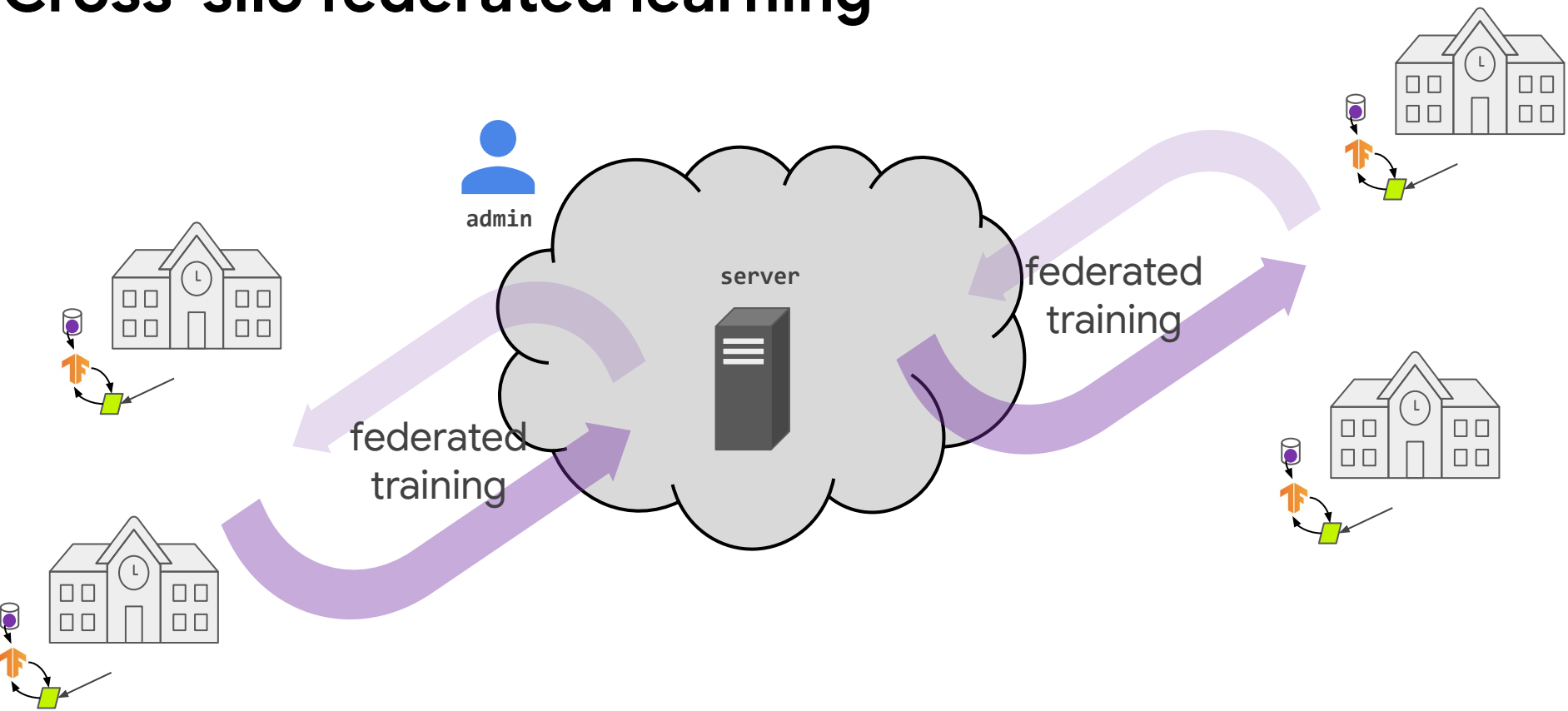
definition proposed in
Advances and Open Problems in Federated Learning ([arxiv/1912.04977](https://arxiv.org/abs/1912.04977))

Federated learning - defining characteristics

Data is generated locally and remains decentralized. Each client stores its own data and cannot read the data of other clients. Data is not independently or identically distributed.



Cross-silo federated learning



Cross-silo federated learning from Intel

ARTIFICIAL INTELLIGENCE, DIAGNOSTICS

UPenn, Intel partner to use federated learning AI for early brain tumor detection

The project will bring in 29 institutions from North America, Europe and India and will use privacy-preserved data to train AI models. Federated learning has been described as being born at the intersection of AI, blockchain, edge computing and the Internet of Things.

By ALARIC DEARMENT

Post a comment / May 11, 2020 at 10:03 AM

"The University of Pennsylvania and chipmaker Intel are forming a partnership to enable 29 healthcare and medical research institutions around the world to train artificial intelligence models to detect brain tumors early."

"The program will rely on a technique known as federated learning, which enables institutions to collaborate on deep learning projects without sharing patient data. The partnership will bring in institutions in the U.S., Canada, U.K., Germany, Switzerland and India. The centers – which include Washington University of St. Louis; Queen's University in Kingston, Ontario; University of Munich; Tata Memorial Hospital in Mumbai and others – will use Intel's federated learning hardware and software."



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Truth Challenge v2: Latest Challenge Results From Genome In A Bottle | Sep 03, 2020

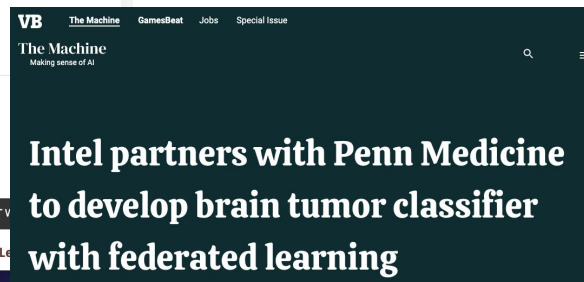
Intel, Penn Medicine Launch Federated Learning Model For Brain Tumors



By Allison Proffitt

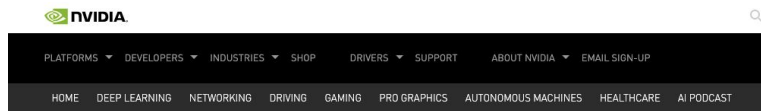
May 28, 2020 | The University of Pennsylvania and Intel have built a federation of 30 institutions to use federated learning to train artificial intelligence (AI) models to identify boundaries of brain tumors.

Led by Spyridon Bakas at the Center for Biomedical Image Computing and Analytics (CBICA) at the Perelman School of Medicine at the University of Pennsylvania, the federation is the next step forward in a years-long effort to gather data that would empower AI in brain image analysis.



- [1] <https://medcitynews.com/2020/05/upenn-intel-partner-to-use-federated-learning-ai-for-early-brain-tumor-detection/>
- [2] <https://www.allaboutcircuits.com/news/can-machine-learning-keep-patient-privacy-for-tumor-research-intel-says-yes-with-federated-learning/>
- [3] <https://venturebeat.com/2020/05/11/intel-partners-with-penn-medicine-to-develop-brain-tumor-classifier-with-federated-learning/>
- [4] <http://www.bio-itworld.com/2020/05/28/intel-penn-medicine-launch-federated-learning-model-for-brain-tumors.aspx>

Cross-silo federated learning from NVIDIA

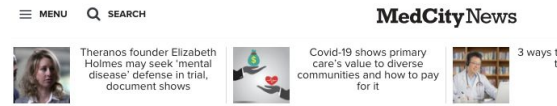
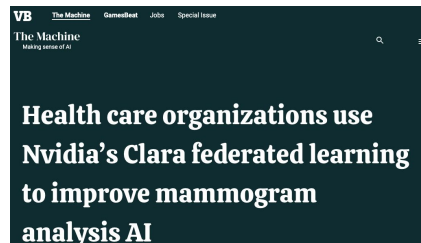


Medical Institutions Collaborate to Improve Mammogram Assessment AI with NVIDIA Clara Federated Learning

In a federated learning collaboration, the American College of Radiology, Diagnosticos da America, Partners HealthCare, Ohio State University and Stanford Medicine developed better predictive models to assess breast tissue density.

April 15, 2020 by MONA FLORES

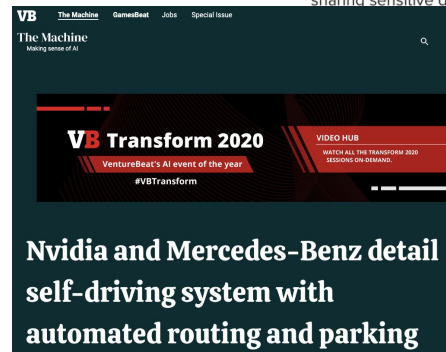
"Federated learning addresses this challenge, enabling different institutions to collaborate on AI model development without sharing sensitive clinical data with each other. The goal is to end up with more generalizable models that perform well on any dataset, instead of an AI biased by the patient demographics or imaging equipment of one specific radiology department."



HOSPITALS, ARTIFICIAL INTELLIGENCE, HEALTH TECH

Nvidia says it has a solution for healthcare's data problems

The chipmaker touted a new framework that would allow hospitals and pharmaceutical companies to collaborate on AI projects without sharing sensitive data. Nvidia said the framework is already gaining traction among hospitals and drug developers.



[1] <https://blogs.nvidia.com/blog/2020/04/15/federated-learning-mammogram-assessment/>

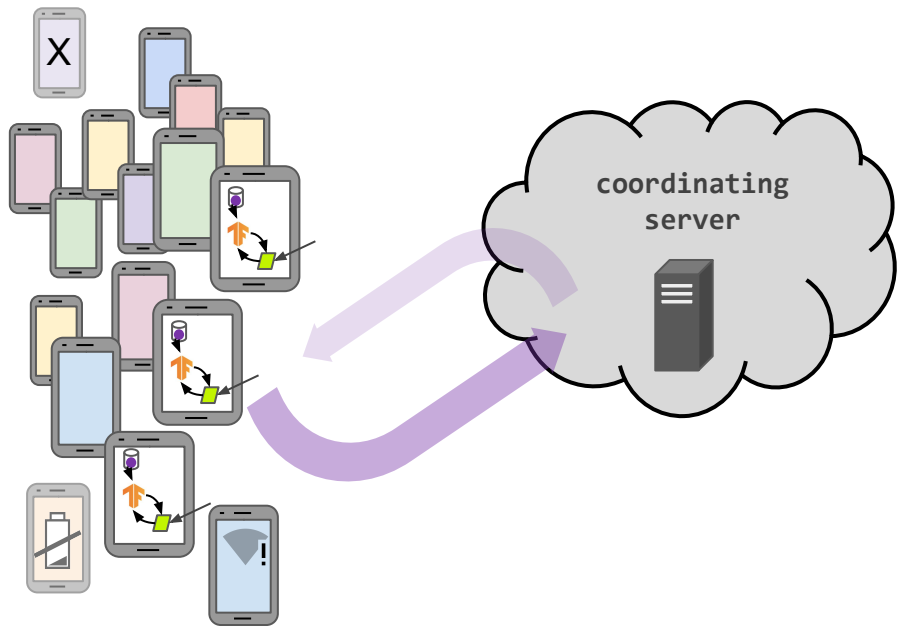
[2] <https://venturebeat.com/2020/04/15/healthcare-organizations-use-nvidias-clara-federated-learning-to-improve-mammogram-analysis-ai/>

[3] <https://medcitynews.com/2020/01/nvidia-says-it-has-a-solution-for-healthcares-data-problems/>

[4] <https://venturebeat.com/2020/06/23/nvidia-and-mercedes-benz-detail-self-driving-system-with-automated-routing-and-parking/>

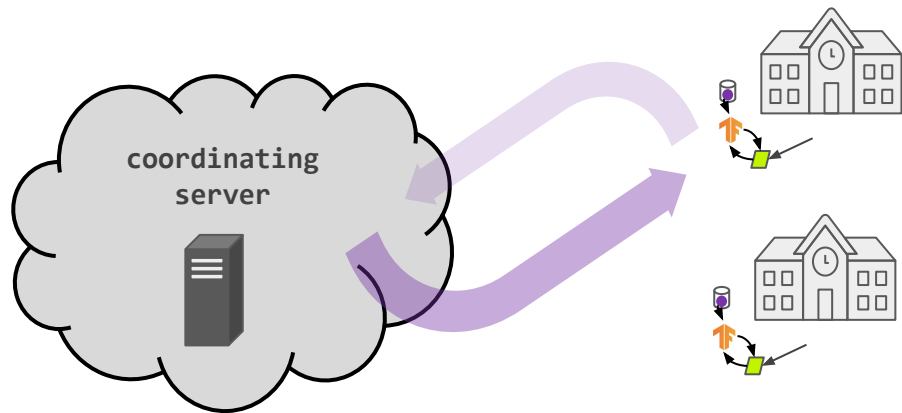
Cross-device federated learning

millions of intermittently
available client devices



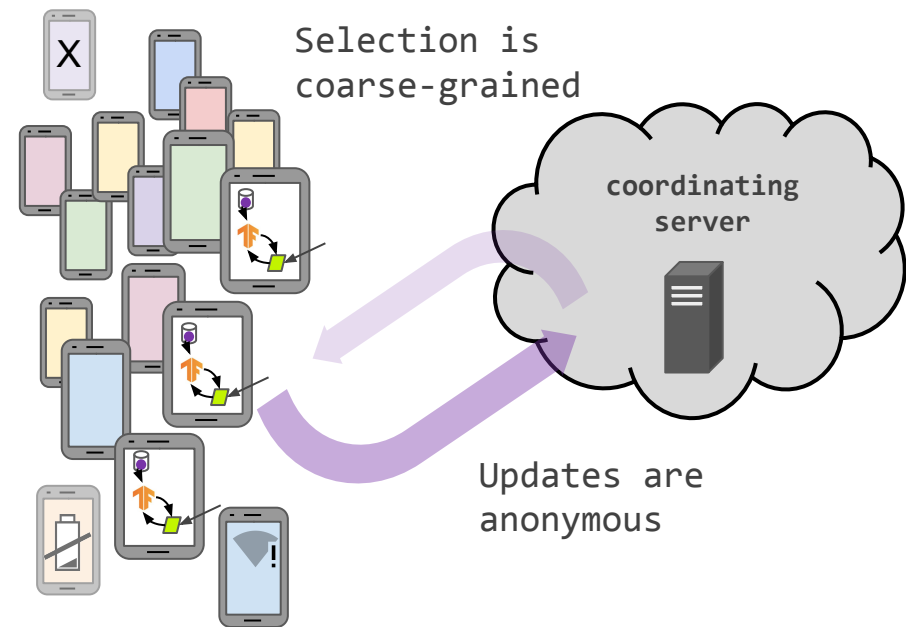
Cross-silo federated learning

small number of clients
(institutions, data silos),
high availability



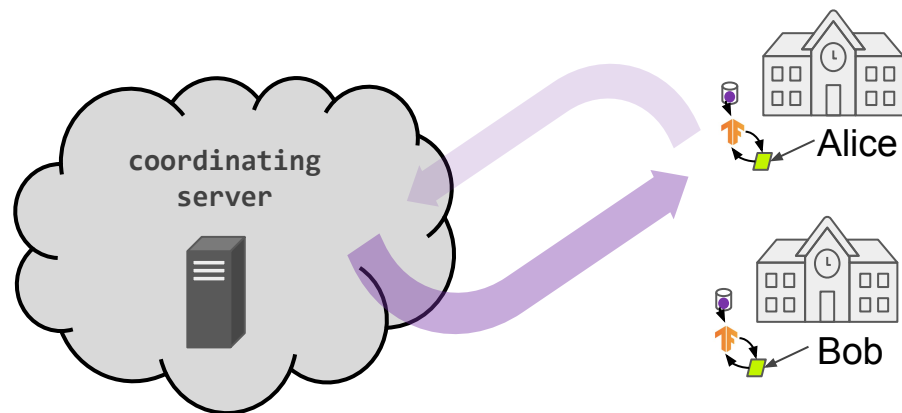
Cross-device federated learning

clients cannot be indexed directly (i.e., no use of client identifiers)



Cross-silo federated learning

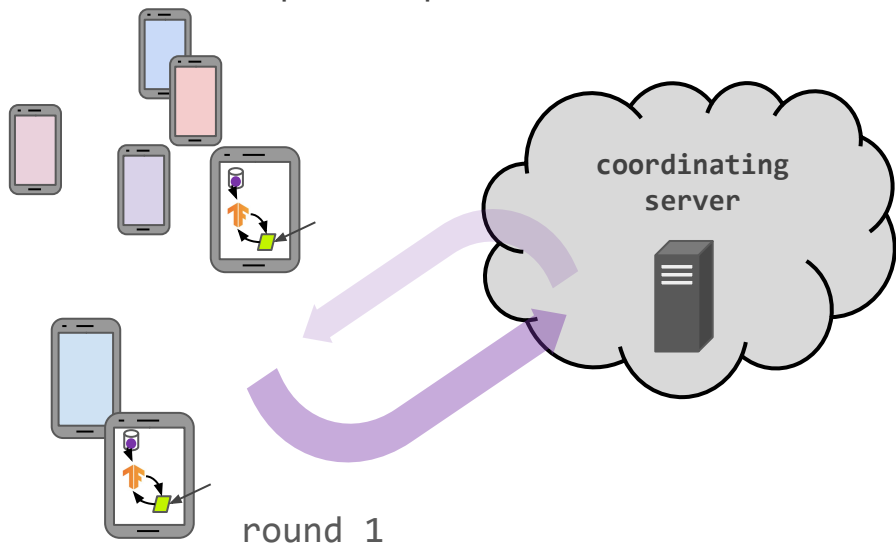
each client has an identity or name that allows the system to access it specifically



Cross-device federated learning

Server can only access a (possibly biased) random sample of clients on each round.

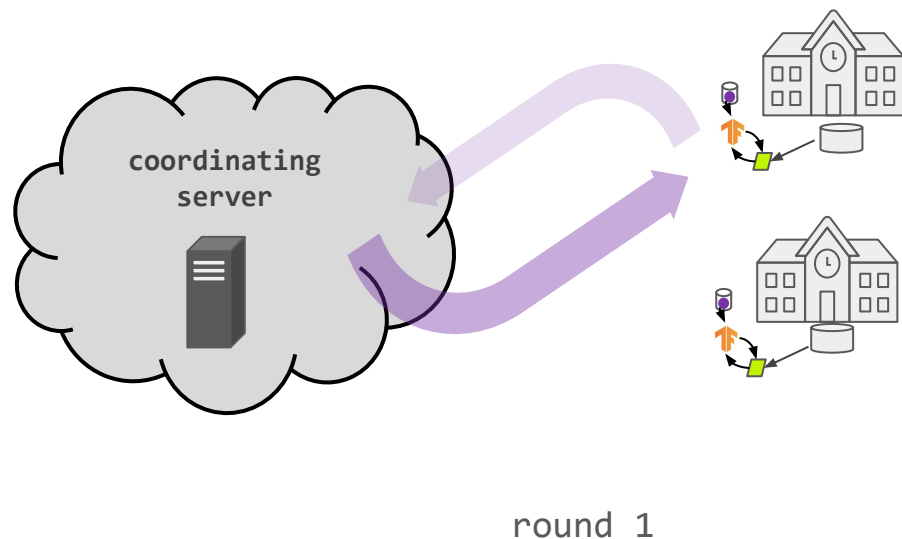
Large population => most clients only participate once.



Cross-silo federated learning

Most clients participate in every round.

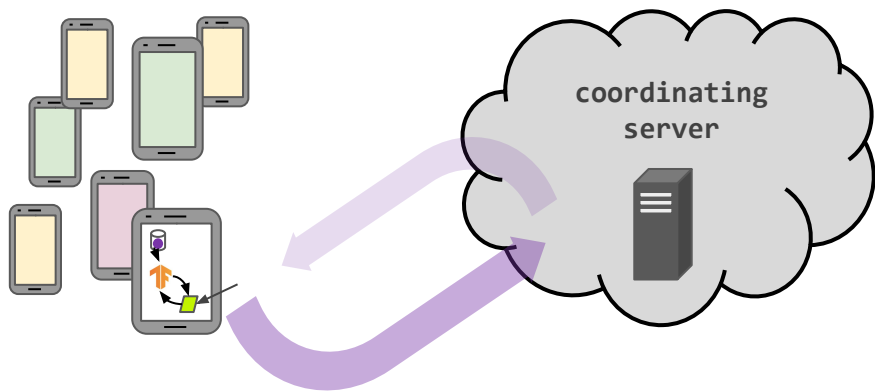
Clients can run algorithms that maintain local state across rounds.



Cross-device federated learning

Server can only access a (possibly biased) random sample of clients on each round.

Large population => most clients only participate once.

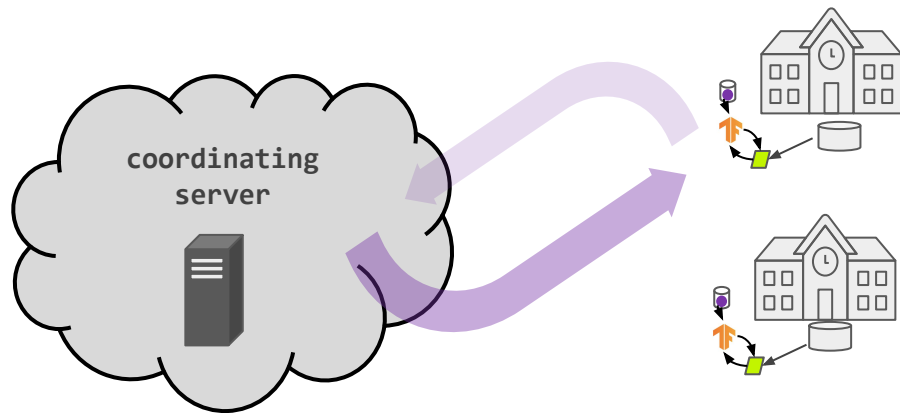


round 2
(completely new set of devices participate)

Cross-silo federated learning

Most clients participate in every round.

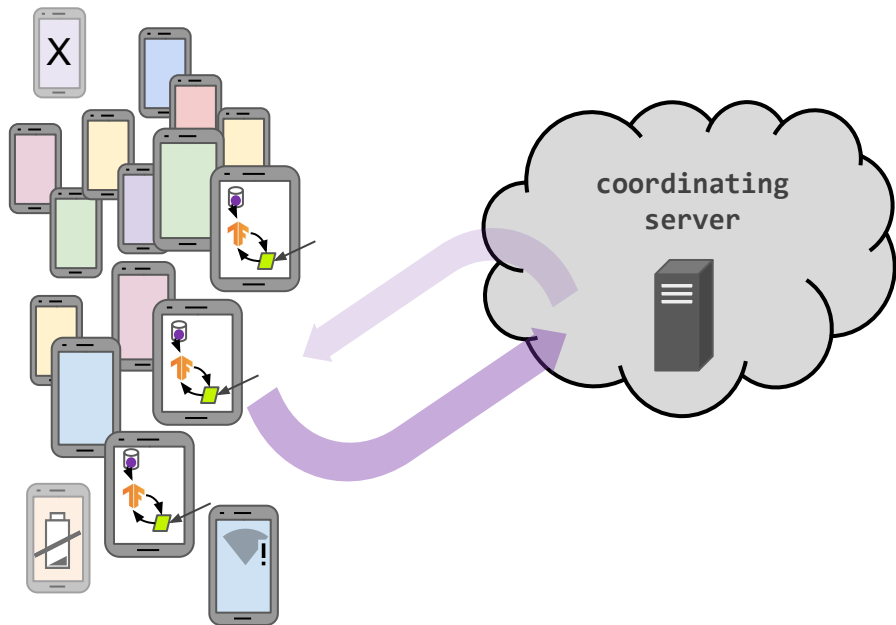
Clients can run algorithms that maintain local state across rounds.



round 2
(same clients)

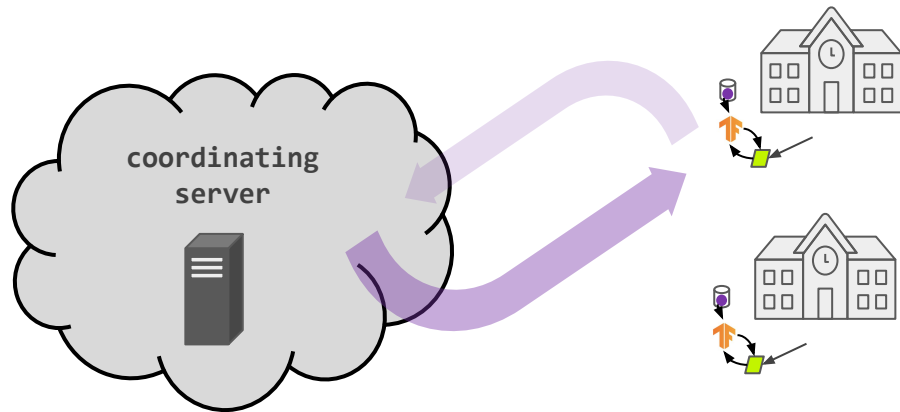
Cross-device federated learning

communication is often the
primary bottleneck



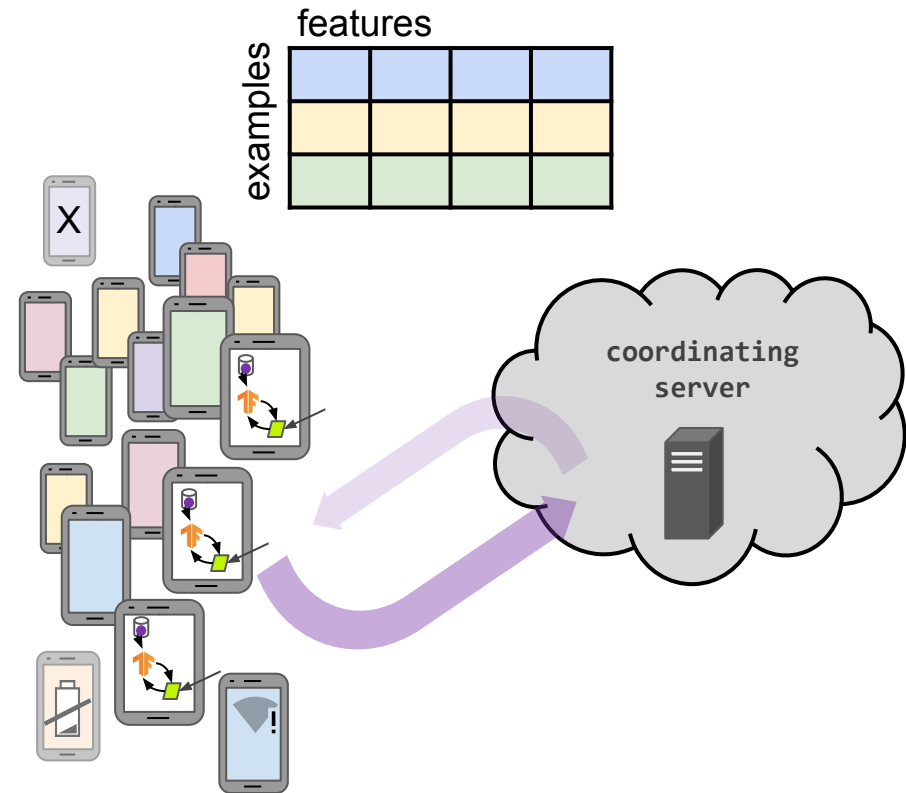
Cross-silo federated learning

communication or computation
might be the primary
bottleneck



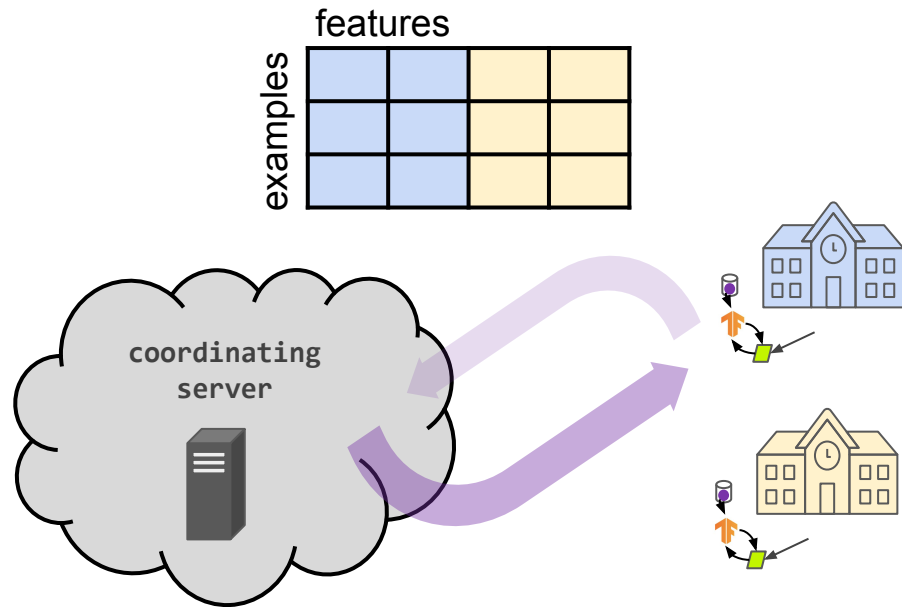
Cross-device federated learning

horizontally partitioned data

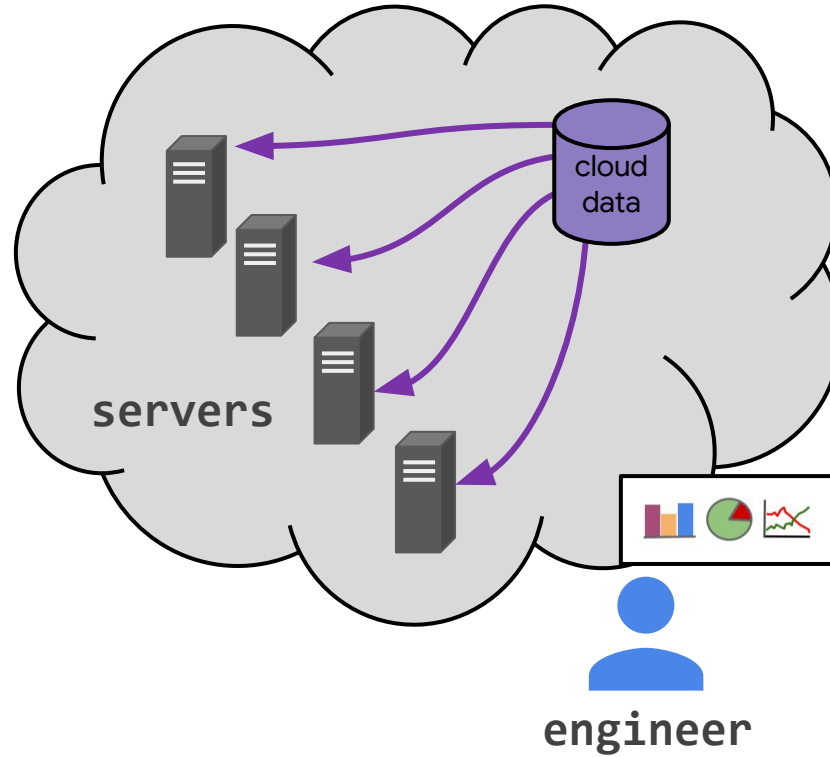


Cross-silo federated learning

horizontal or
vertically partitioned data



Distributed datacenter machine learning



FL terminology

- **Clients** - Compute nodes also holding local data, usually belonging to one entity:
 - IoT devices
 - Mobile devices
 - Data silos
 - Data centers in different geographic regions
- **Server** - Additional compute nodes that coordinate the FL process but don't access raw data. Usually not a single physical machine.

Characteristics of the federated learning setting

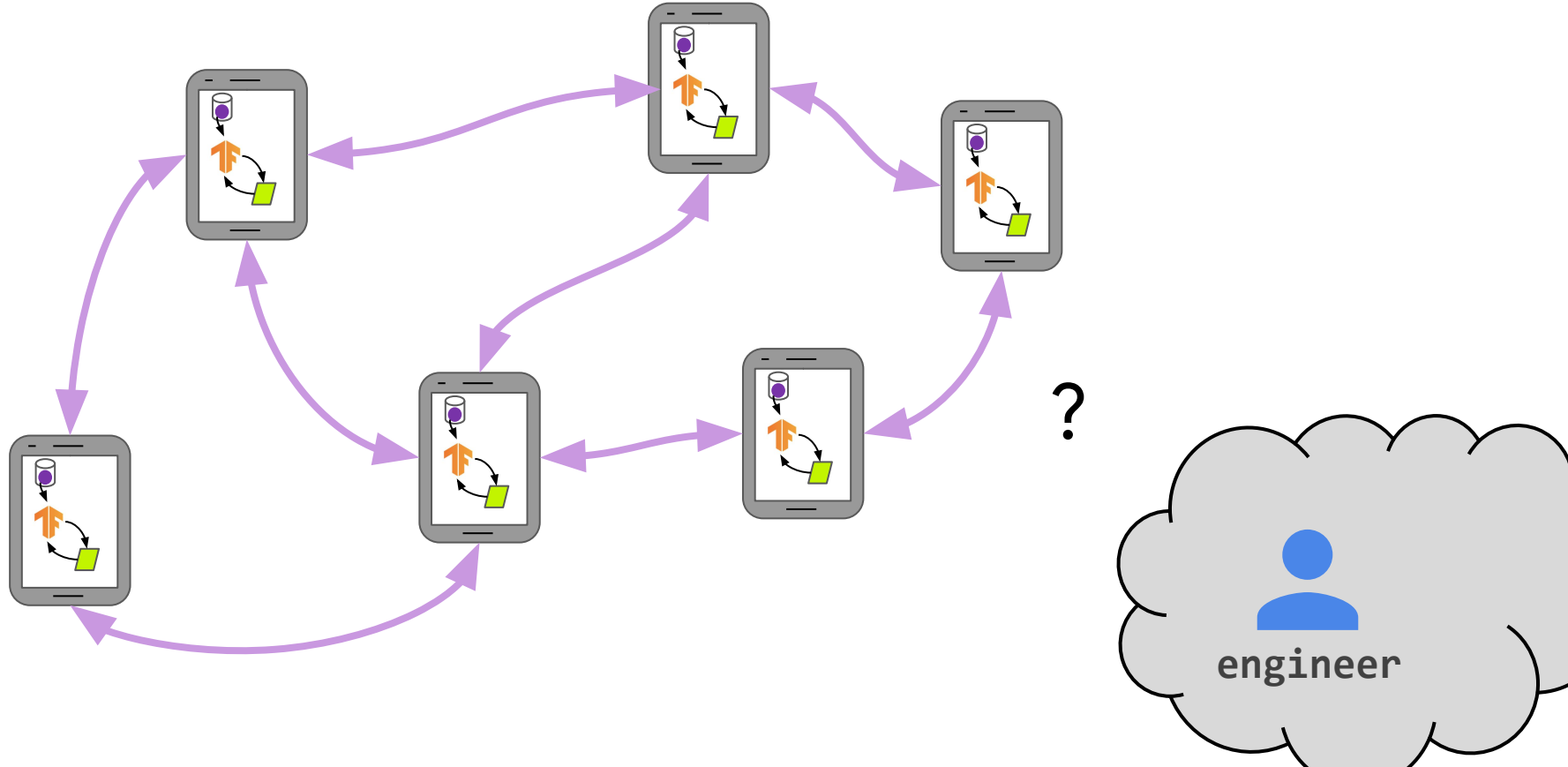
	Datacenter distributed learning	Cross-silo federated learning	Cross-device federated learning
Setting	Training a model on a large but "flat" dataset. Clients are compute nodes in a single cluster or datacenter.	Training a model on siloed data. Clients are different organizations (e.g., medical or financial) or datacenters in different geographical regions.	The clients are a very large number of mobile or IoT devices.
Data distribution	Data is centrally stored, so it can be shuffled and balanced across clients. Any client can read any part of the dataset.	Data is generated locally and remains decentralized. Each client stores its own data and cannot read the data of other clients. Data is not independently or identically distributed.	
Orchestration	Centrally orchestrated.	A central orchestration server/service organizes the training, but never sees raw data.	
Wide-area communication	None (fully connected clients in one datacenter/cluster).	Typically hub-and-spoke topology, with the hub representing a coordinating service provider (typically without data) and the spokes connecting to clients.	
Data availability	All clients are almost always available.		Only a fraction of clients are available at any one time, often with diurnal and other variations.
Distribution scale	Typically 1 - 1000 clients.	Typically 2 - 100 clients.	Massively parallel, up to 10^{10} clients.

Characteristics of the federated learning setting

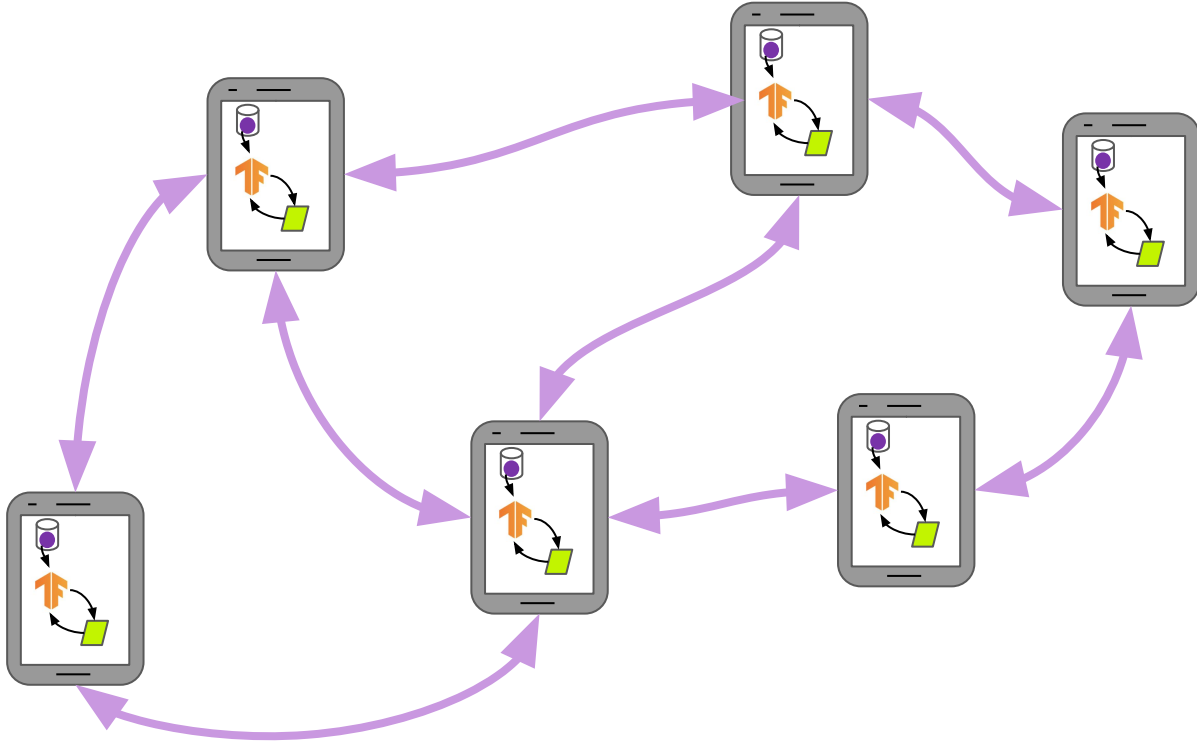
	Datacenter distributed learning	Cross-silo federated learning	Cross-device federated learning
Addressability	Each client has an identity or name that allows the system to access it specifically.		Clients cannot be indexed directly (i.e., no use of client identifiers)
Client statefulness	Stateful --- each client may participate in each round of the computation, carrying state from round to round.		Generally stateless --- each client will likely participate only once in a task, so generally we assume a fresh sample of never before seen clients in each round of computation.
Primary bottleneck	Computation is more often the bottleneck in the datacenter, where very fast networks can be assumed.	Might be computation or communication.	Communication is often the primary bottleneck, though it depends on the task. Generally, federated computations uses wi-fi or slower connections.
Reliability of clients	Relatively few failures.		Highly unreliable --- 5% or more of the clients participating in a round of computation are expected to fail or drop out (e.g., because the device becomes ineligible when battery, network, or idleness requirements for training/computation are violated).
Data partition axis	Data can be partitioned / re-partitioned arbitrarily across clients.	Partition is fixed. Could be example-partitioned (horizontal) or feature-partitioned (vertical).	Fixed partitioning by example (horizontal).

Adapted from Table 1 in *Advances and Open Problems in Federated Learning* ([arxiv/1912.04977](https://arxiv.org/abs/1912.04977))

Fully decentralized (peer-to-peer) learning



Fully decentralized (peer-to-peer) learning

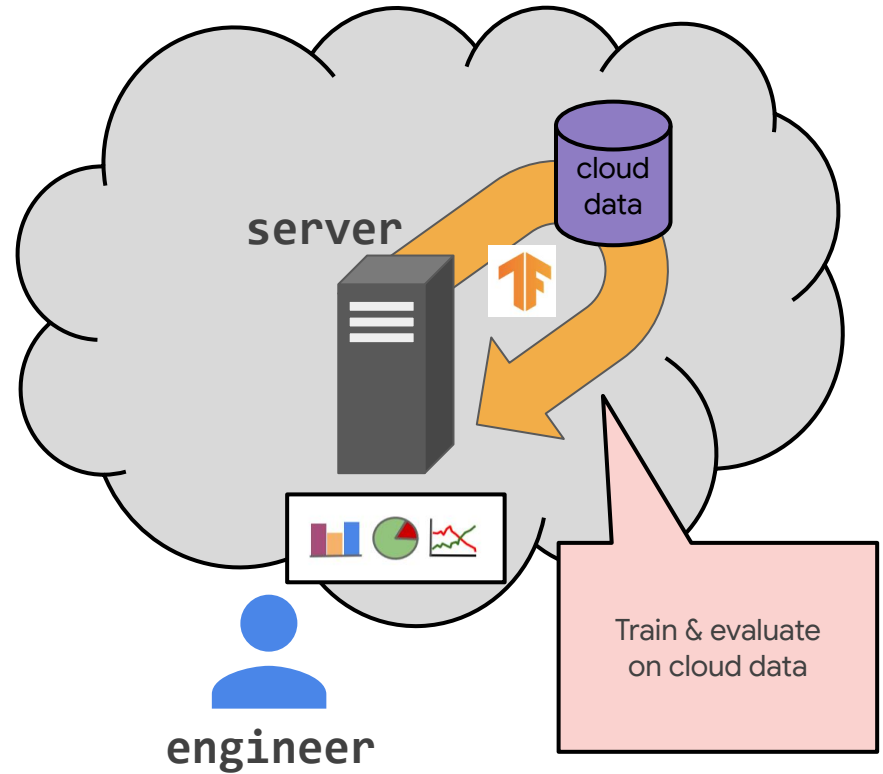


Characteristics of FL vs decentralized learning

	Federated learning	Fully decentralized (peer-to-peer) learning
Orchestration	A central orchestration server/service organizes the training, but never sees raw data.	No centralized orchestration.
Wide-area communication pattern	Typically hub-and-spoke topology, with the hub representing a coordinating service provider (typically without data) and the spokes connecting to clients.	Peer-to-peer topology.

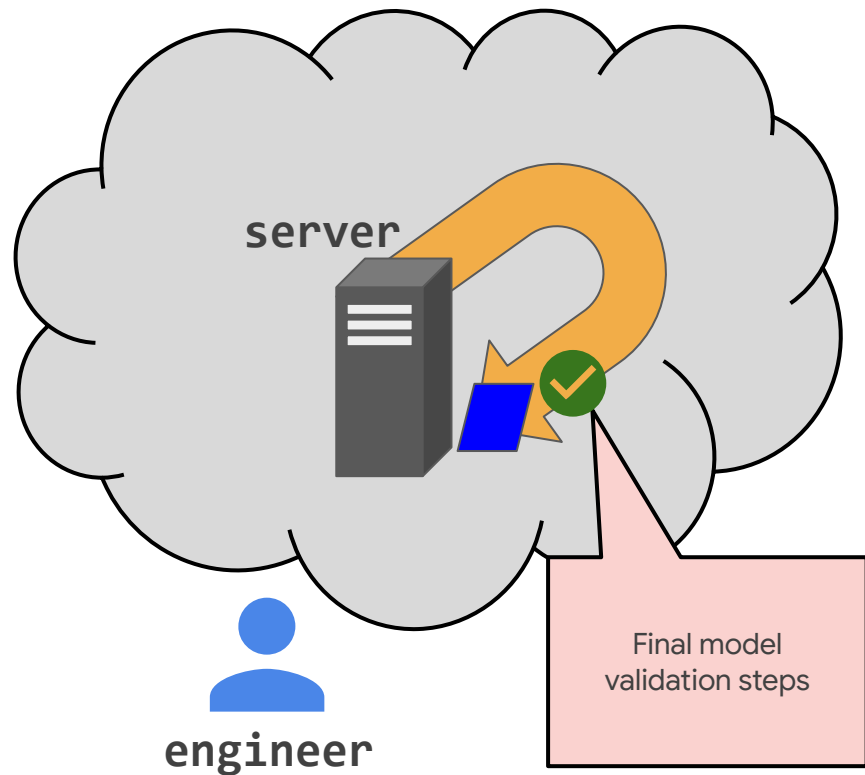
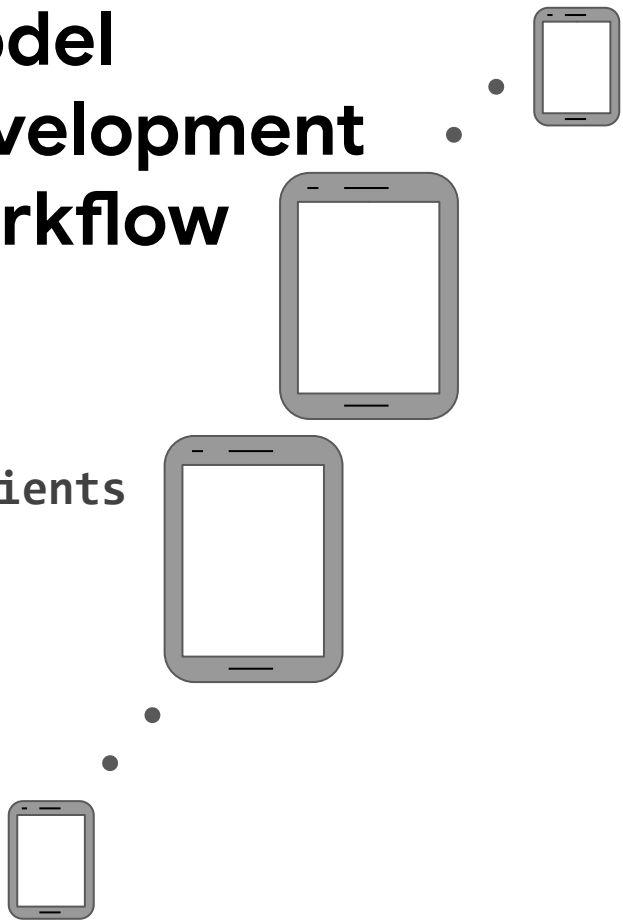
Cross-Device Federated Learning

Model development workflow

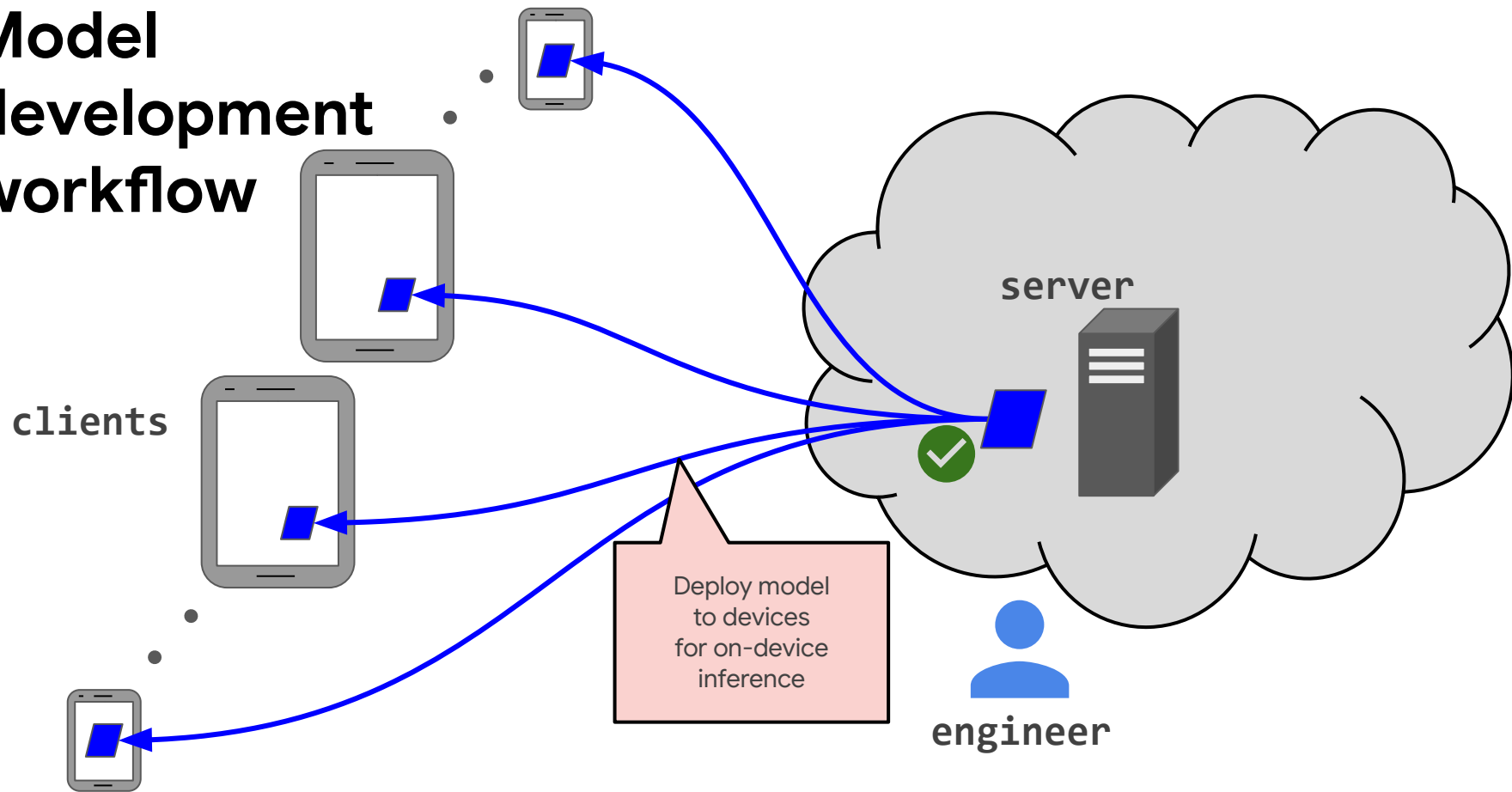


Model development workflow

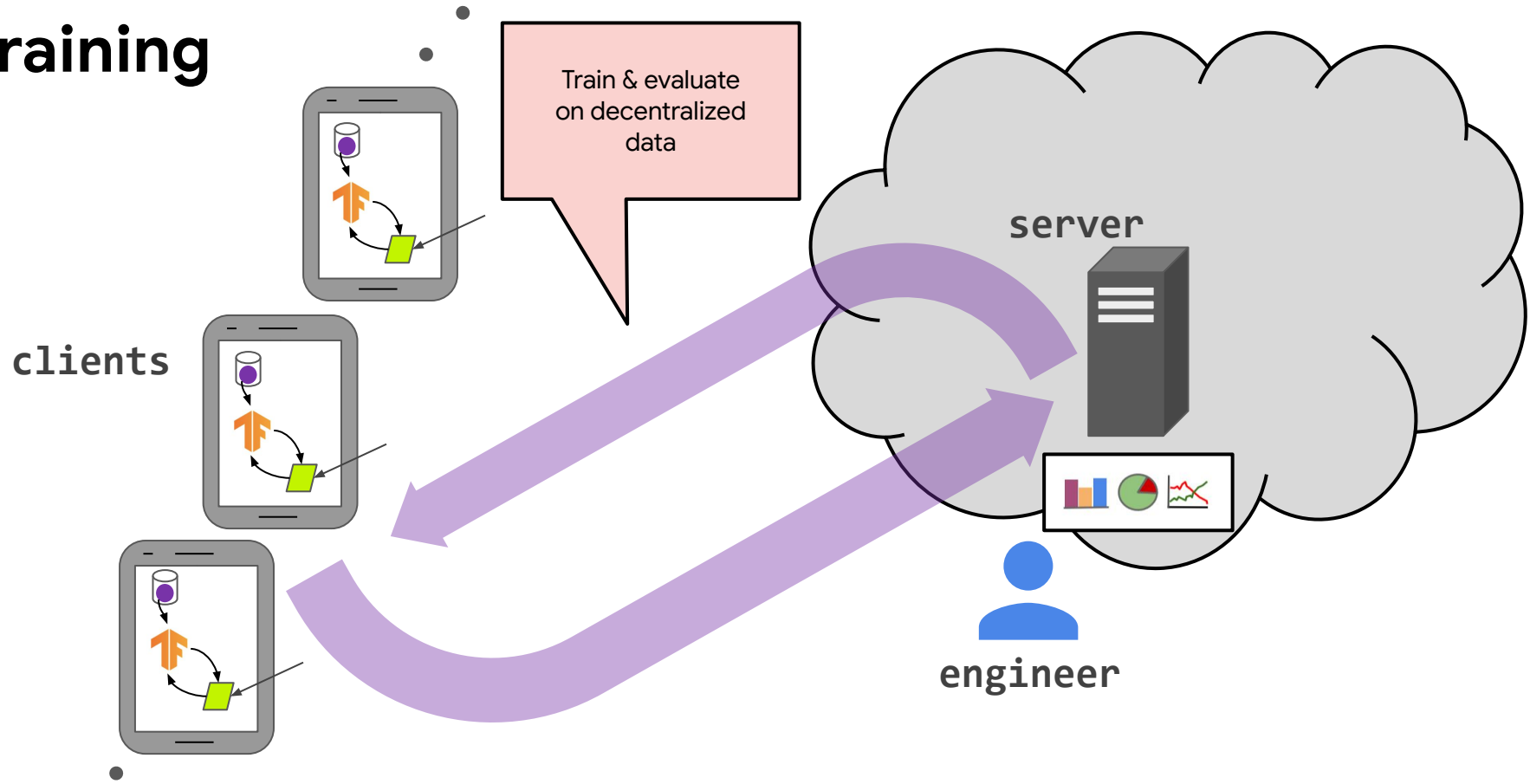
clients



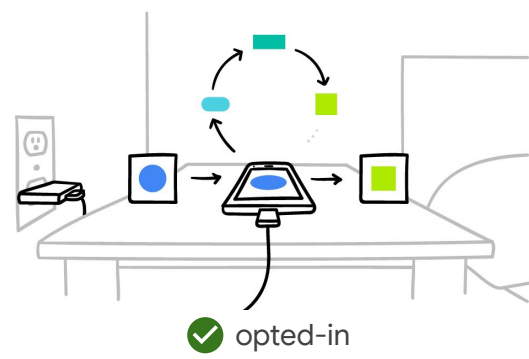
Model development workflow



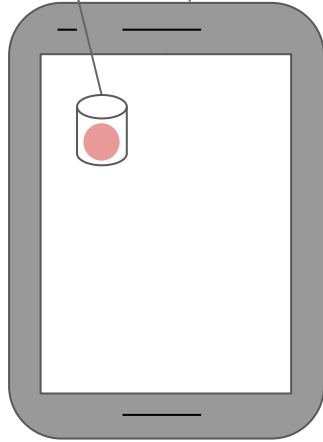
Federated training



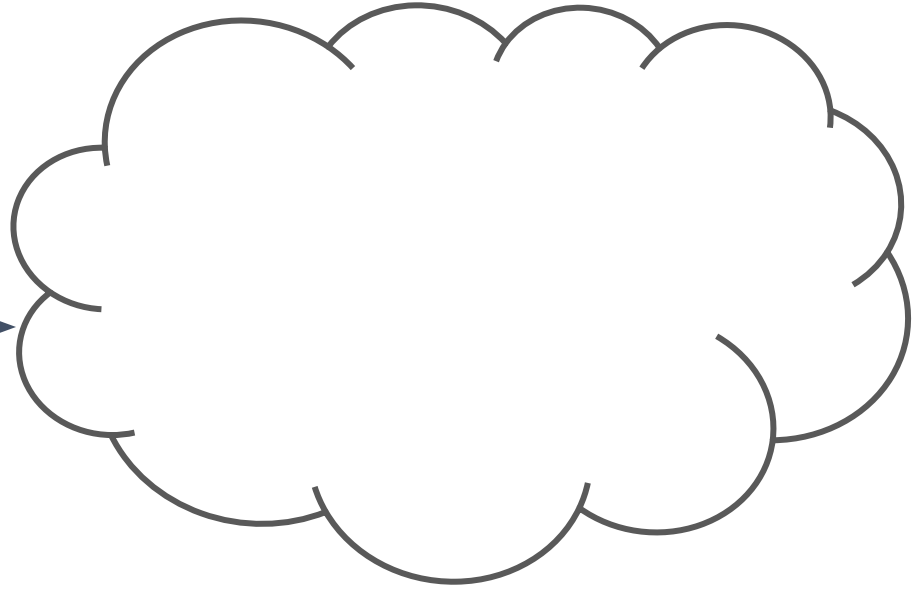
Federated learning



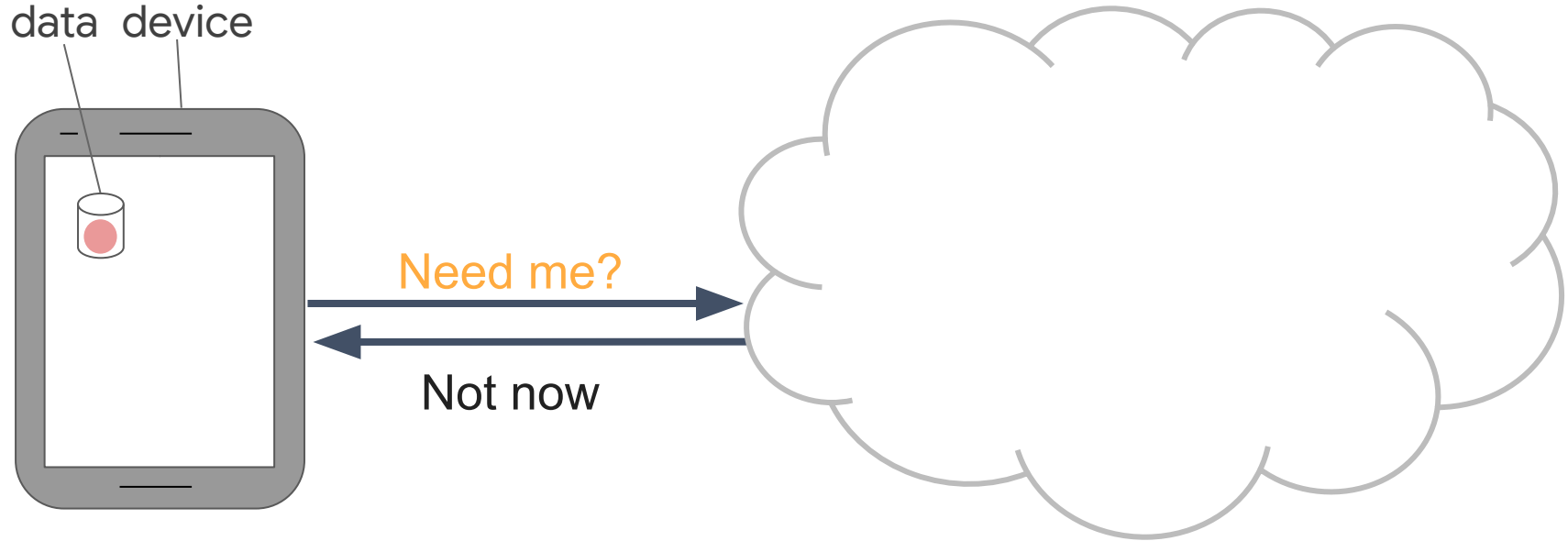
data device



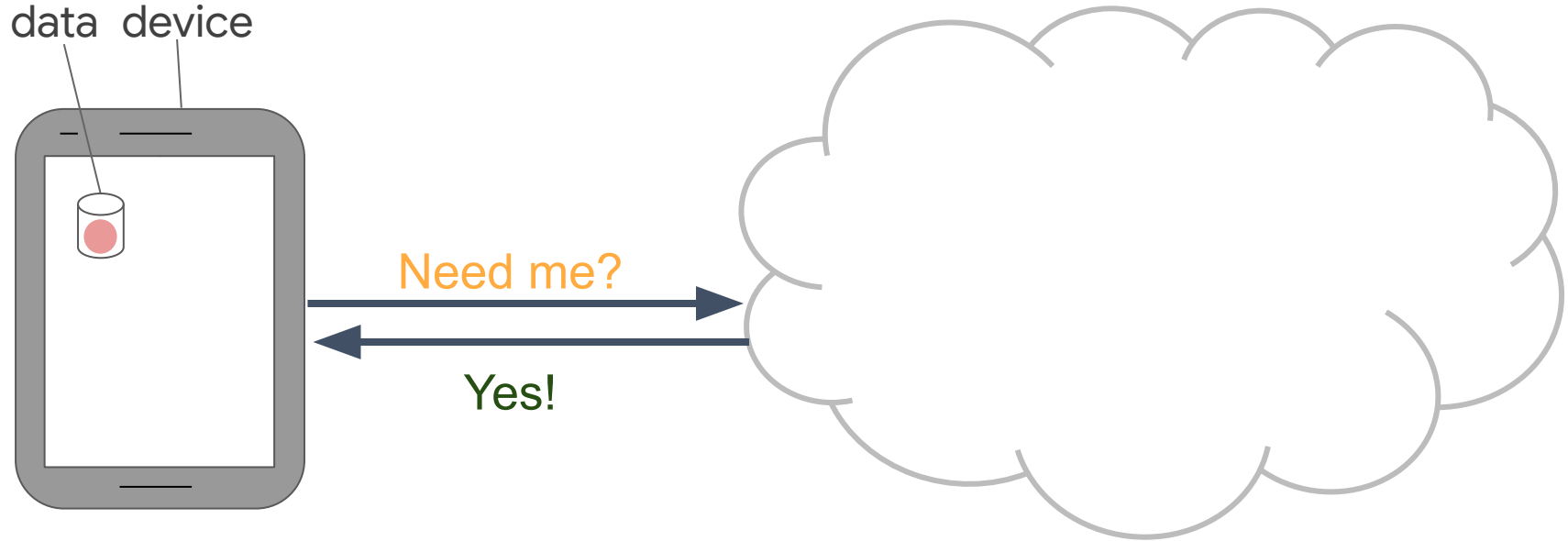
Need me?



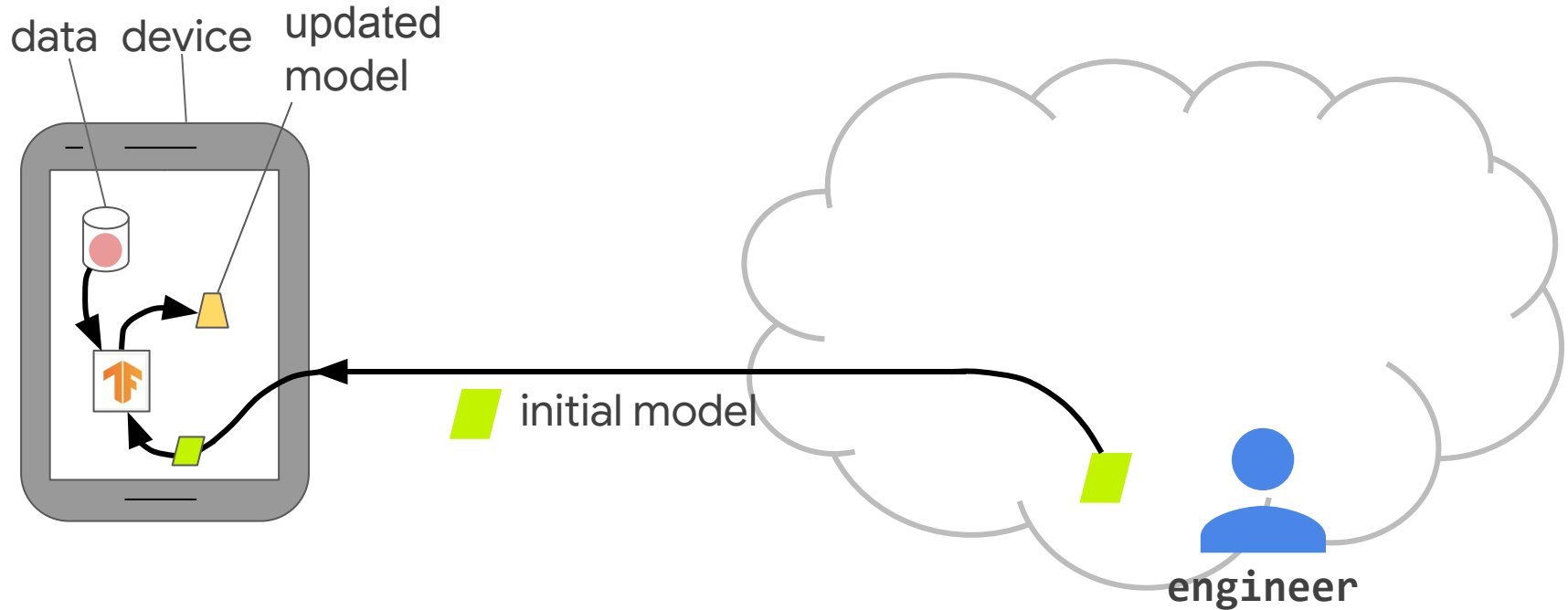
Federated learning



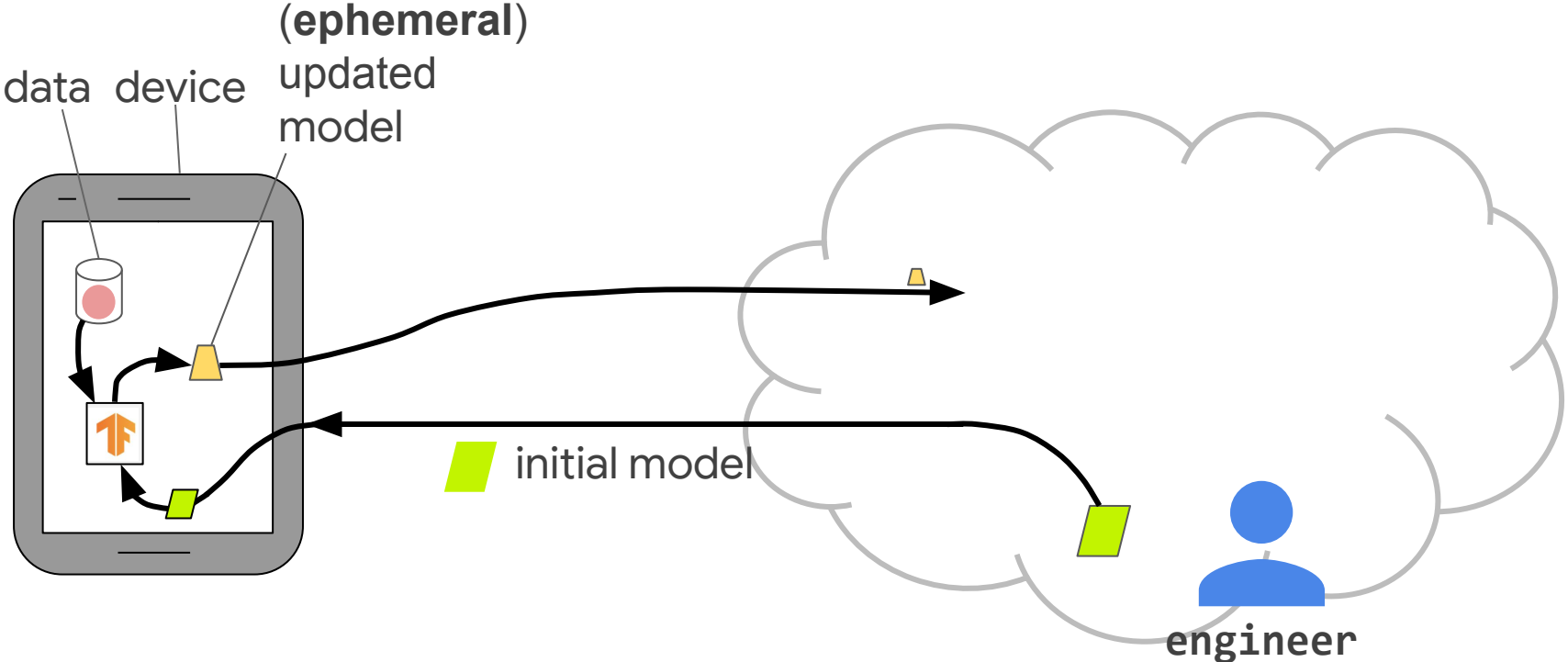
Federated learning



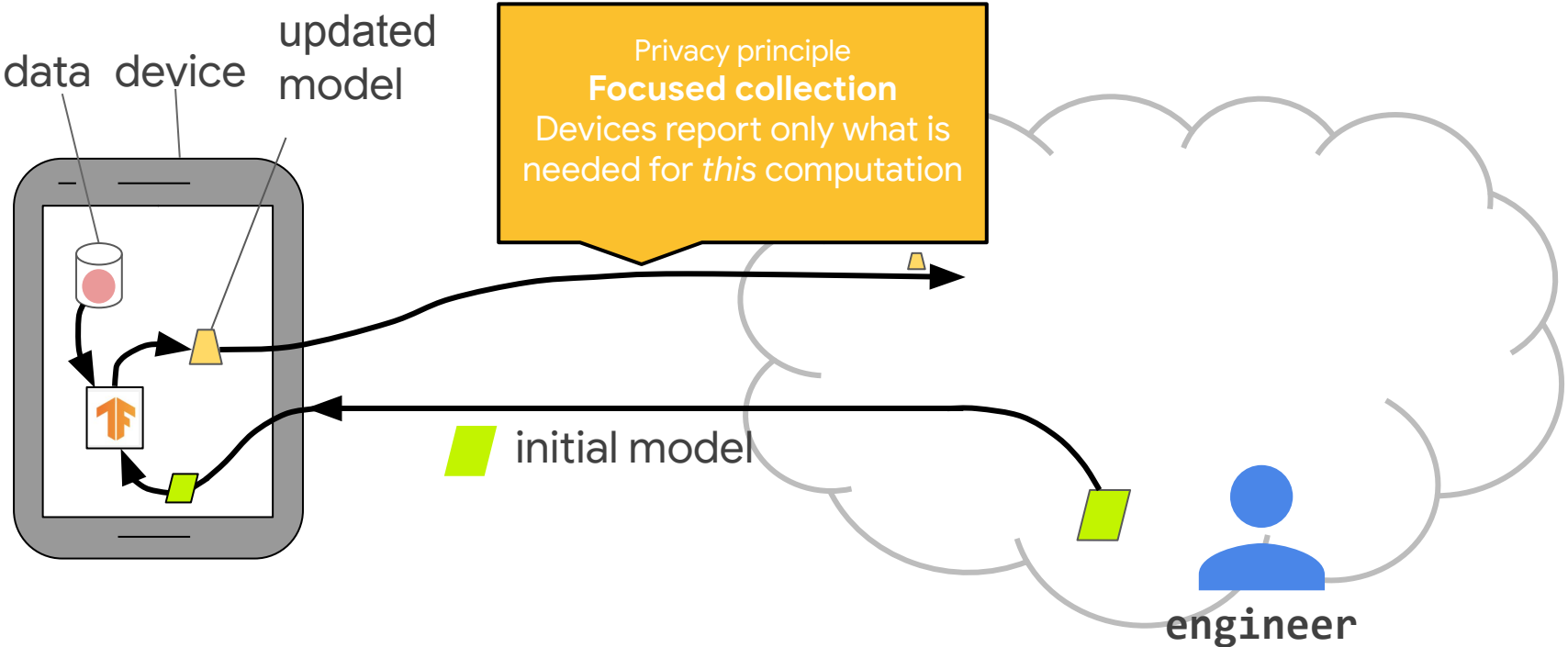
Federated learning



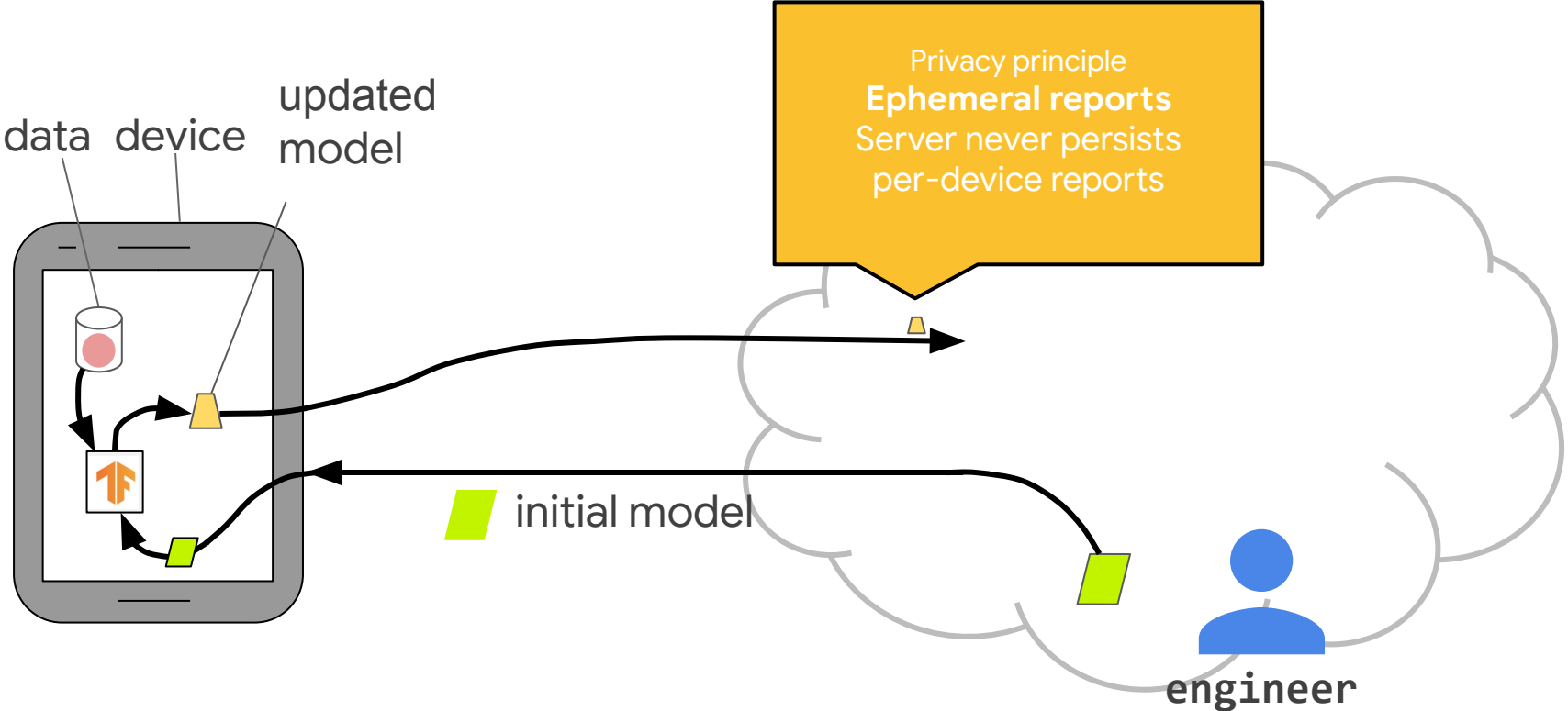
Federated learning



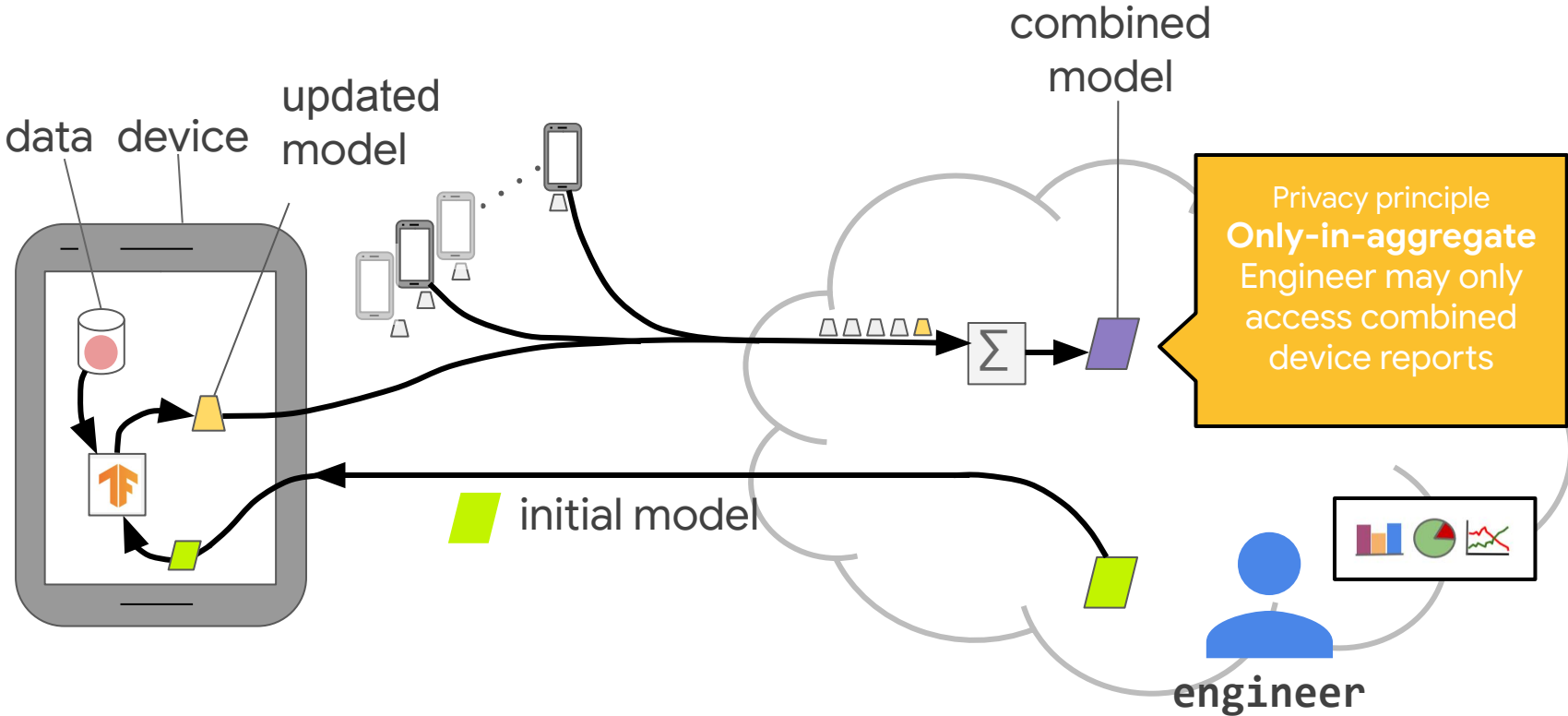
Federated learning



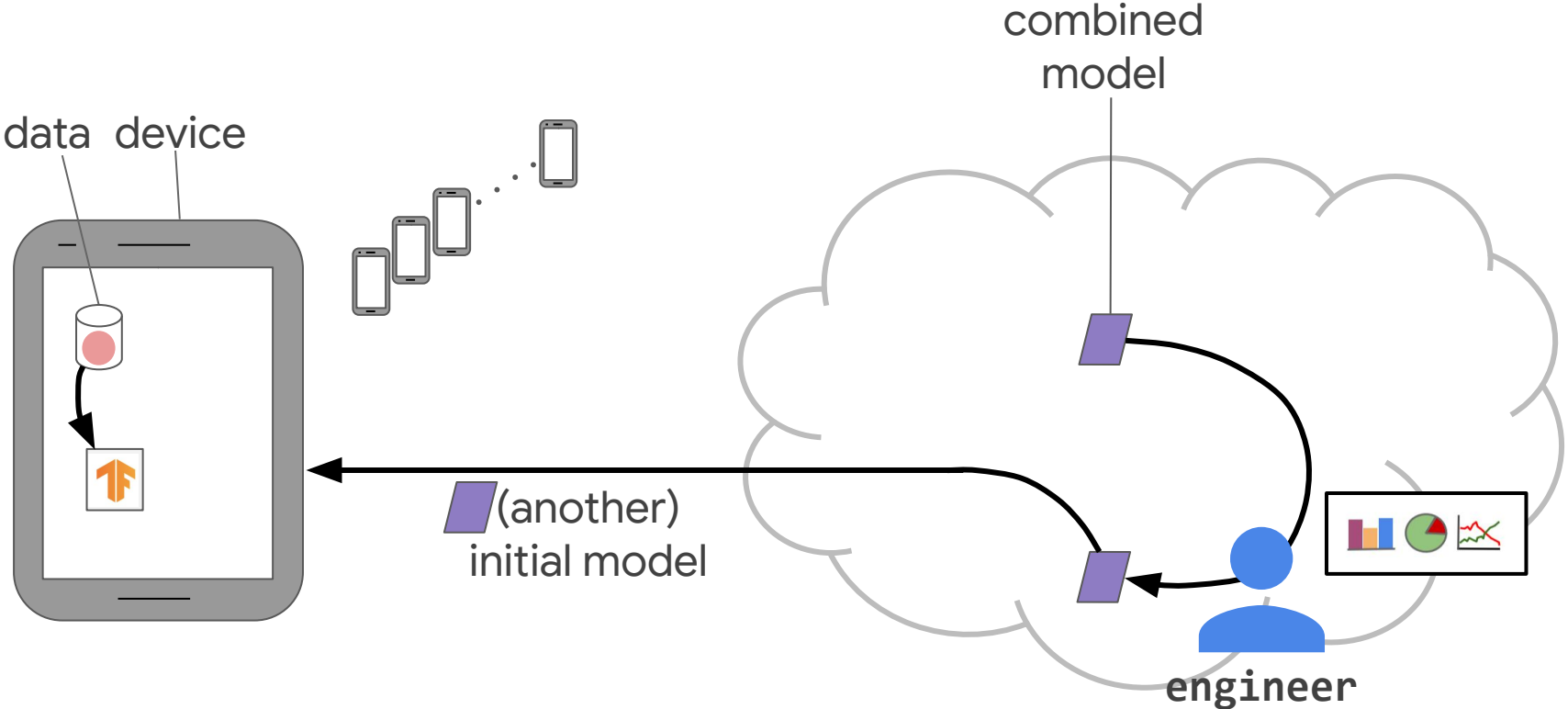
Federated learning



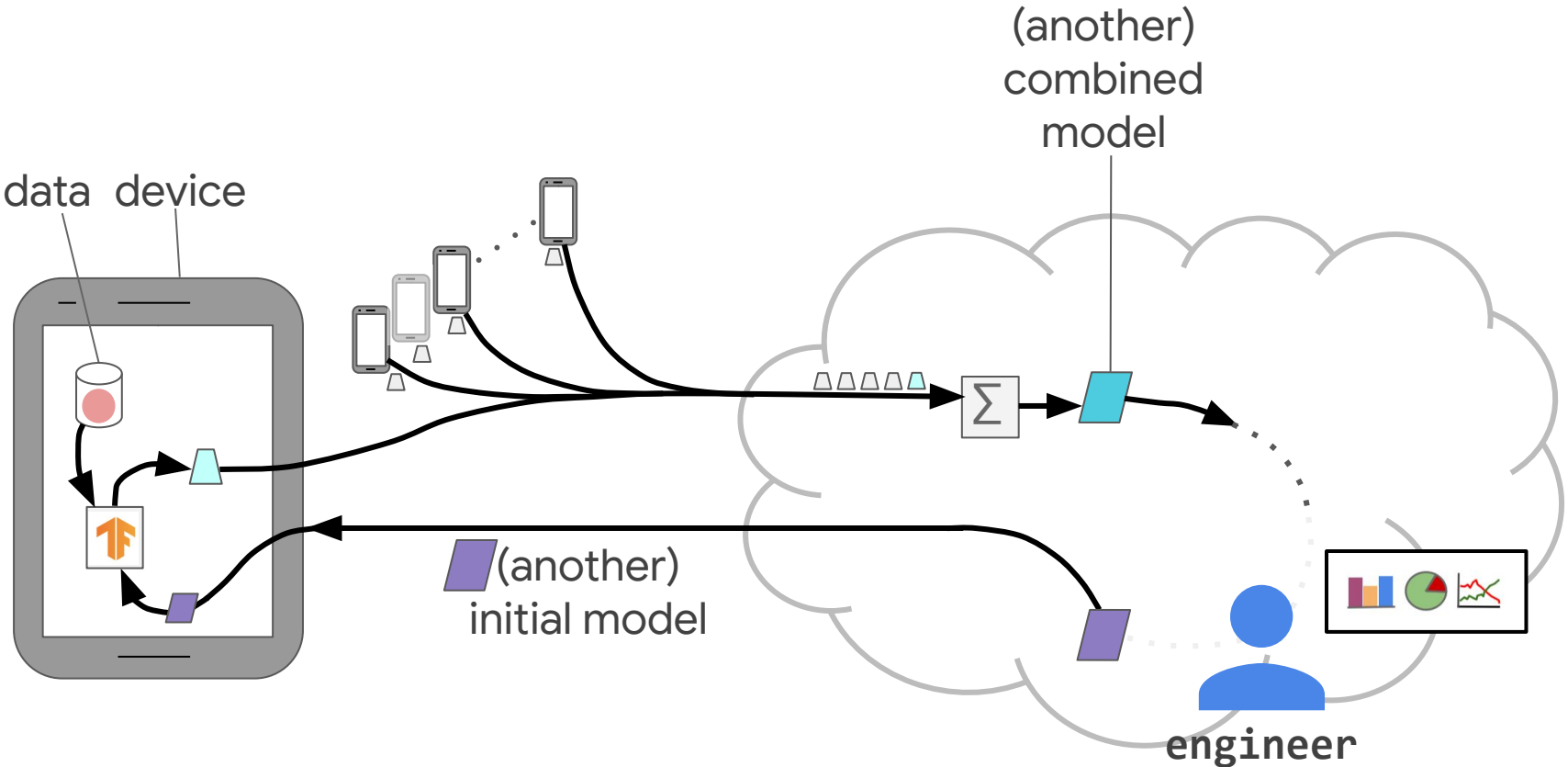
Federated learning



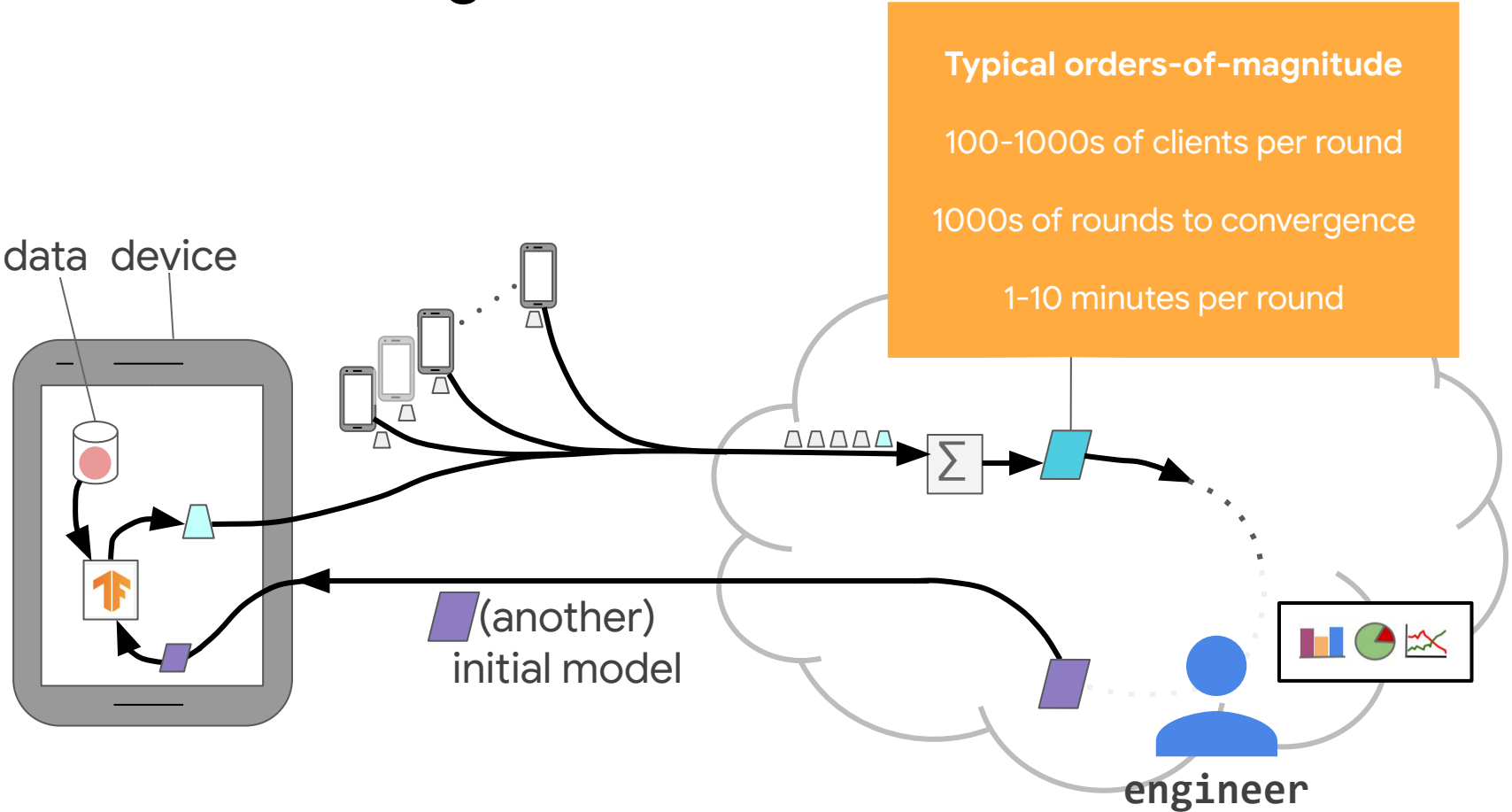
Federated learning



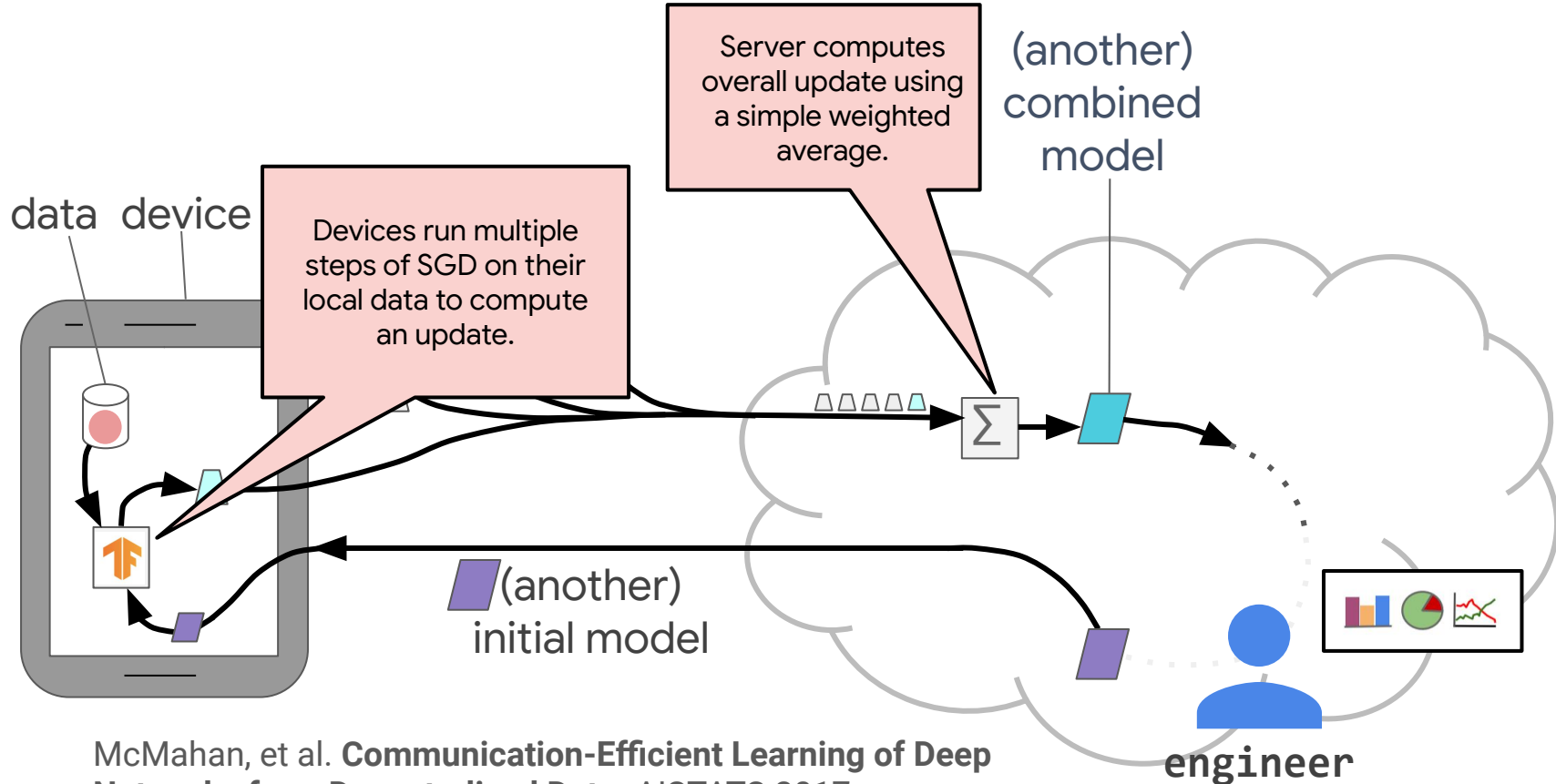
Federated learning



Federated learning



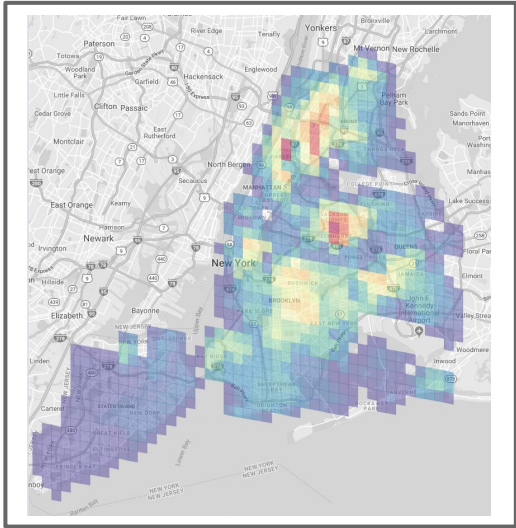
Federated Averaging (FedAvg) algorithm



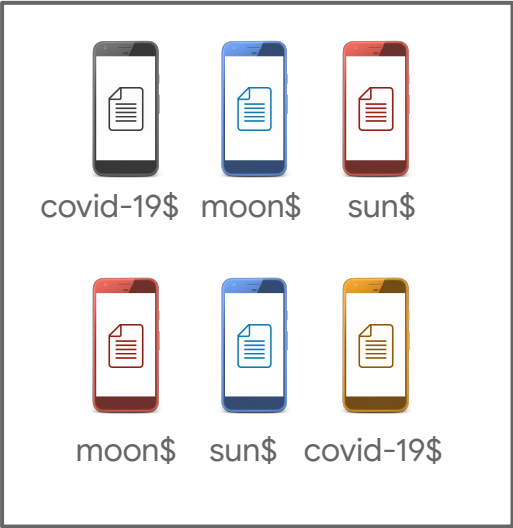
McMahan, et al. **Communication-Efficient Learning of Deep Networks from Decentralized Data**. AISTATS 2017.

Beyond Learning: Federated Analytics

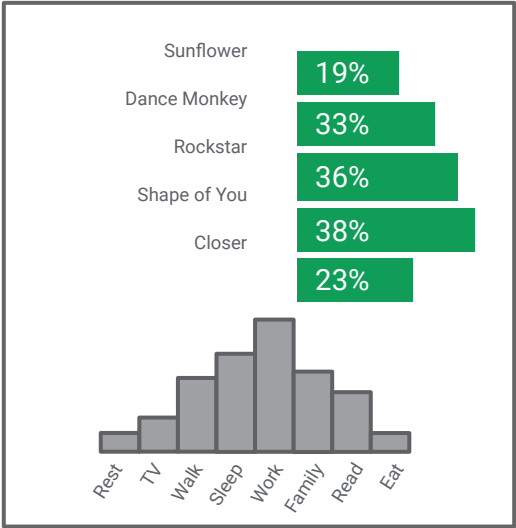
Beyond learning: federated analytics



Geo-location heatmaps



Frequently typed
out-of-dictionary words



Popular songs, trends,
and activities

Federated analytics

Federated analytics is the practice of applying data science methods to the analysis of raw data that is stored locally on users' devices. Like federated learning, it works by running local computations over each device's data, and only making the aggregated results — and never any data from a particular device — available to product engineers. Unlike federated learning, however, federated analytics aims to support basic data science needs.

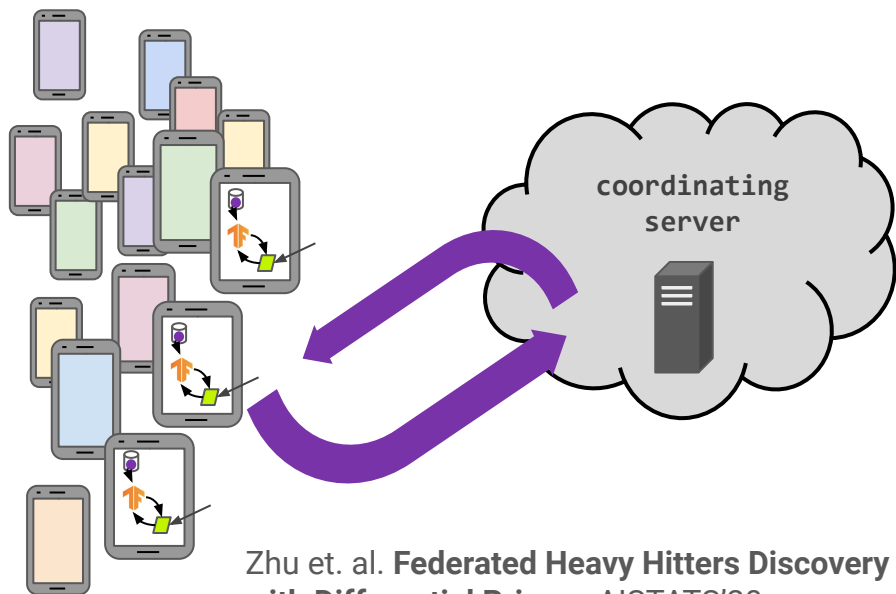
definition proposed in <https://ai.googleblog.com/2020/05/federated-analytics-collaborative-data.html>

Federated analytics

- Federated histograms over closed sets
- Federated quantiles and distinct element counts
- Federated heavy hitters discovery over open sets
- Federated density of vector spaces
- Federated selection of random data subsets
- Federated SQL
- Federated computations?
- etc...

Interactive algorithms

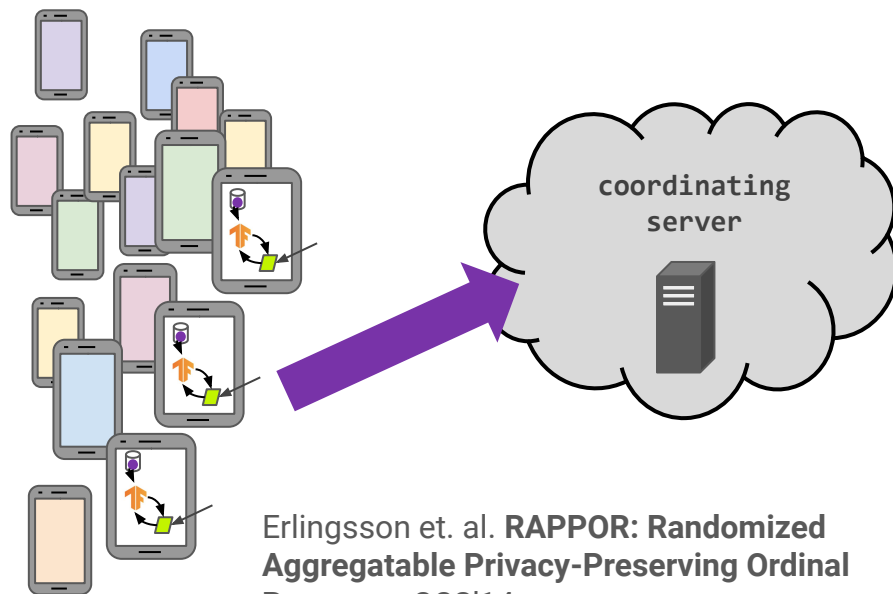
Similar to learning, the on-device computation is a function of a server state



Zhu et. al. **Federated Heavy Hitters Discovery with Differential Privacy** AISTATS'20.

Non-interactive algorithms

Unlike learning, the on-device computation does not depend on a server state



Erlingsson et. al. **RAPPOR: Randomized Aggregatable Privacy-Preserving Ordinal Response** CCS'14.

Part II: Privacy for Federated Learning and Analytics

Aspects of Privacy

The why, what, and how of using data.

Why?

Transparency & consent

Why use this data? The user understands and supports the intended use of the data.

What?

Limited influence of any individual

What is computed? When data is released, ensure it does not reveal any user's private information.

How?

Security & data minimization

How and where does the computation happen? Release data to as few parties as possible. Minimize the attack surface where private information could be accessed.

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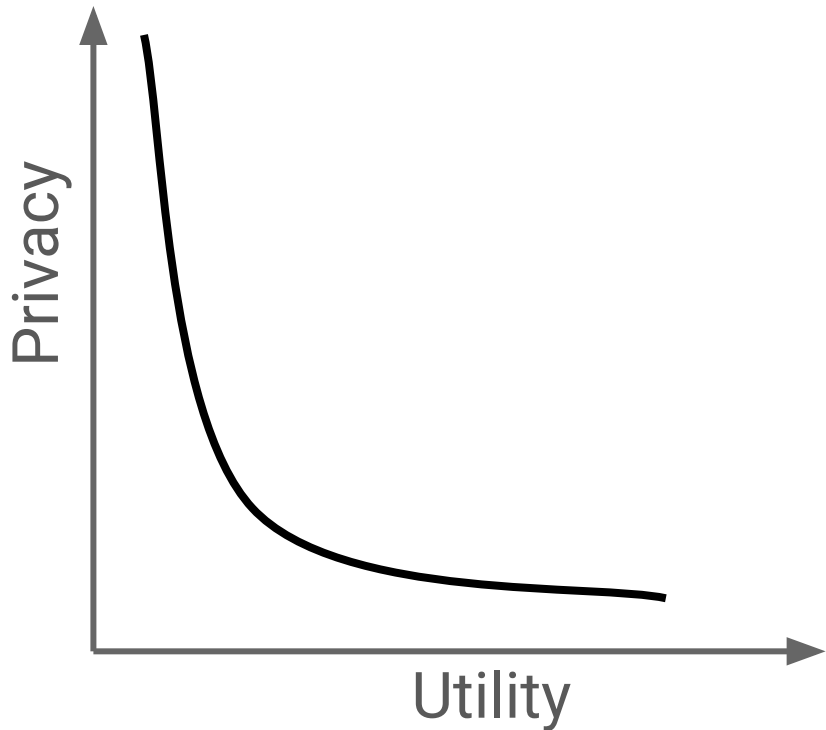
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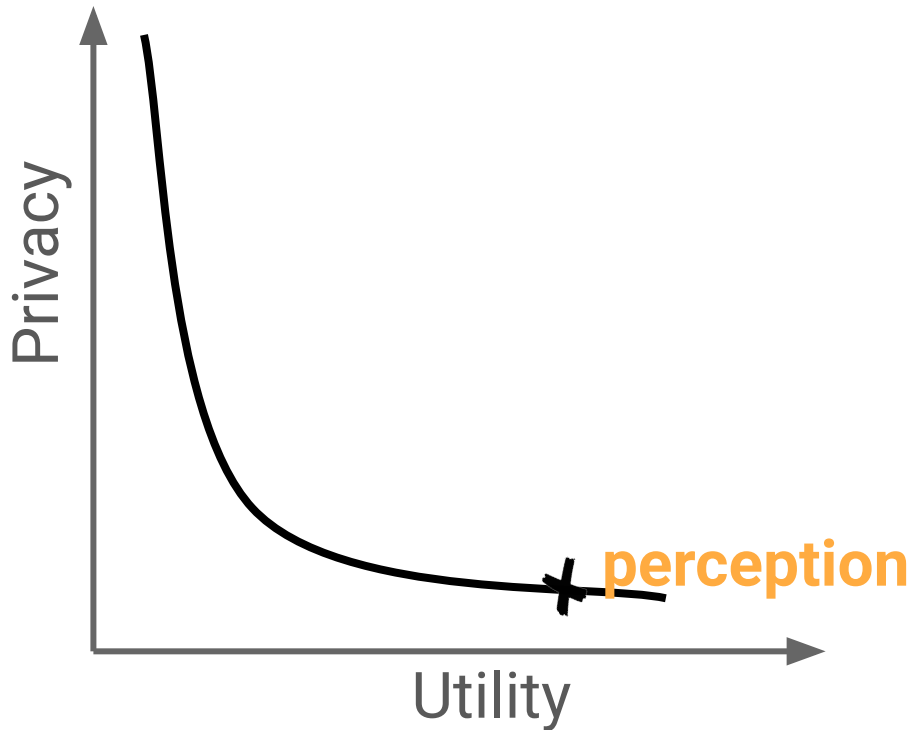
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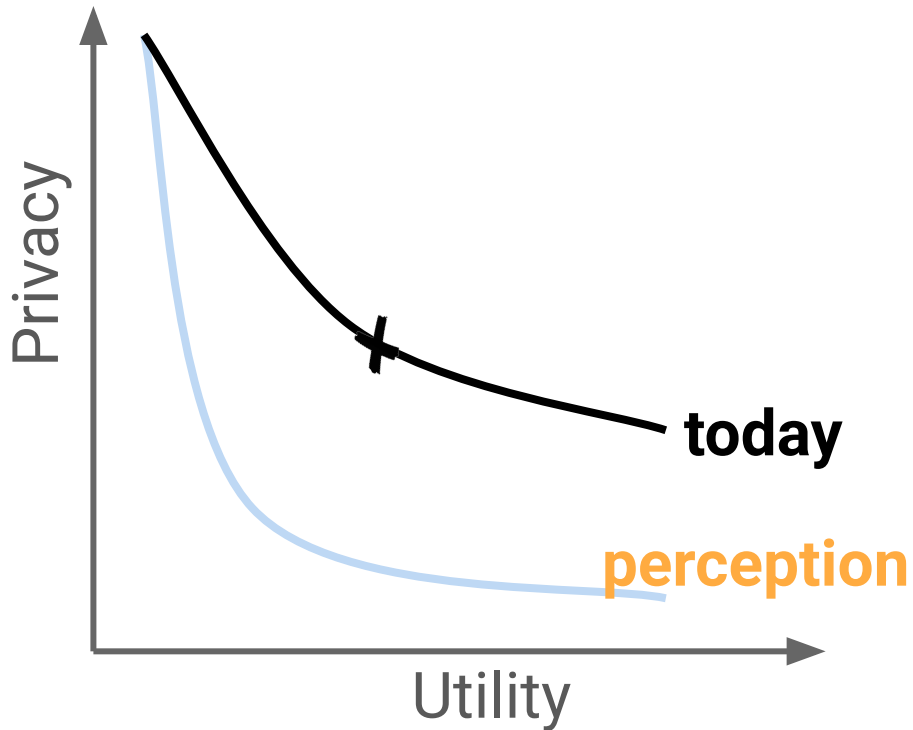
ML on sensitive data: privacy vs. utility



ML on sensitive data: privacy vs. utility

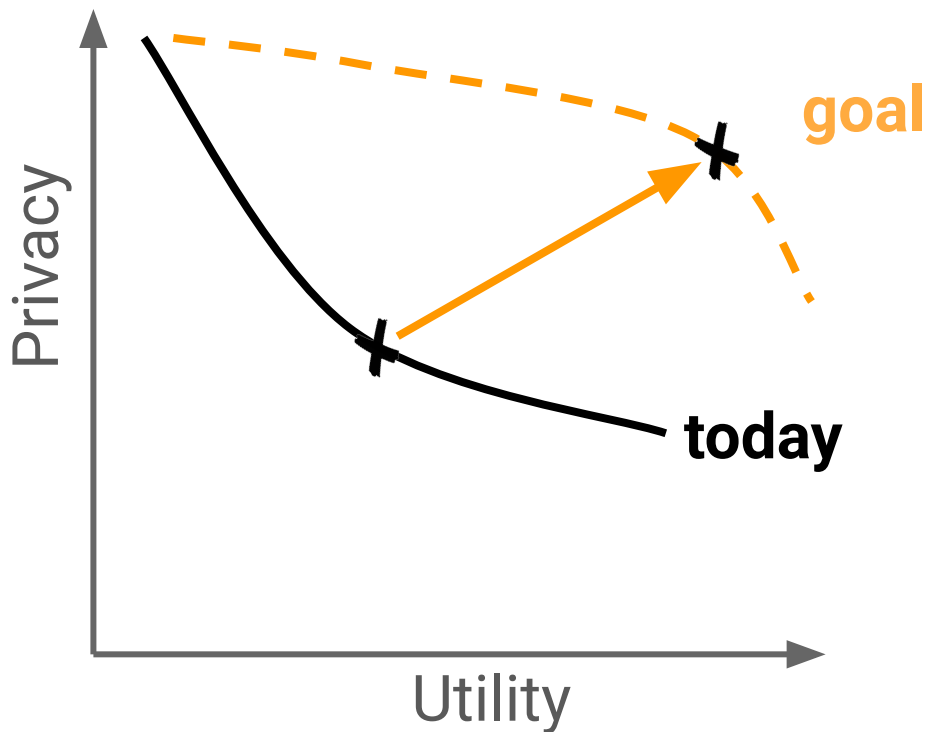


ML on sensitive data: privacy vs. utility



- 1. Policy
- 2. Technology

ML on sensitive data: privacy vs. utility (?)



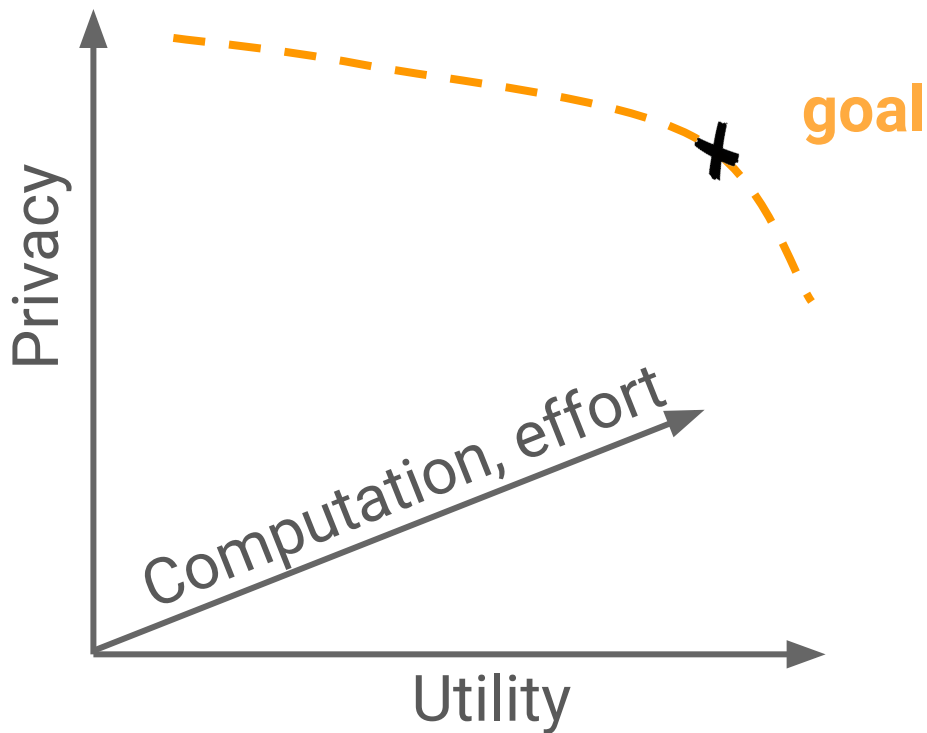
1. Policy

2. New Technology

Push the pareto frontier
with better technology.

Make achieving high
privacy and utility
possible.

ML on sensitive data: privacy vs. utility (?)

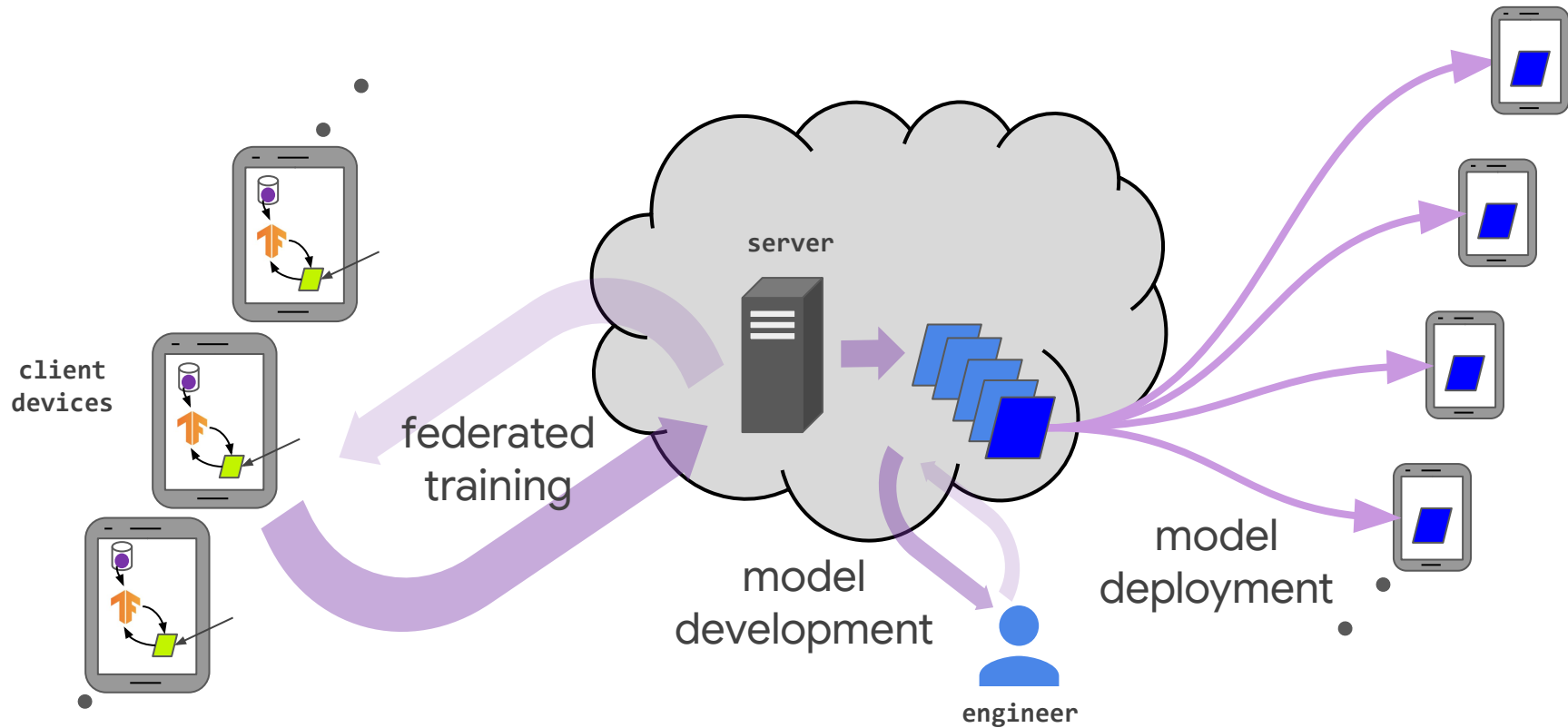


1. Policy

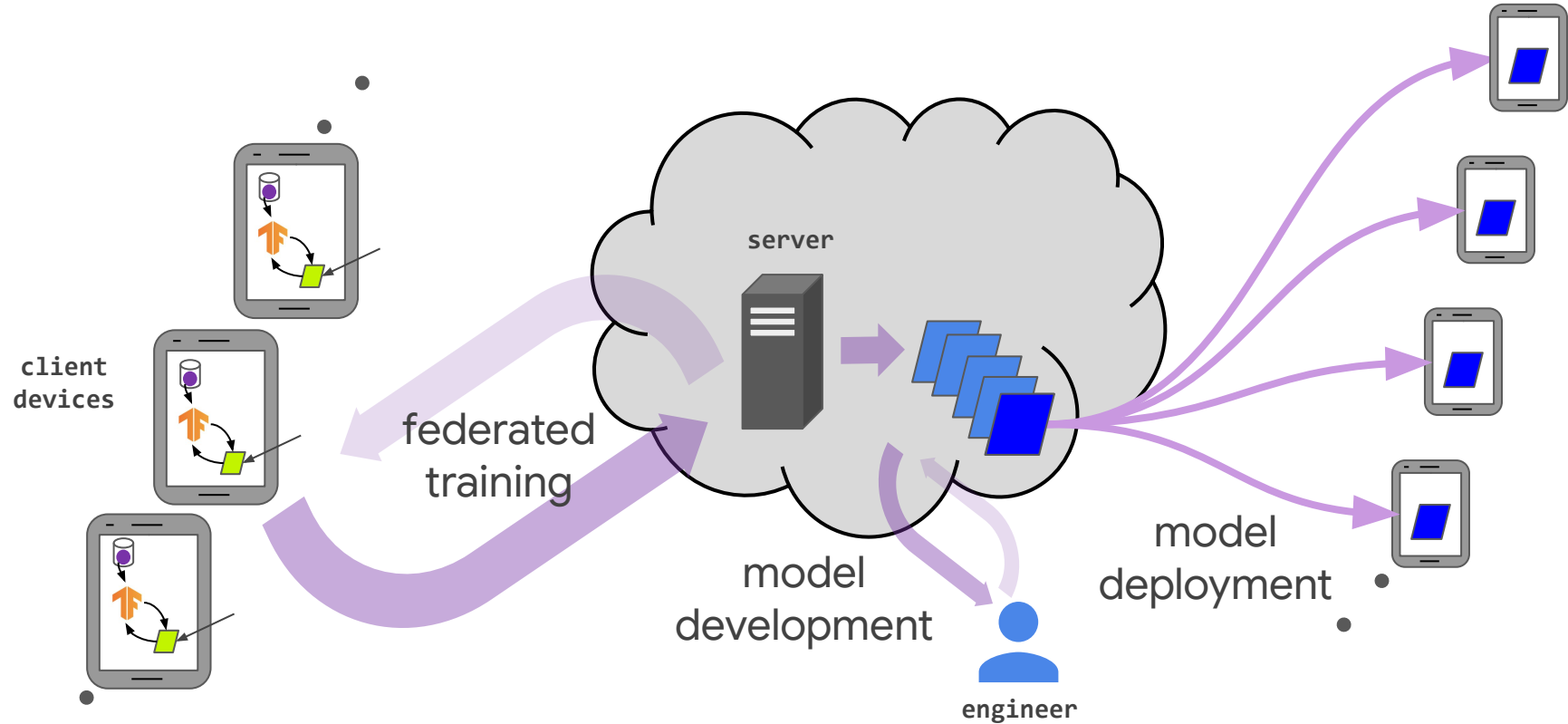
2. **New Technology**

Push the pareto frontier
with better technology.

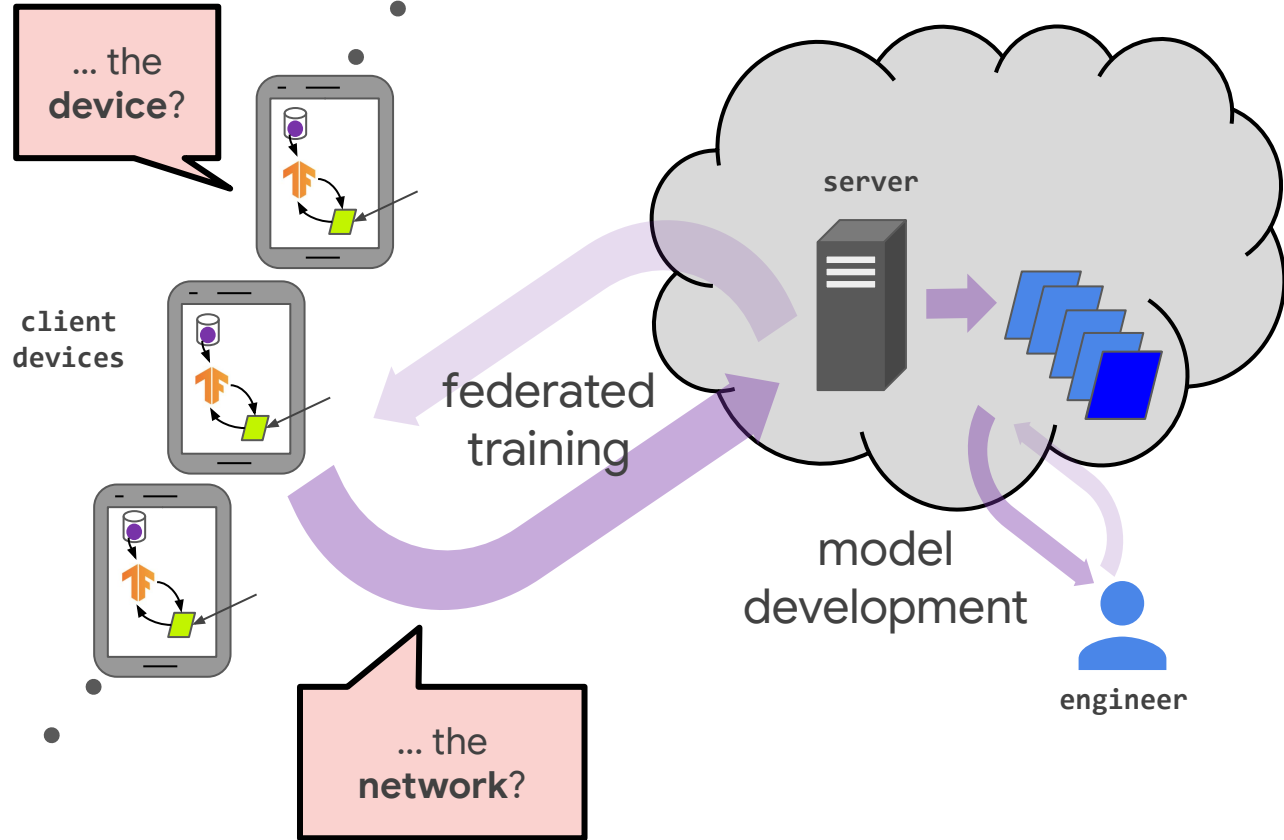
Make achieving high
privacy and utility
possible **with less work.**



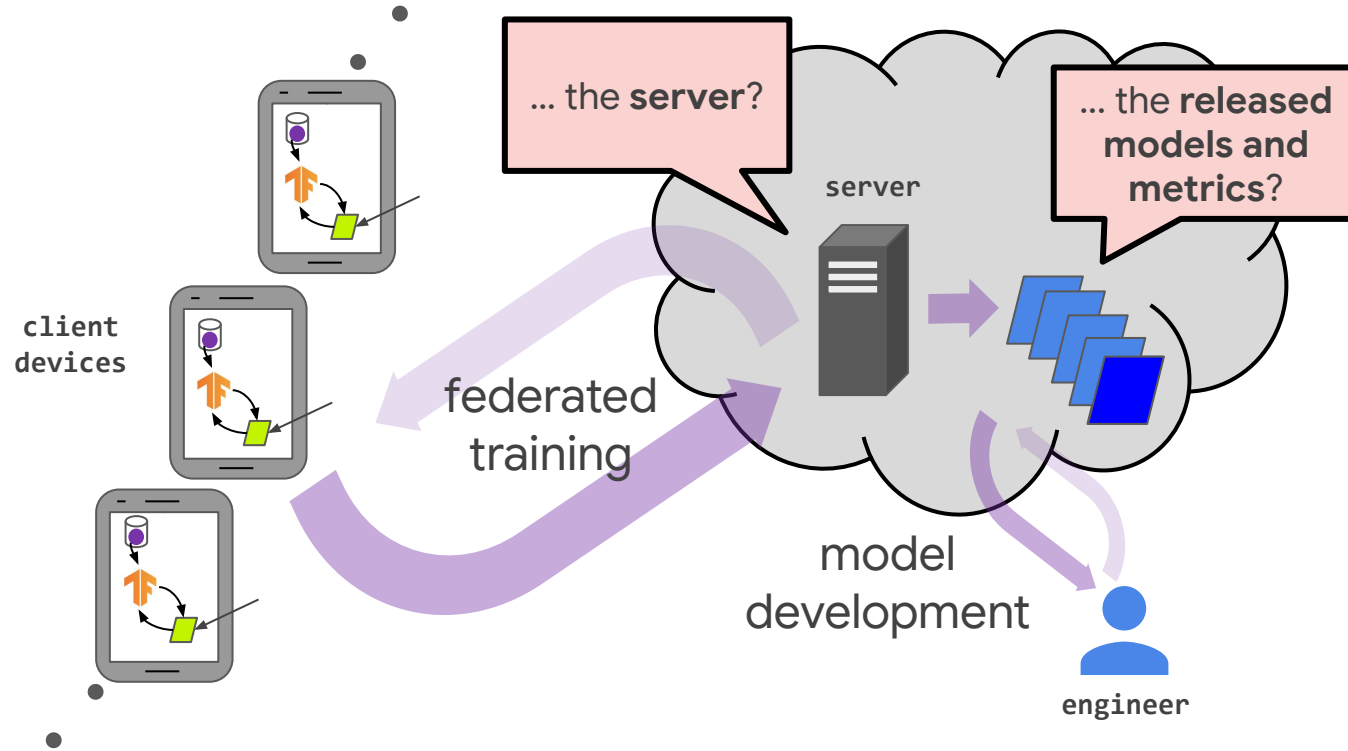
What private information might an actor learn?



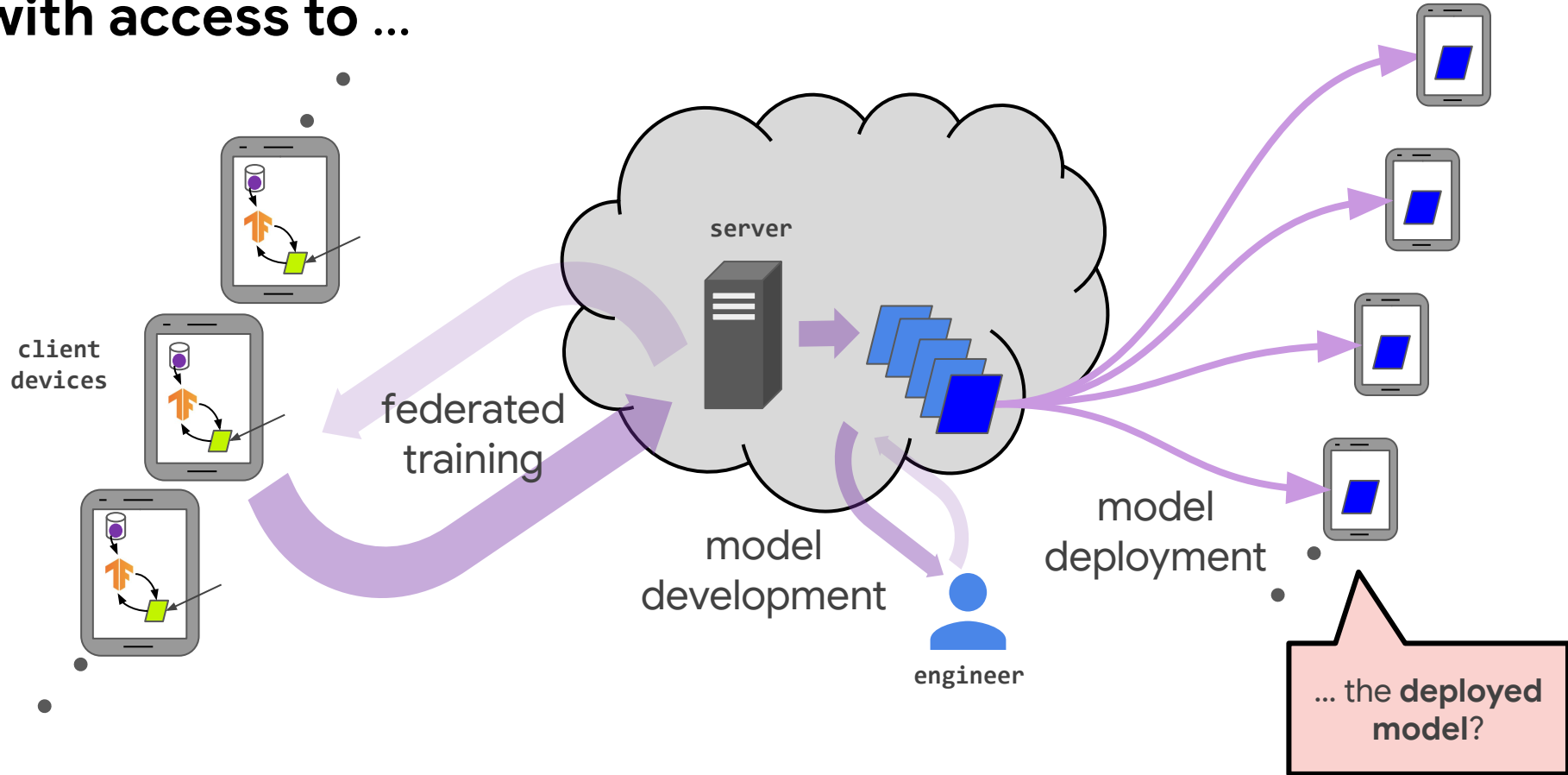
What private information might an actor learn with access to ...



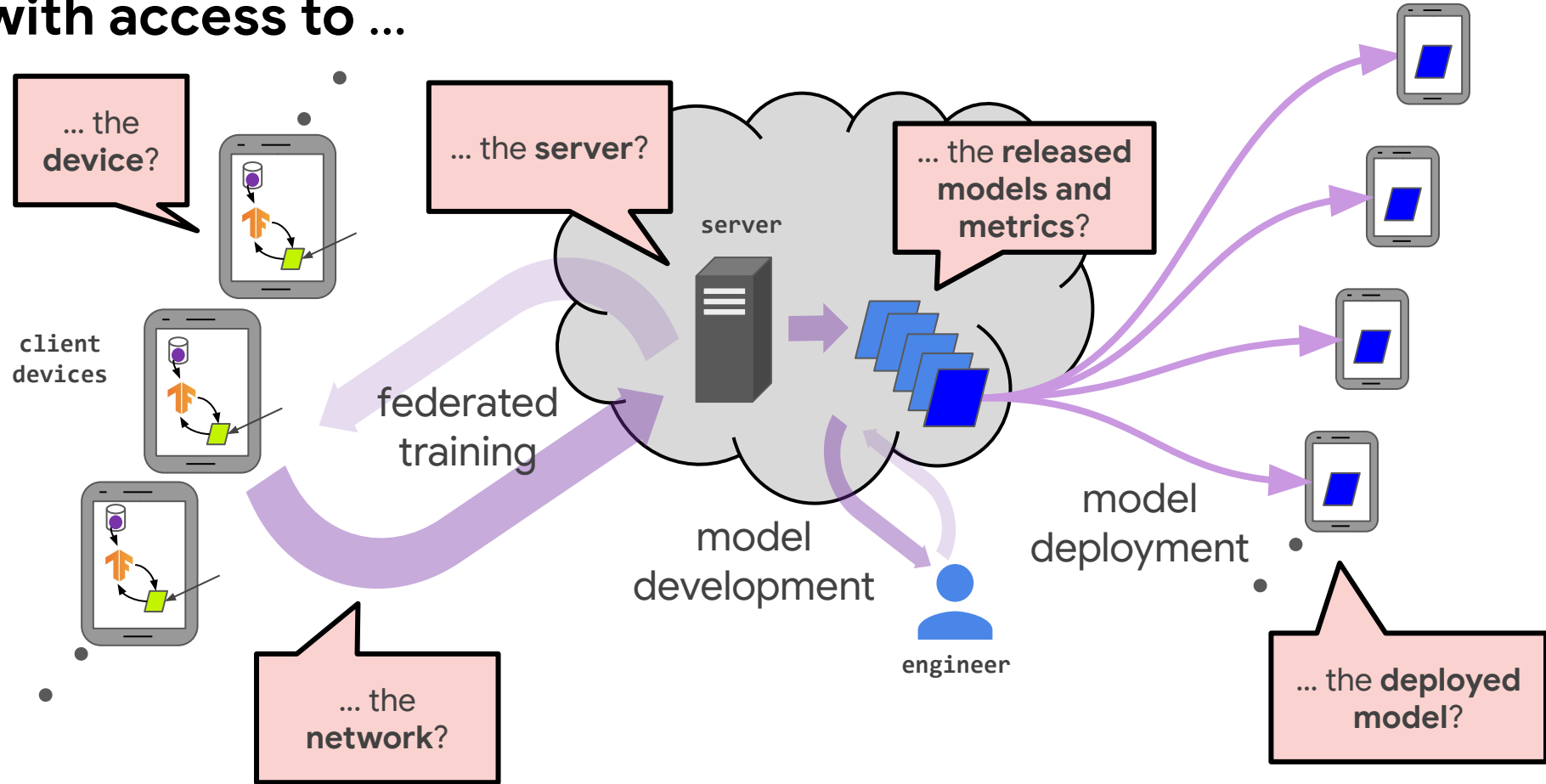
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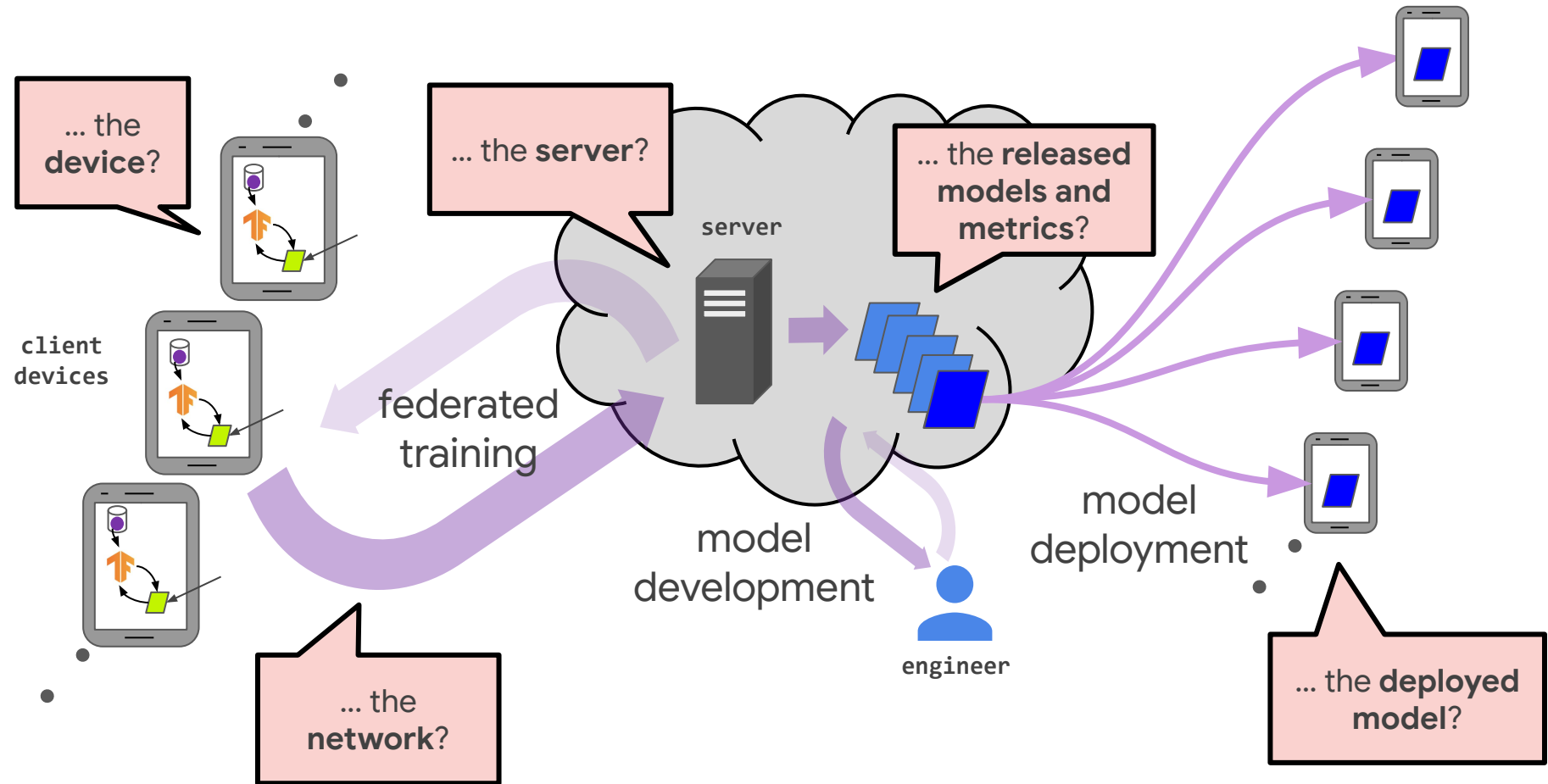
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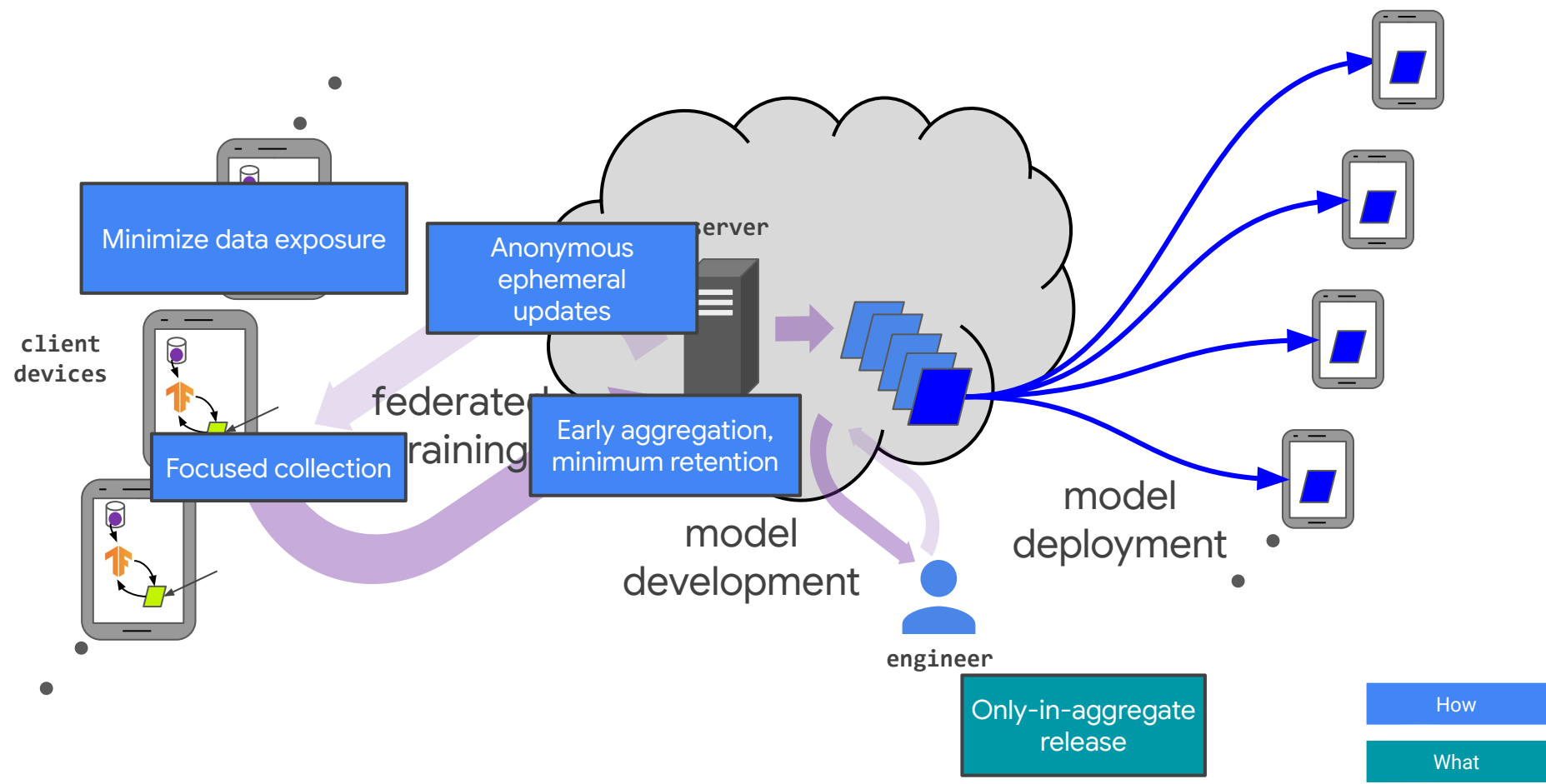
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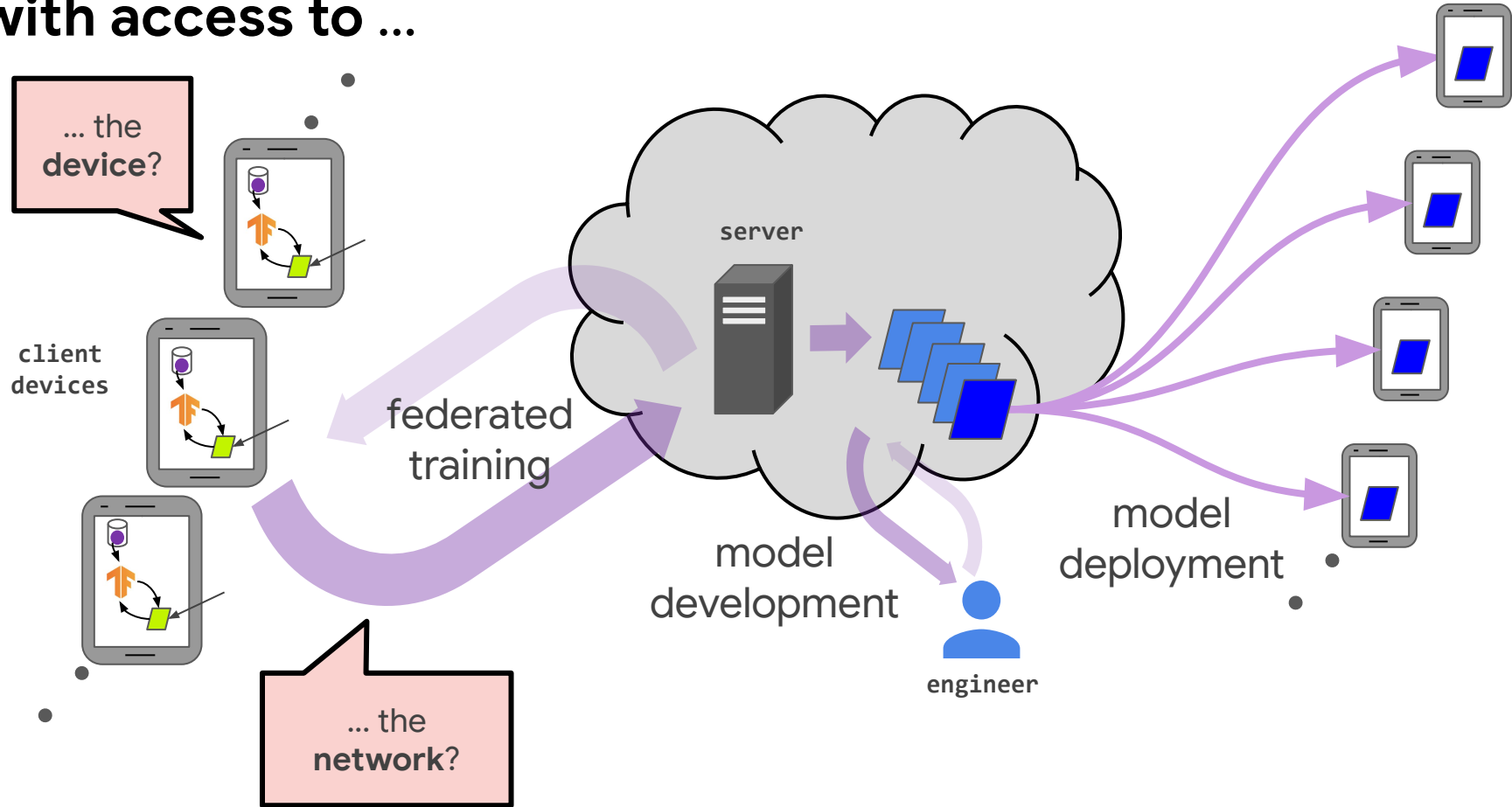
How much do I need to trust ...



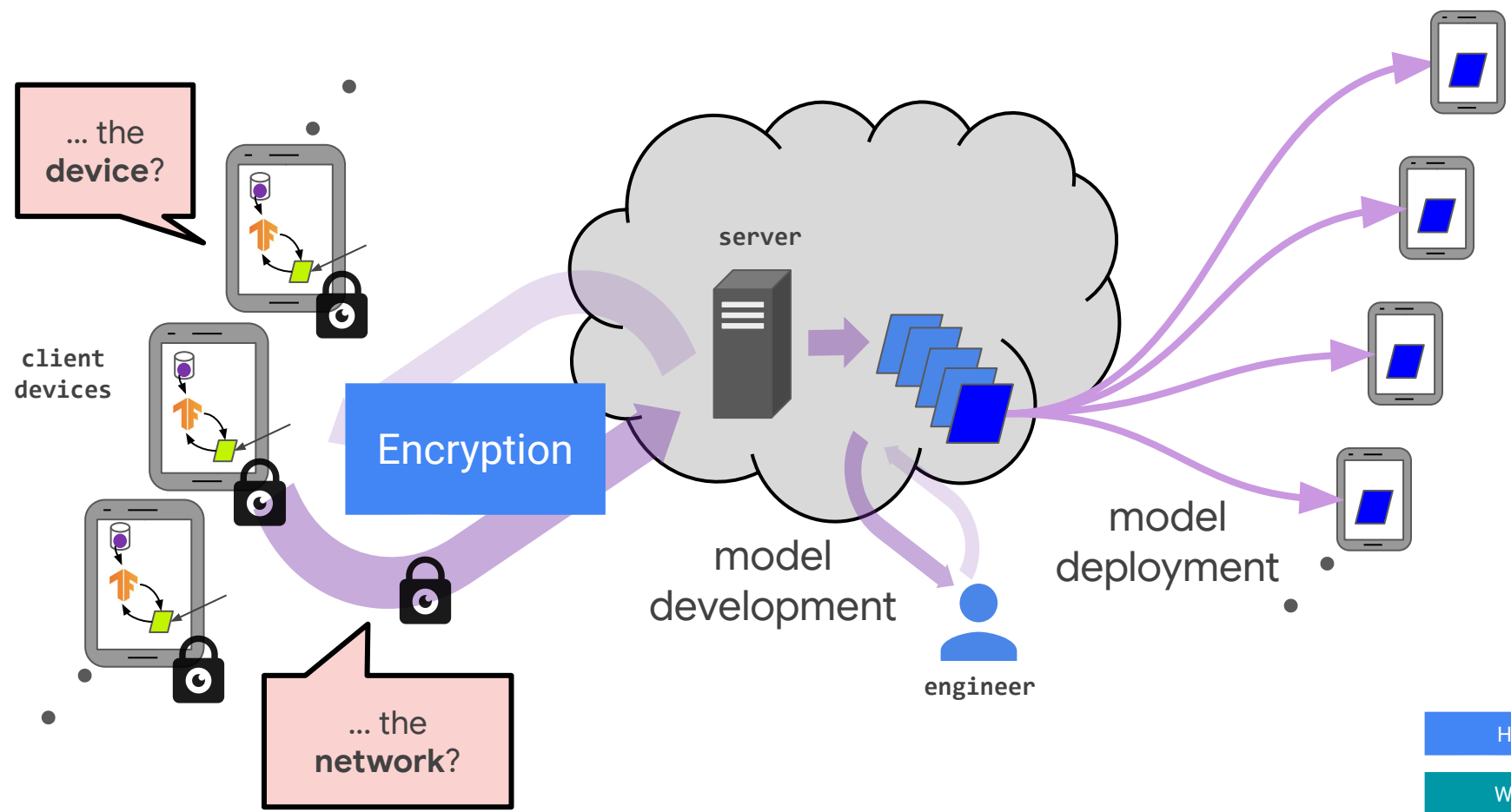
Privacy principles guiding FL



What private information might an actor learn with access to ...



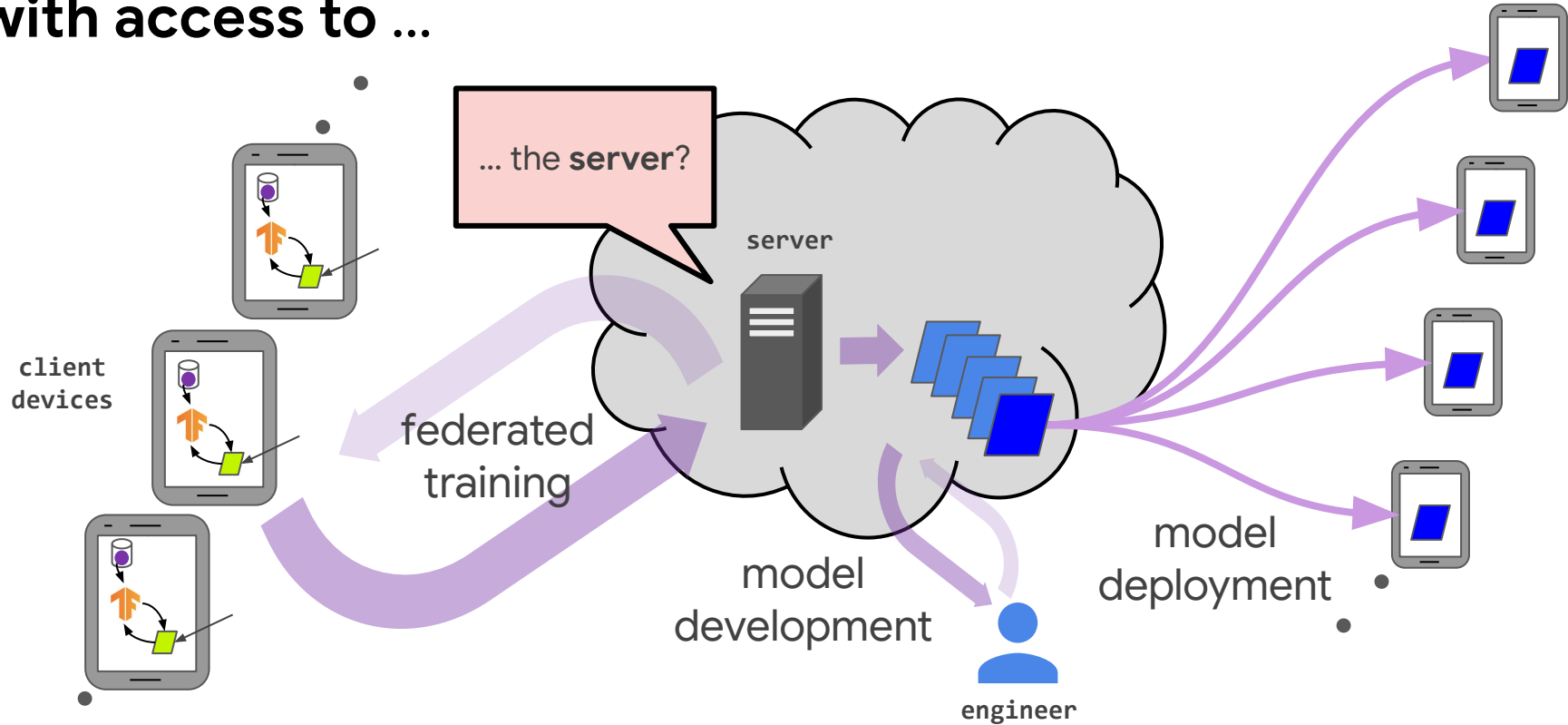
Encryption, at rest and on the wire



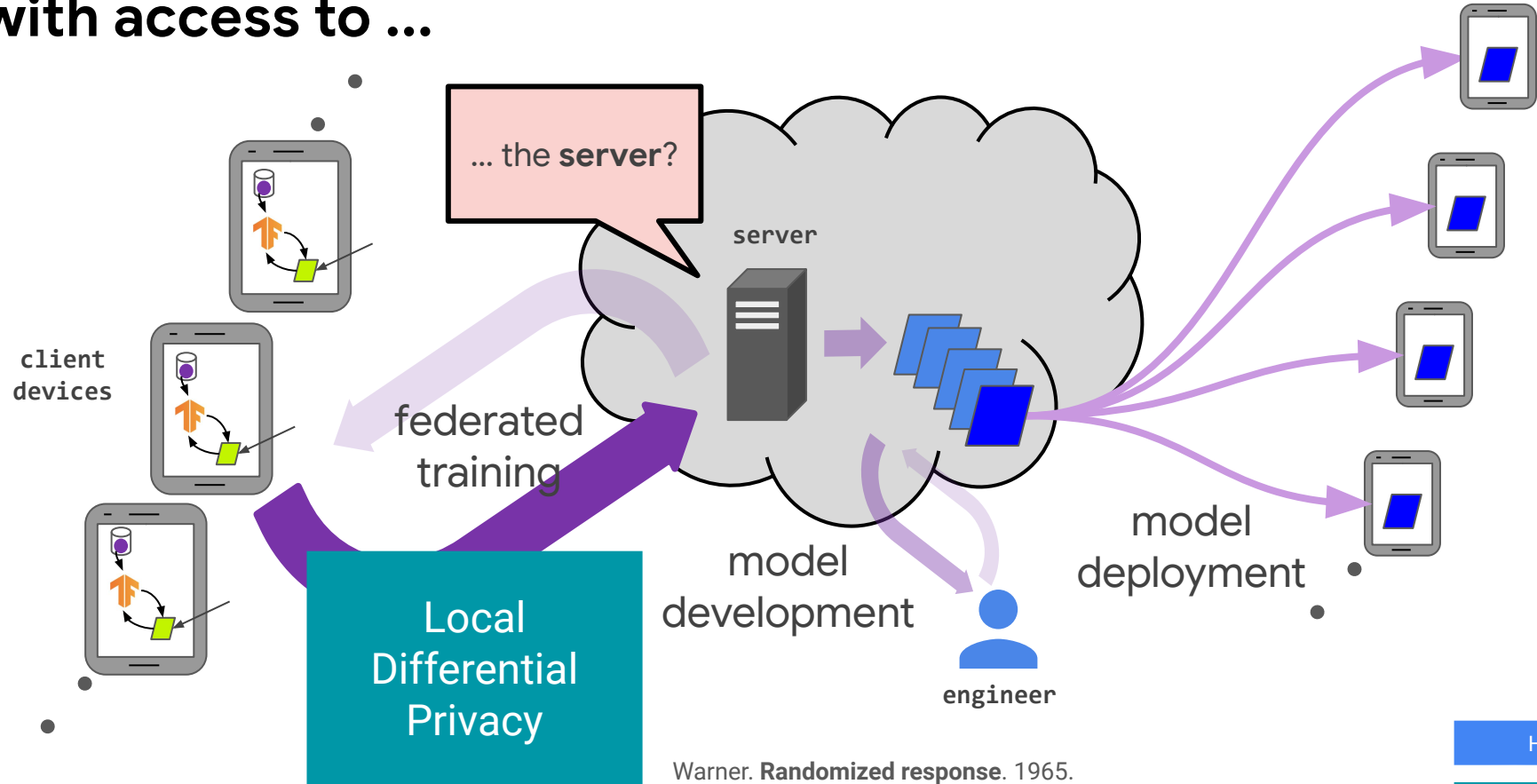
How

What

What private information might an actor learn with access to ...



What private information might an actor learn with access to ...



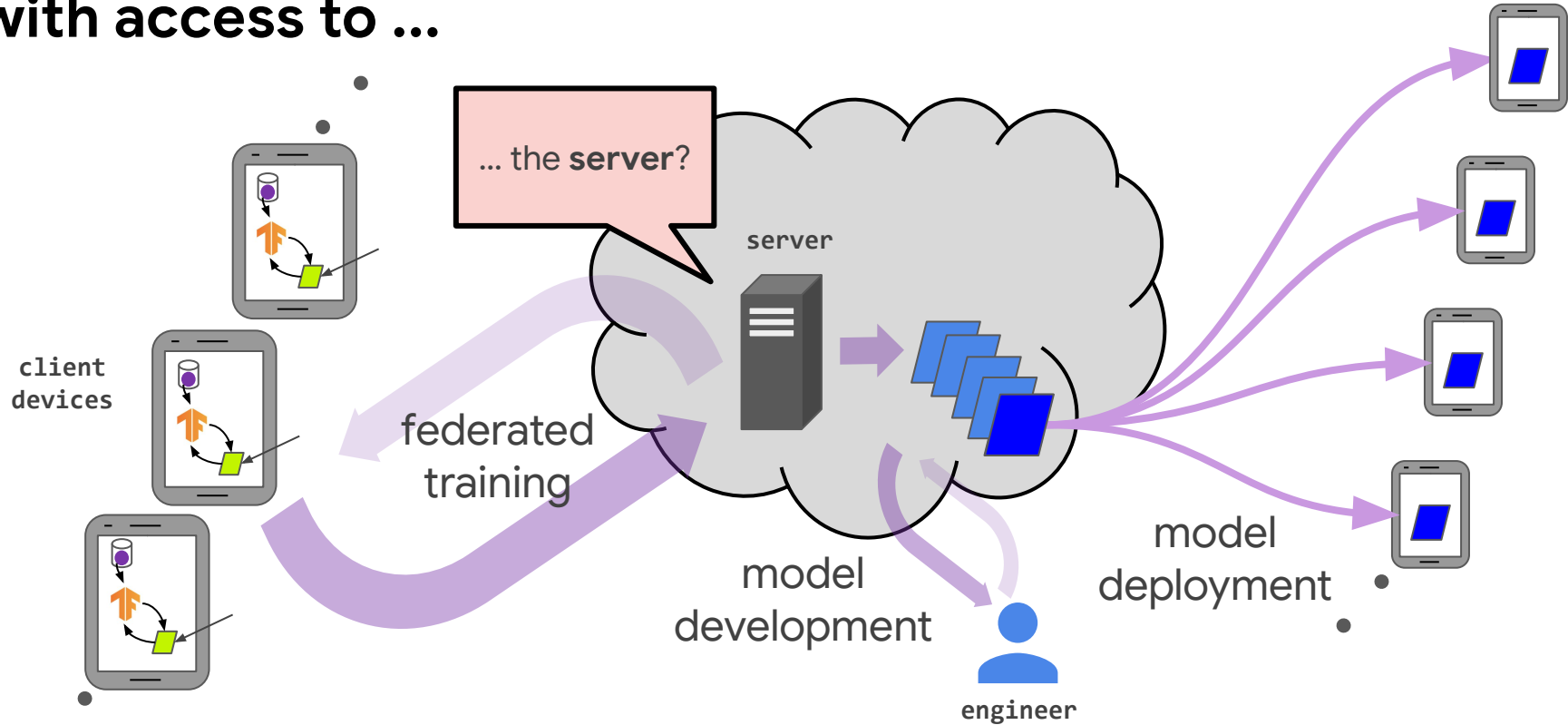
Warner. **Randomized response**. 1965.

Kasiviswanathan, et. al. **What can we learn privately?** 2011.

How

What

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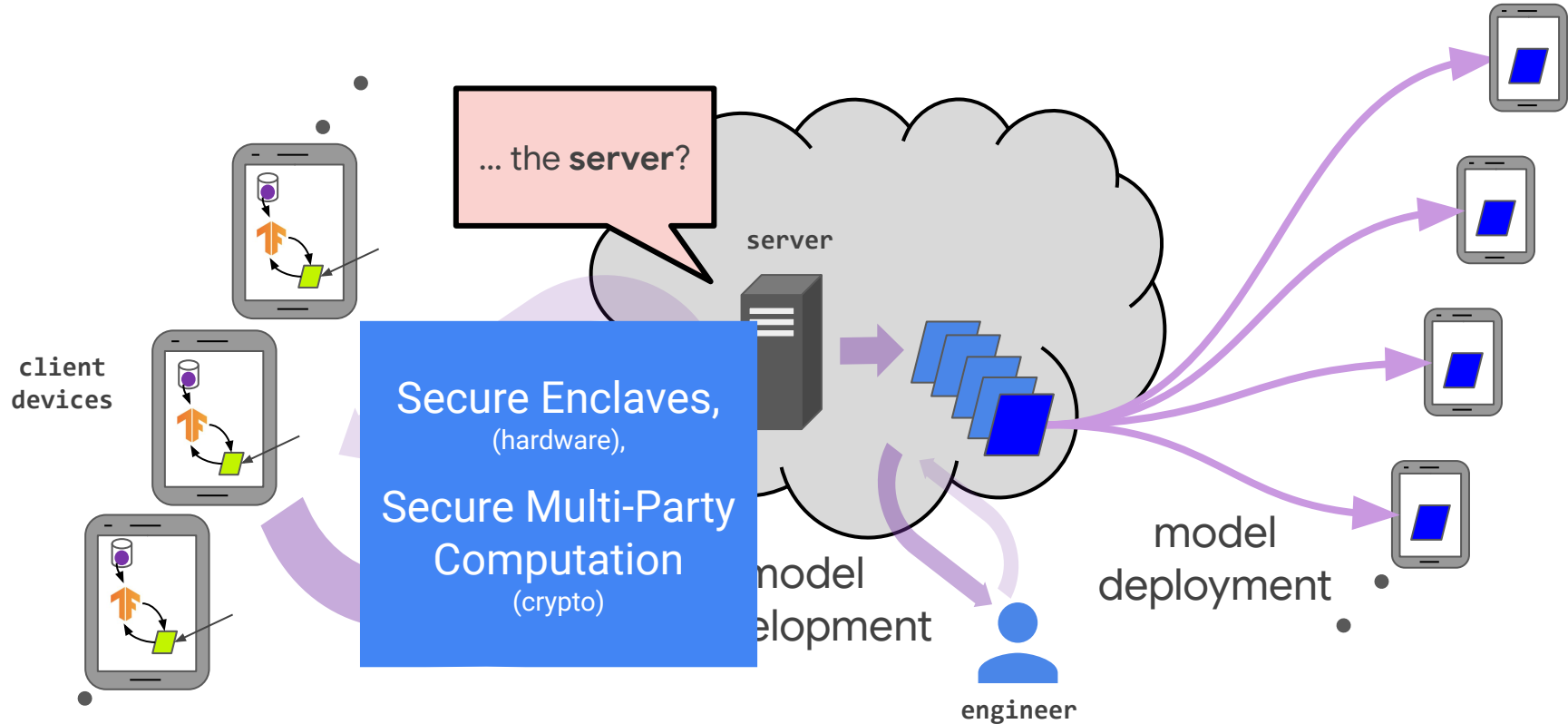


Ideally, **nothing**, even with root access.

How

What

What private information might an actor learn

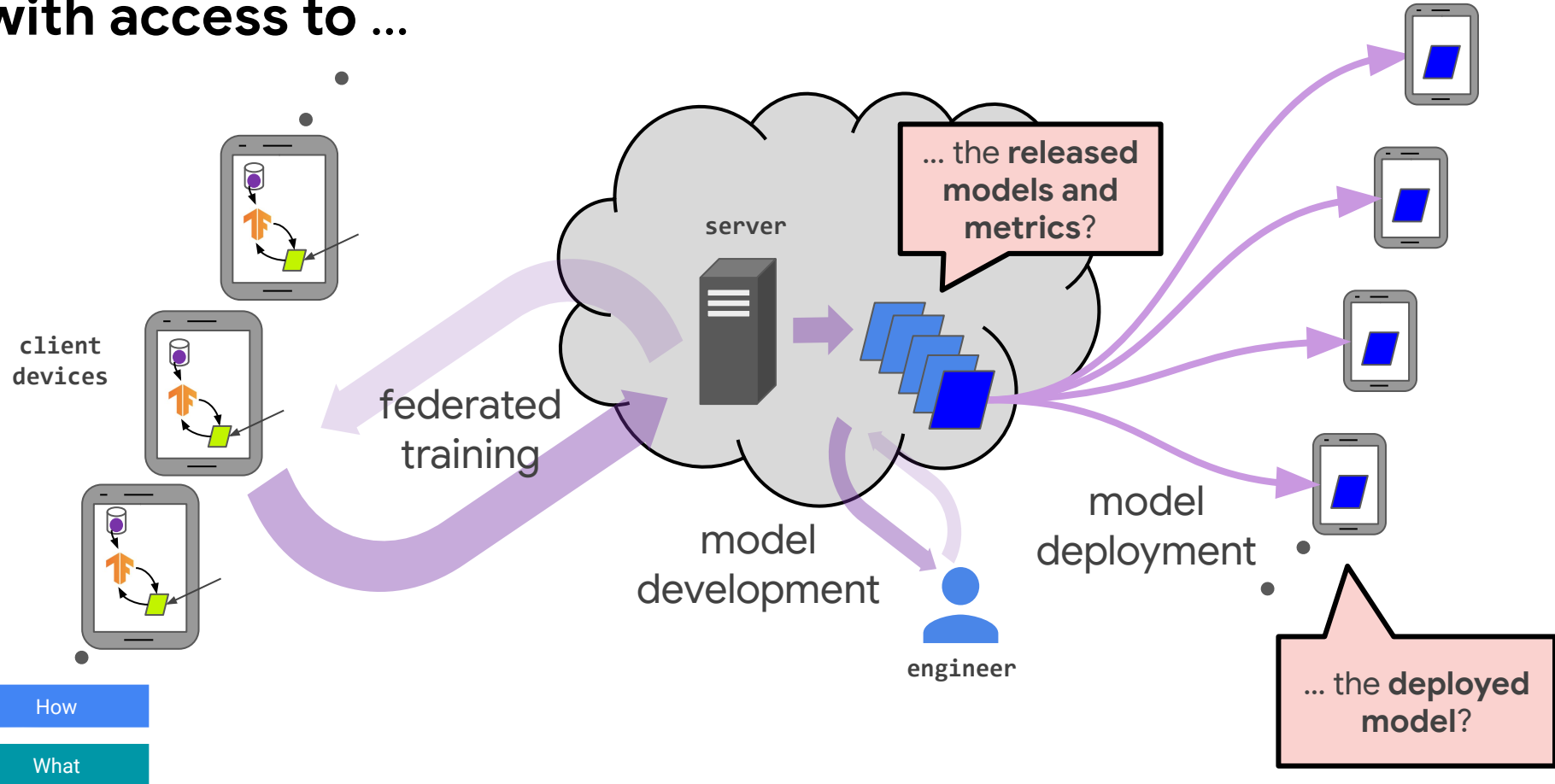


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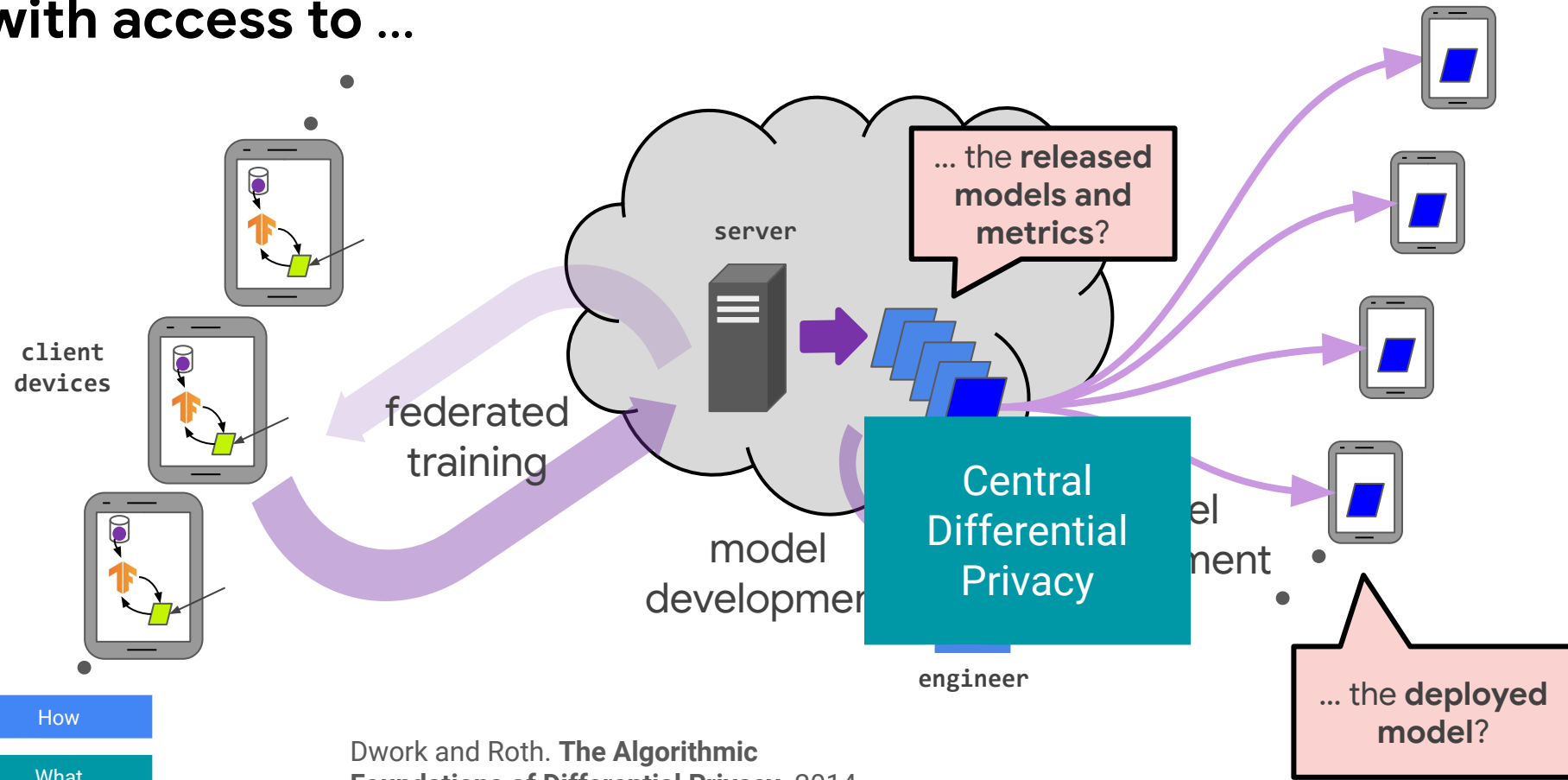
How

What

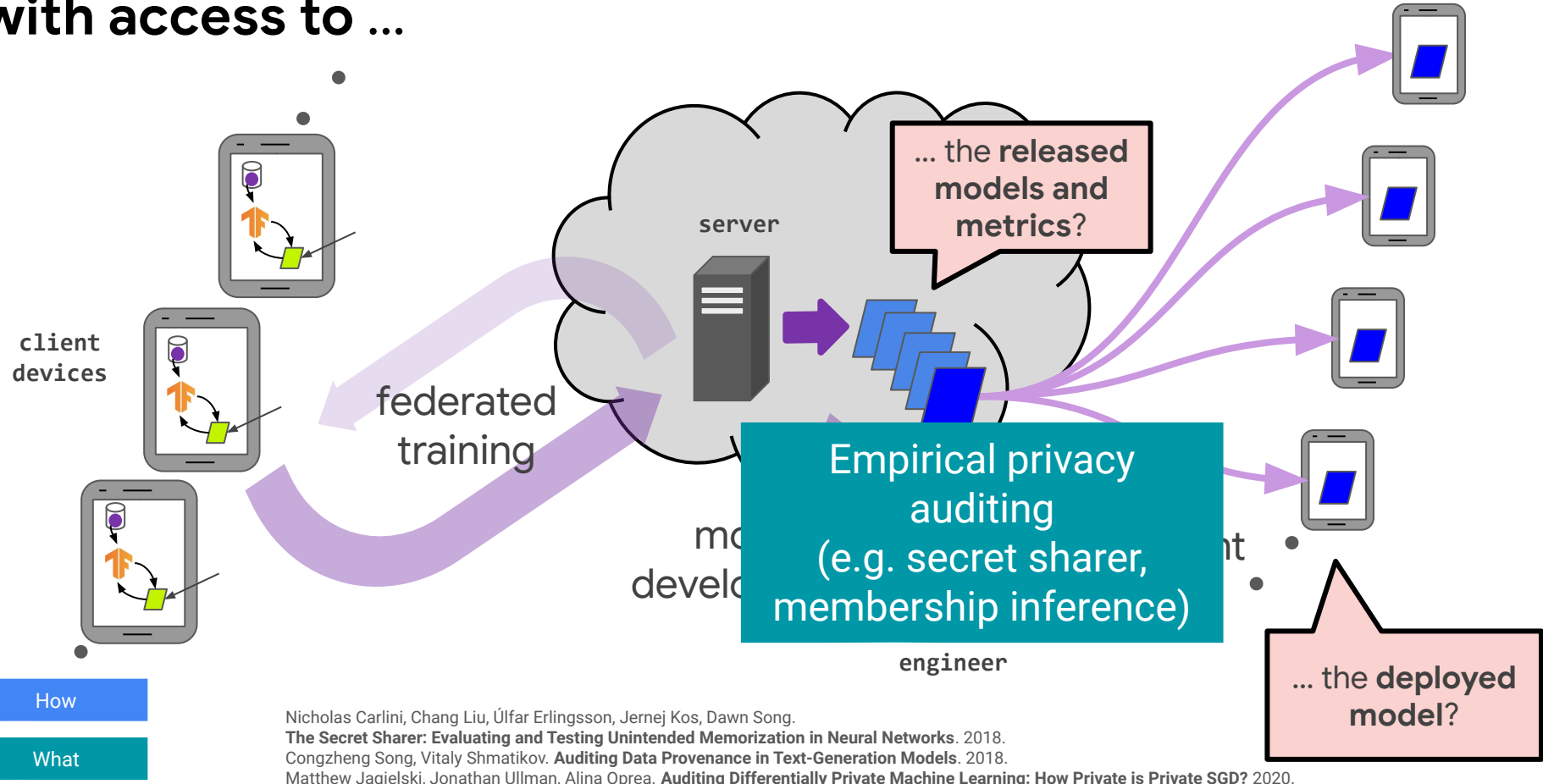
What private information might an actor learn with access to ...



What private information might an actor learn with access to ...



What private information might an actor learn with access to ...



How

What

Nicholas Carlini, Chang Liu, Úlfar Erlingsson, Jernej Kos, Dawn Song.

The Secret Sharer: Evaluating and Testing Unintended Memorization in Neural Networks. 2018.

Congzheng Song, Vitaly Shmatikov. **Auditing Data Provenance in Text-Generation Models.** 2018.

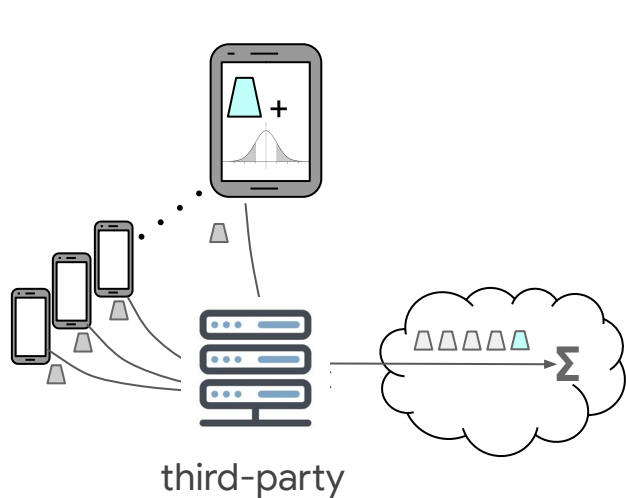
Matthew Jagielski, Jonathan Ullman, Alina Oprea. **Auditing Differentially Private Machine Learning: How Private is Private SGD?** 2020.

Private Aggregation & Trust

Distributing Trust for Private Aggregation

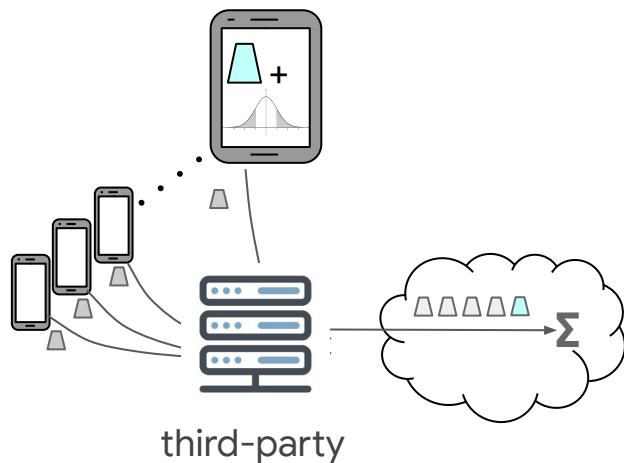
Distributing Trust for Private Aggregation

1 Trusted “third party”

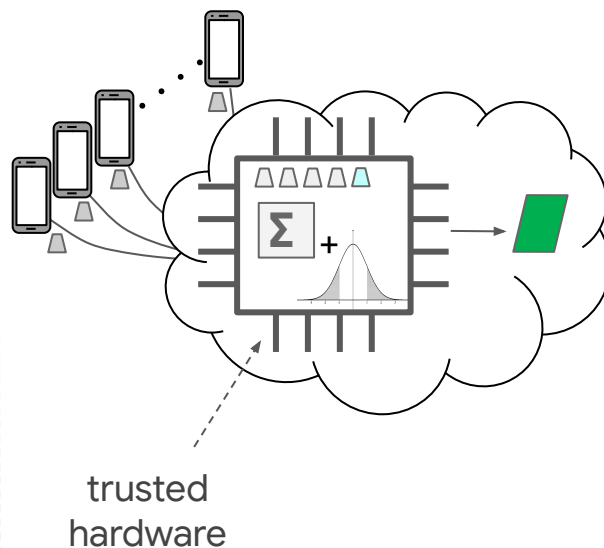


Distributing Trust for Private Aggregation

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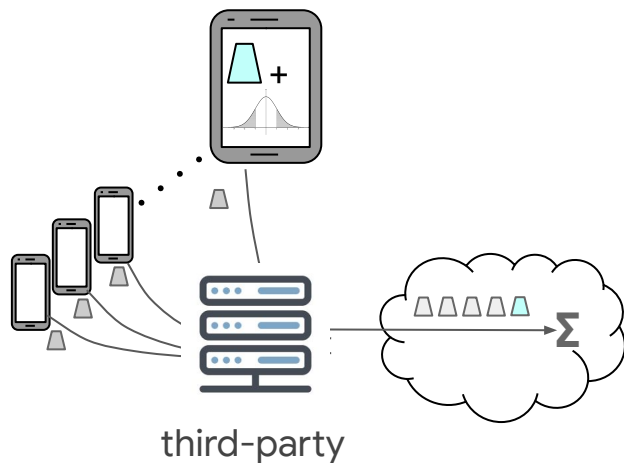


2 Trusted Execution Environments

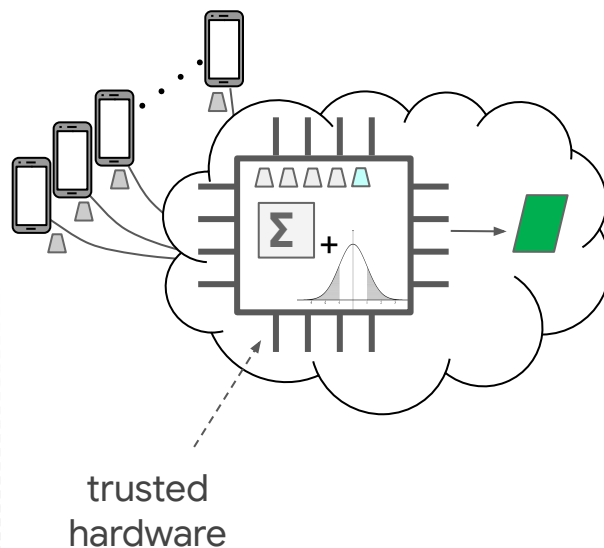


Distributing Trust for Private Aggregation

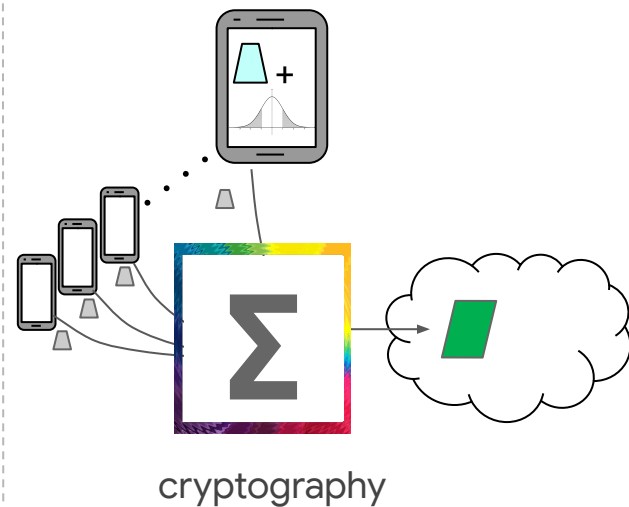
1 Trusted “third party”



2 Trusted Execution Environments



3 Trust via Cryptography



Differentially Private Federated Training

Differential Privacy



Differential privacy is the statistical science of trying to learn **as much as possible about a group** while learning **as little as possible about any individual in it.**

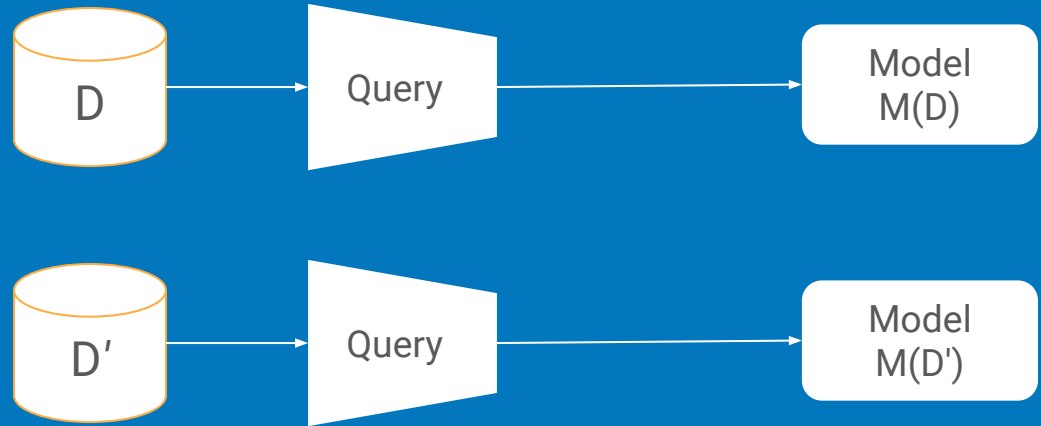


Andy Greenberg
Wired 2016.06.13

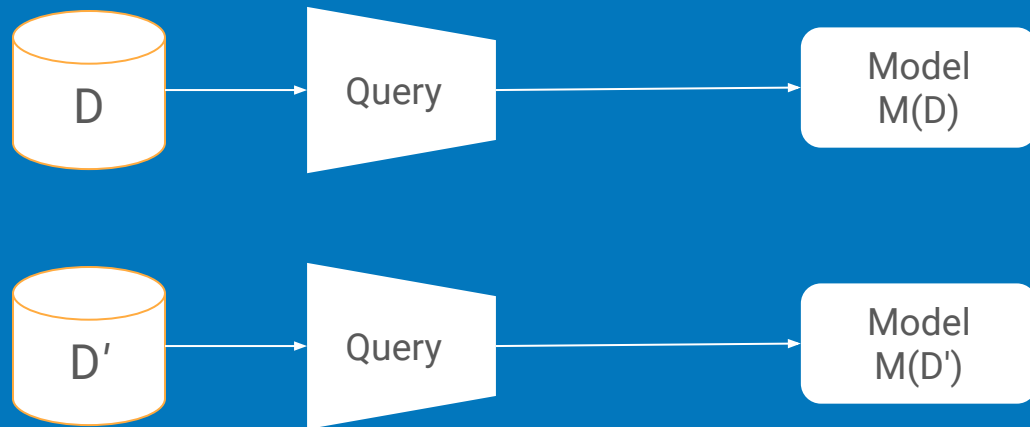
Differential Privacy



Differential Privacy



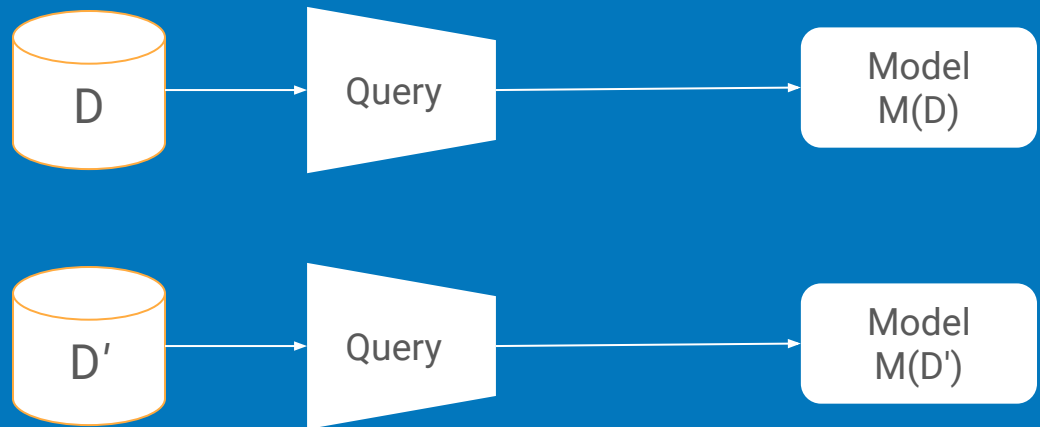
Differential Privacy



(ϵ, δ) -Differential Privacy: The distribution of the output $M(D)$ (a trained model) on database (training dataset) D is **nearly the same** as $M(D')$ for all adjacent databases D and D'

$$\forall S: \Pr[M(D) \in S] \leq \exp(\epsilon) \cdot \Pr[M(D') \in S] + \delta$$

Record-level Differential Privacy

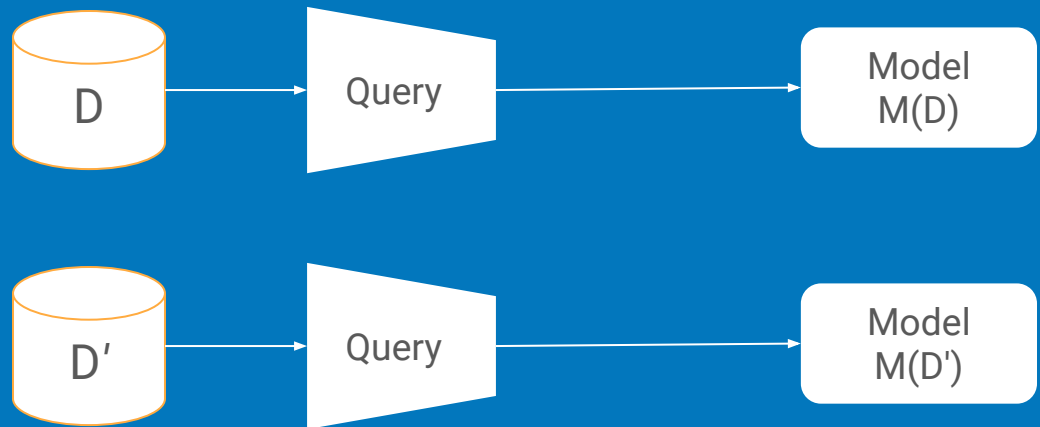


(ϵ, δ) -Differential Privacy: The distribution of the output $M(D)$ (a trained model) on database (training dataset) D is nearly the same as $M(D')$ for all **adjacent** databases D and D'

adjacent: Sets D and D' differ only by presence/absence of one **example**

M. Abadi, A. Chu, I. Goodfellow, H. B. McMahan, I. Mironov, K. Talwar, & L. Zhang. **Deep Learning with Differential Privacy**. CCS 2016.

User-level Differential Privacy

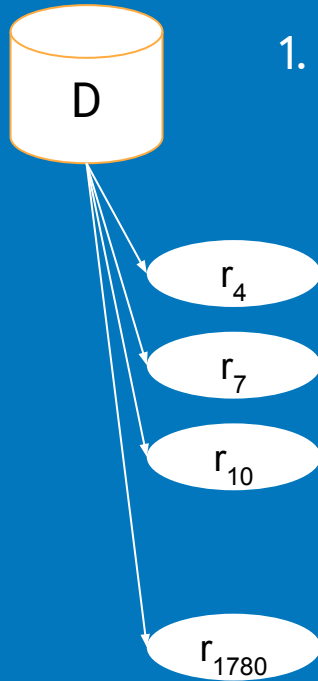


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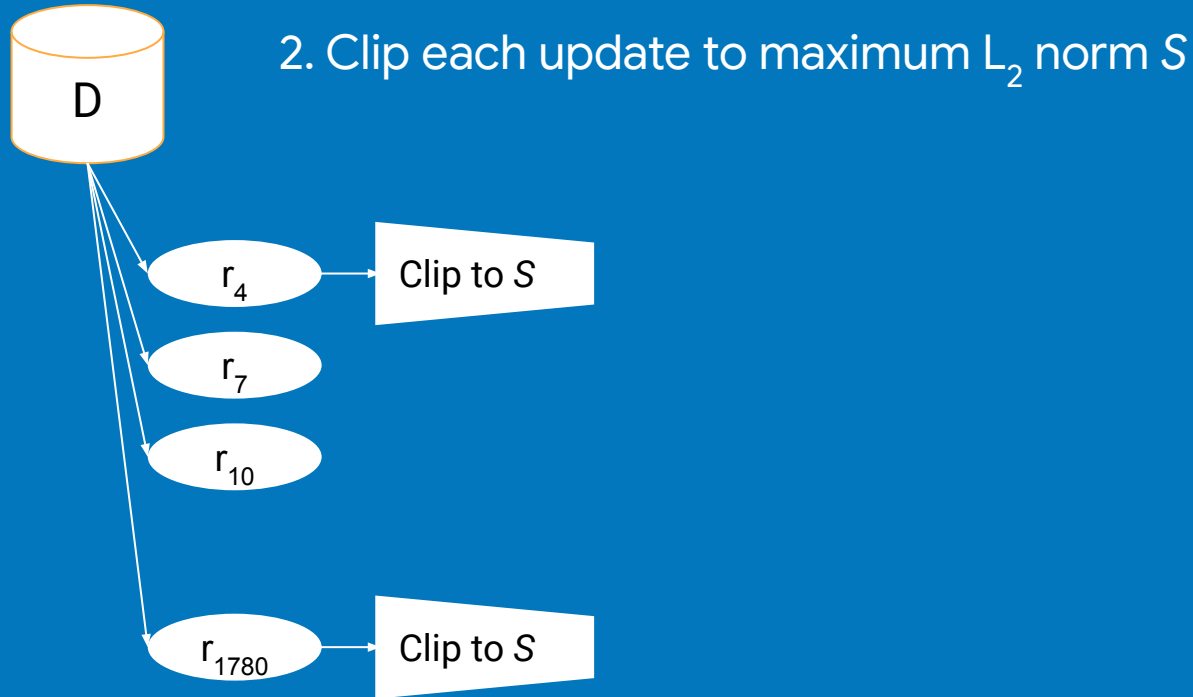
*H. B. McMahan, et al.
Learning Differentially
Private Recurrent
Language Models.
ICLR 2018.*

Iterative training with differential privacy

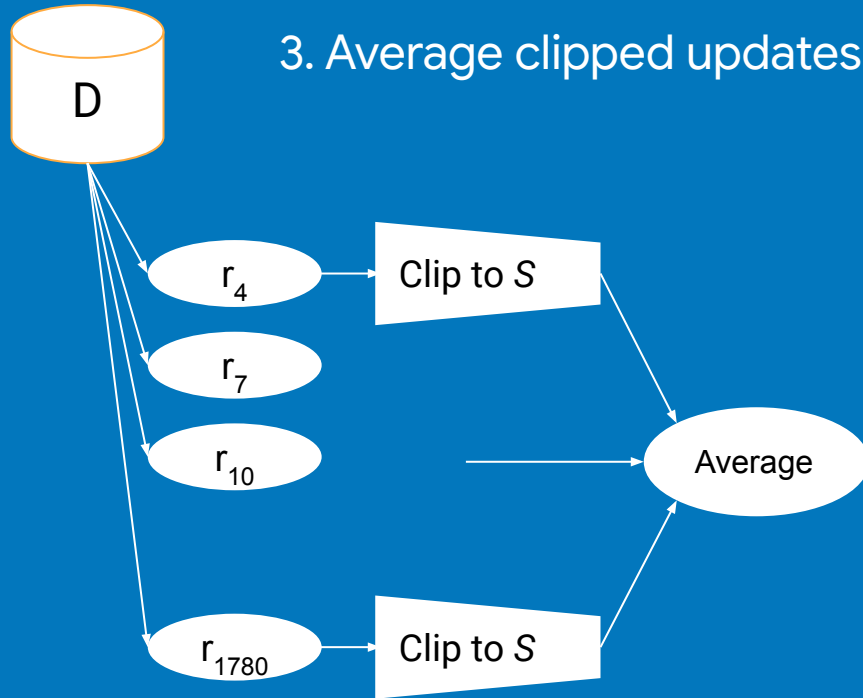


1. Sample a batch of clients *uniformly at random*

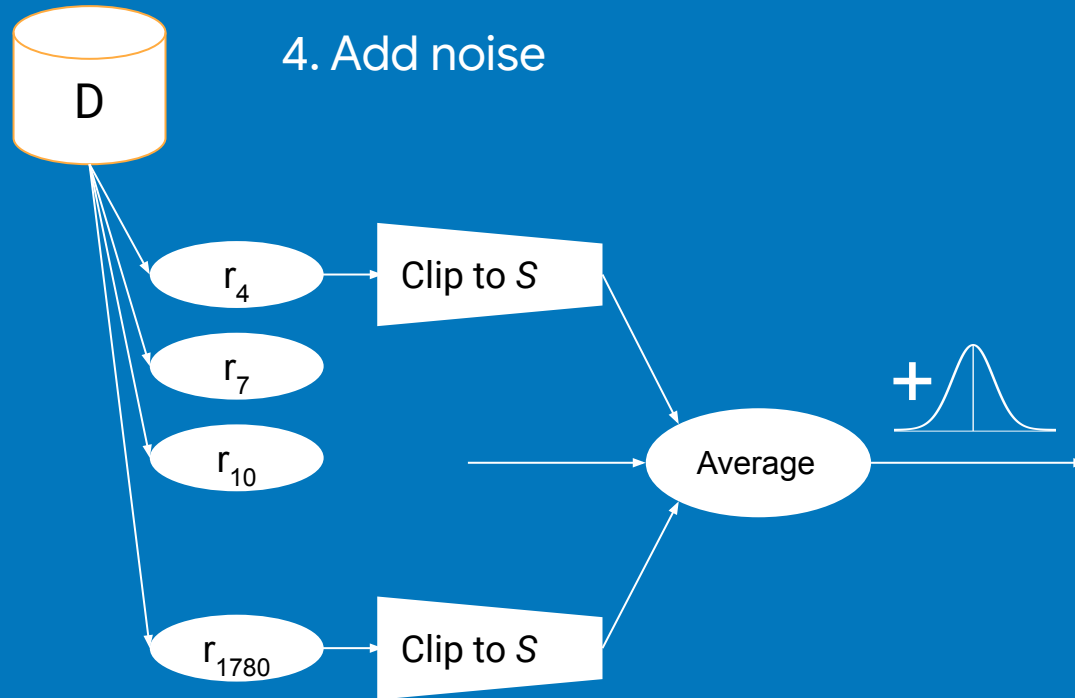
Iterative training with differential privacy



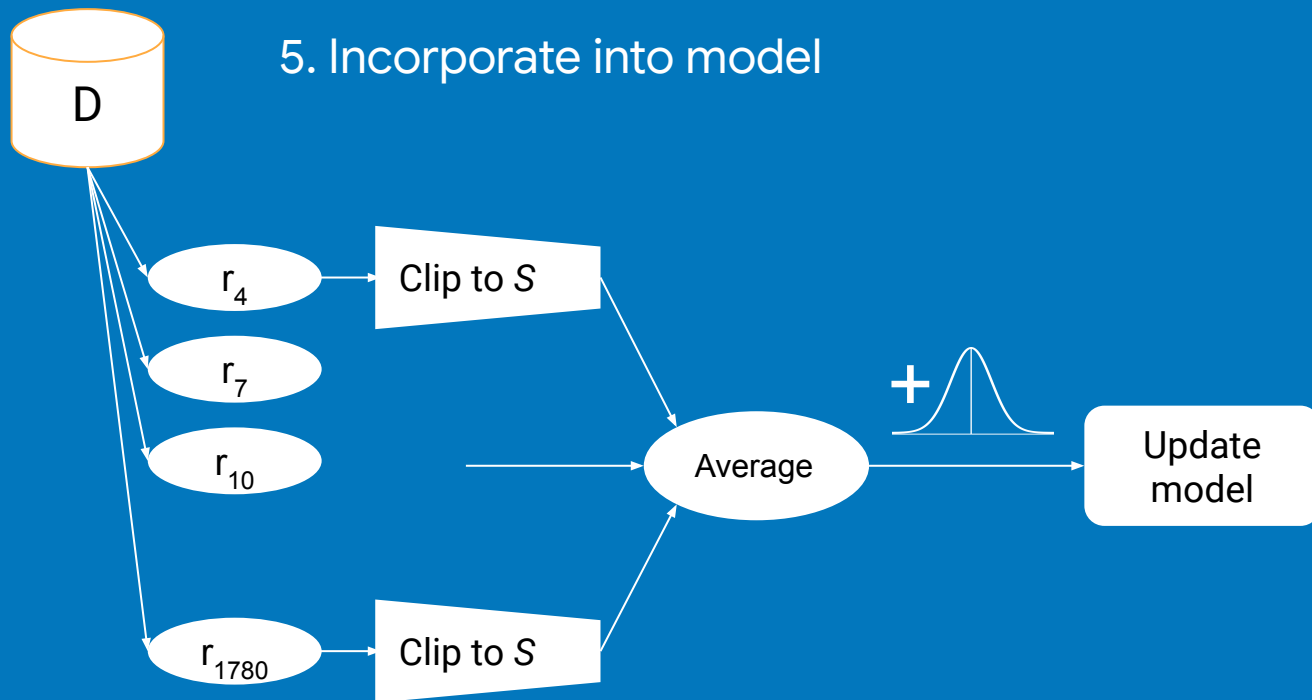
Iterative training with differential privacy



Iterative training with differential privacy



Iterative training with differential privacy



There are many details and possibilities

A General Approach to Adding Differential Privacy to Iterative Training Procedures

H. Brendan McMahan
mcmahan@google.com

Galen Andrew
galenandrew@google.com

Úlfar Erlingsson
ulfar@google.com

Steve Chien
schien@google.com

Ilya Mironov
mironov@google.com

Nicolas Papernot
papernot@google.com

Peter Kairouz
kairouz@google.com

Abstract

In this work we address the practical challenges of training machine learning models on privacy-sensitive datasets by introducing a modular approach that minimizes changes to training algorithms, provides a variety of configuration strategies for the privacy mechanism, and then isolates and simplifies the critical logic that computes the final privacy guarantees. A key challenge is that training algorithms often require estimating many different quantities (vectors) from the same set of examples — for example, gradients of different layers in a deep learning architecture, as well as metrics and batch normalization parameters. Each of these

Introducing TensorFlow Privacy: Learning with Differential Privacy for Training Data



TensorFlow [Follow](#)
Mar 6 · 7 min read

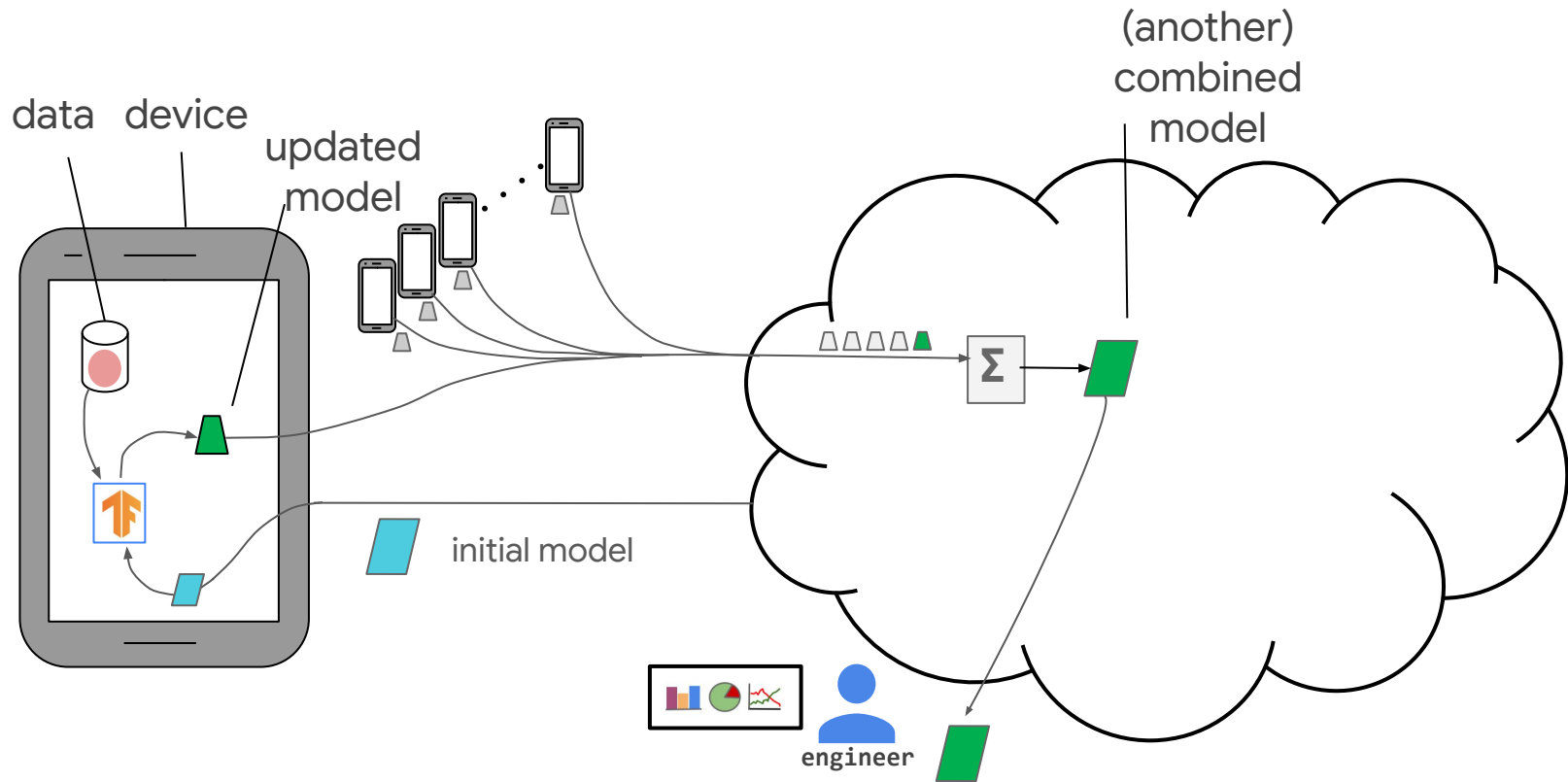


Posted by [Carey Radebaugh](#) (Product Manager) and [Úlfar Erlingsson](#) (Research Scientist)

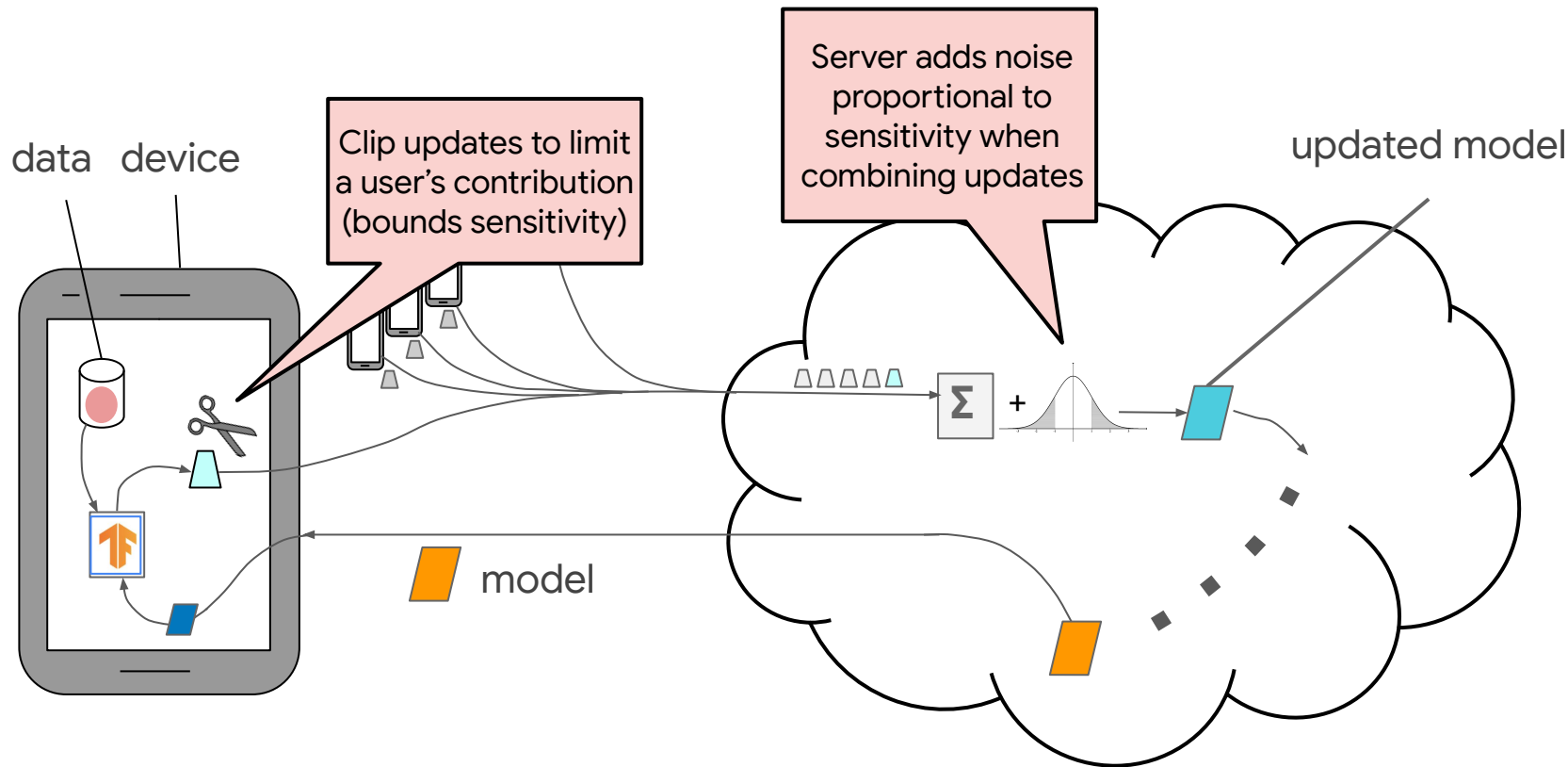
Today, we're excited to announce TensorFlow Privacy ([GitHub](#)), an open source library that makes it easier not only for developers to train machine-learning models with privacy, but also for researchers to advance the state of the art in machine learning with strong privacy guarantees.

Modern machine learning is increasingly applied to create amazing new technologies and user experiences, many of which involve training

Back to federated learning



Differentially private federated learning

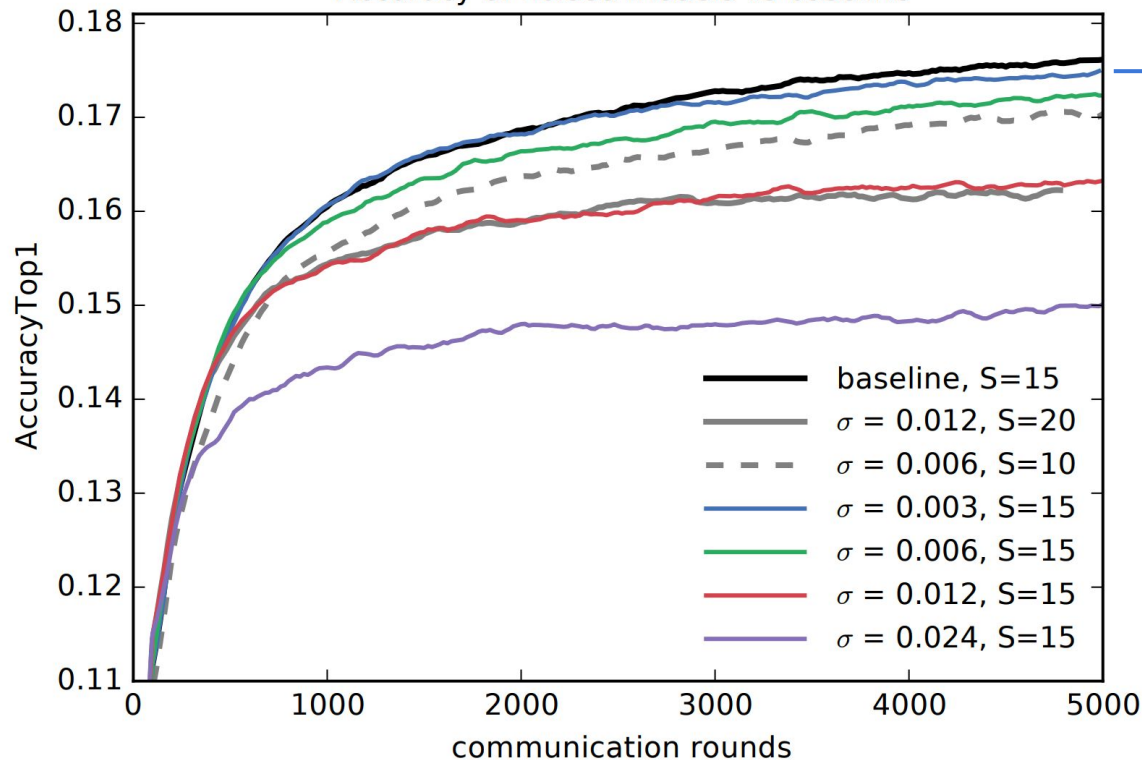


Differential privacy for language models

LSTM-based predictive language model.

10K word dictionary, word embeddings $\in \mathbb{R}^{96}$, state $\in \mathbb{R}^{256}$, parameters: 1.35M. Corpus=Reddit posts, by author.

Accuracy of noised models vs baseline

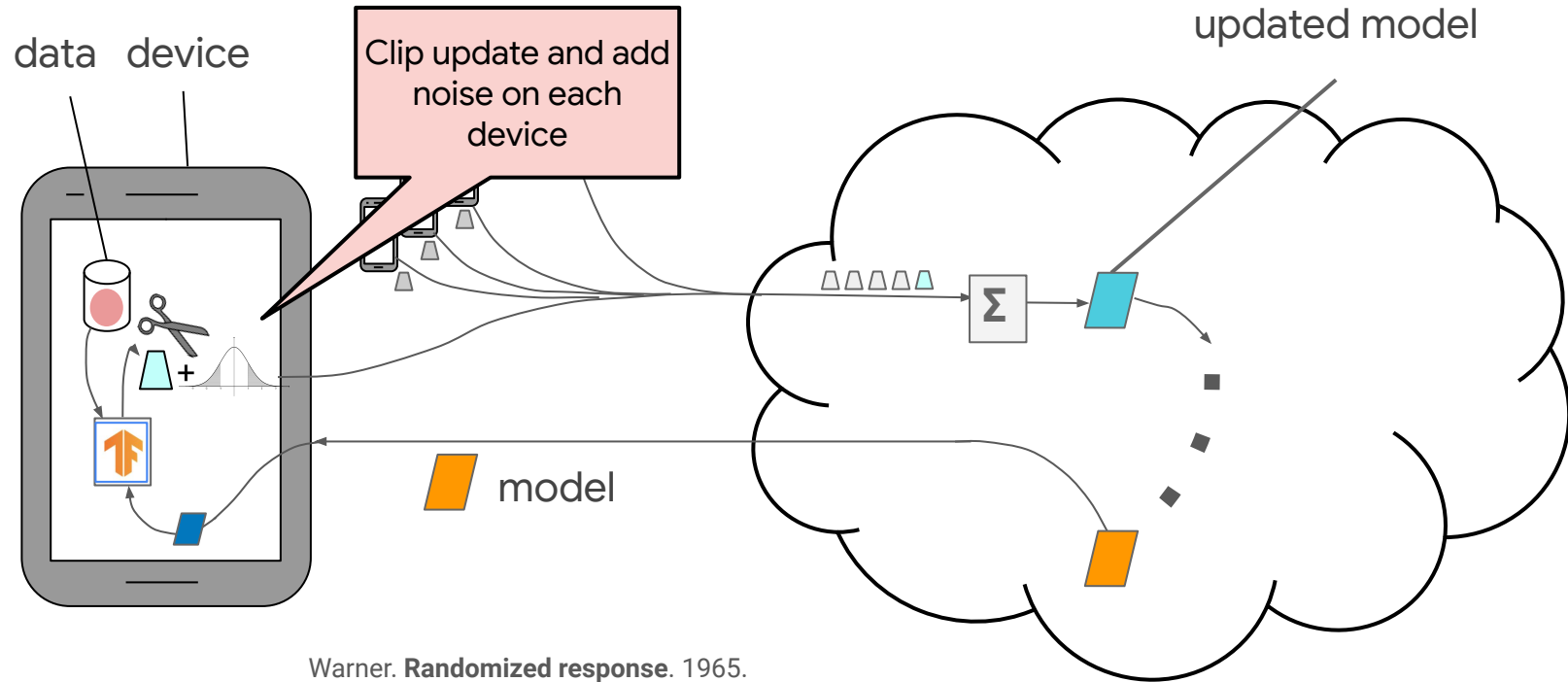


(4.634, $1e-9$)-DP with 763k users
(1.152, $1e-9$)-DP with $1e8$ users

$\mathbb{E}[\text{users per minibatch}] = 5k$
 $\mathbb{E}[\text{tokens per minibatch}] = 8m$

H. B. McMahan, et al. **Learning Differentially Private Recurrent Language Models**. ICLR 2018.

Locally differentially private federated learning



Warner. **Randomized response**. 1965.

Kasiviswanathan, et. al. **What can we learn privately?** 2011.

Central DP:

easier to get high utility with good privacy

Local DP:

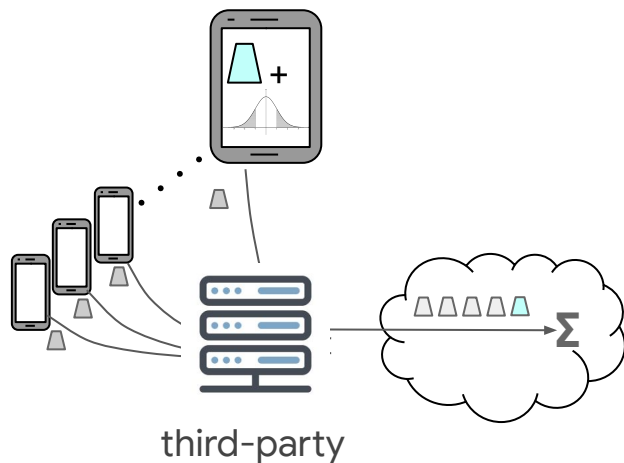
requires much weaker trust assumptions

Can we combine the best of both worlds?

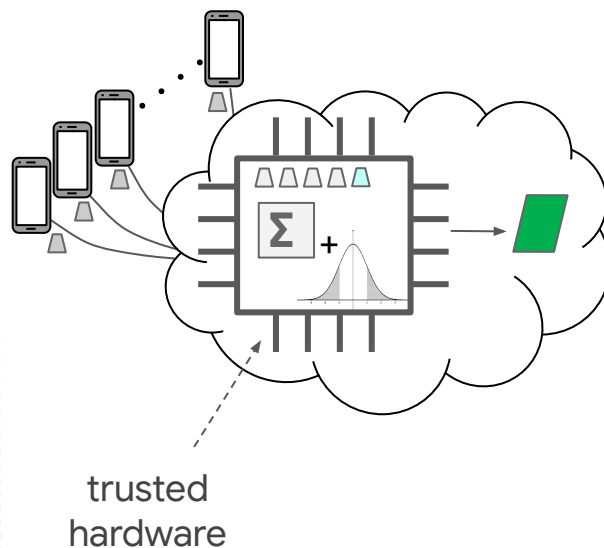
Distributed Differential Privacy

Distributing Trust for Private Aggregation

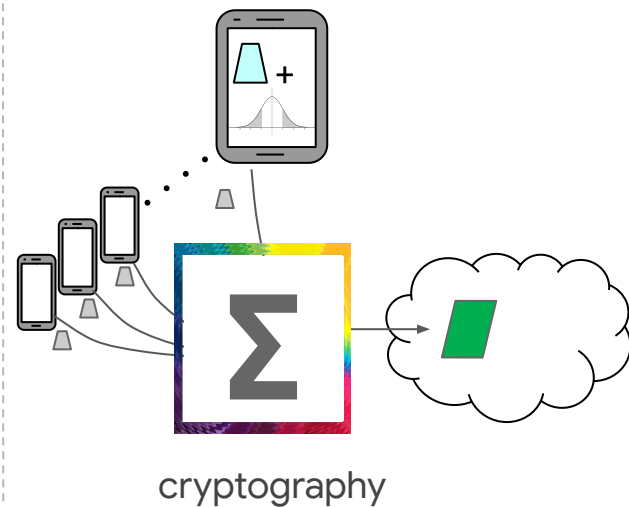
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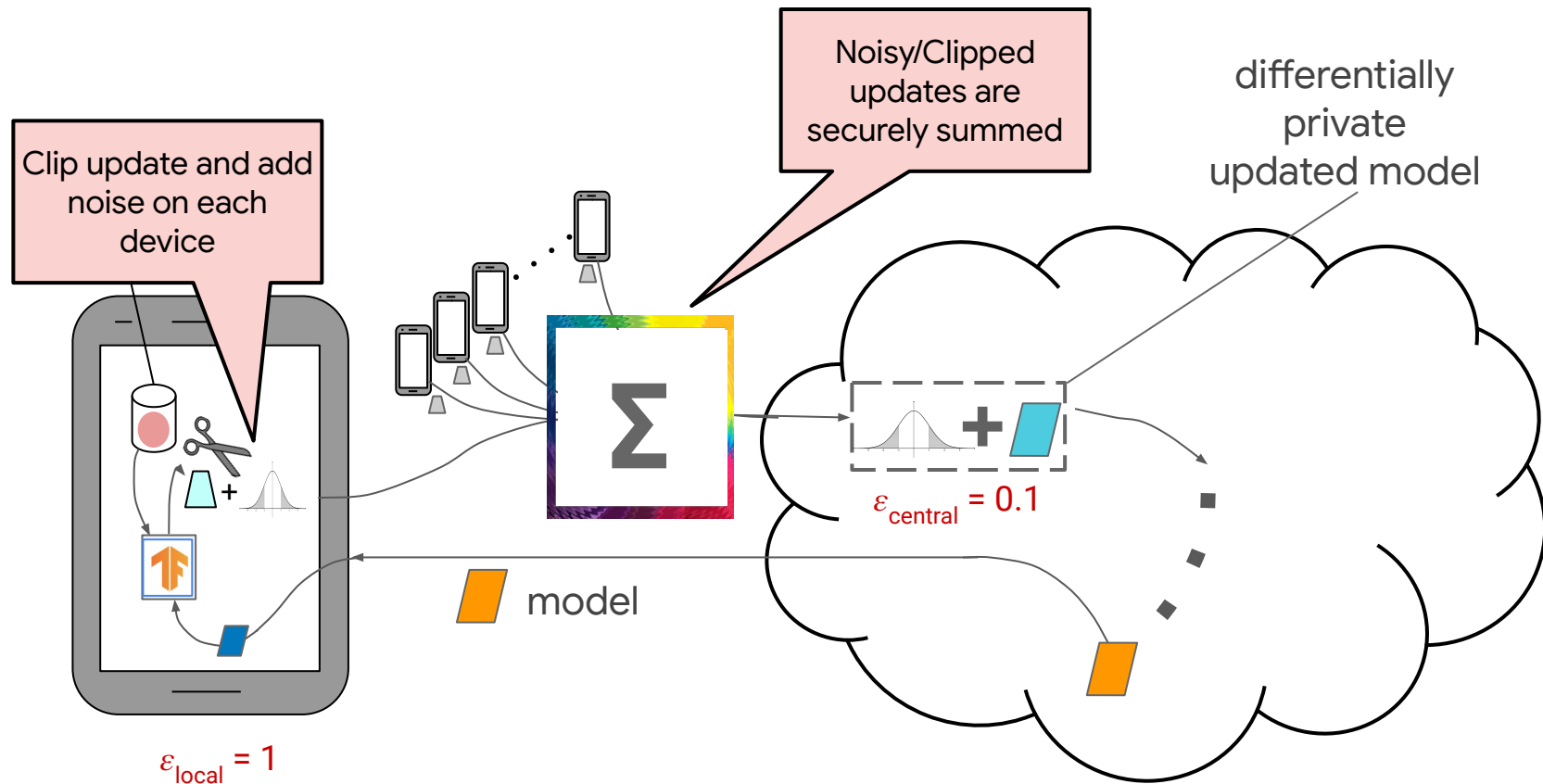
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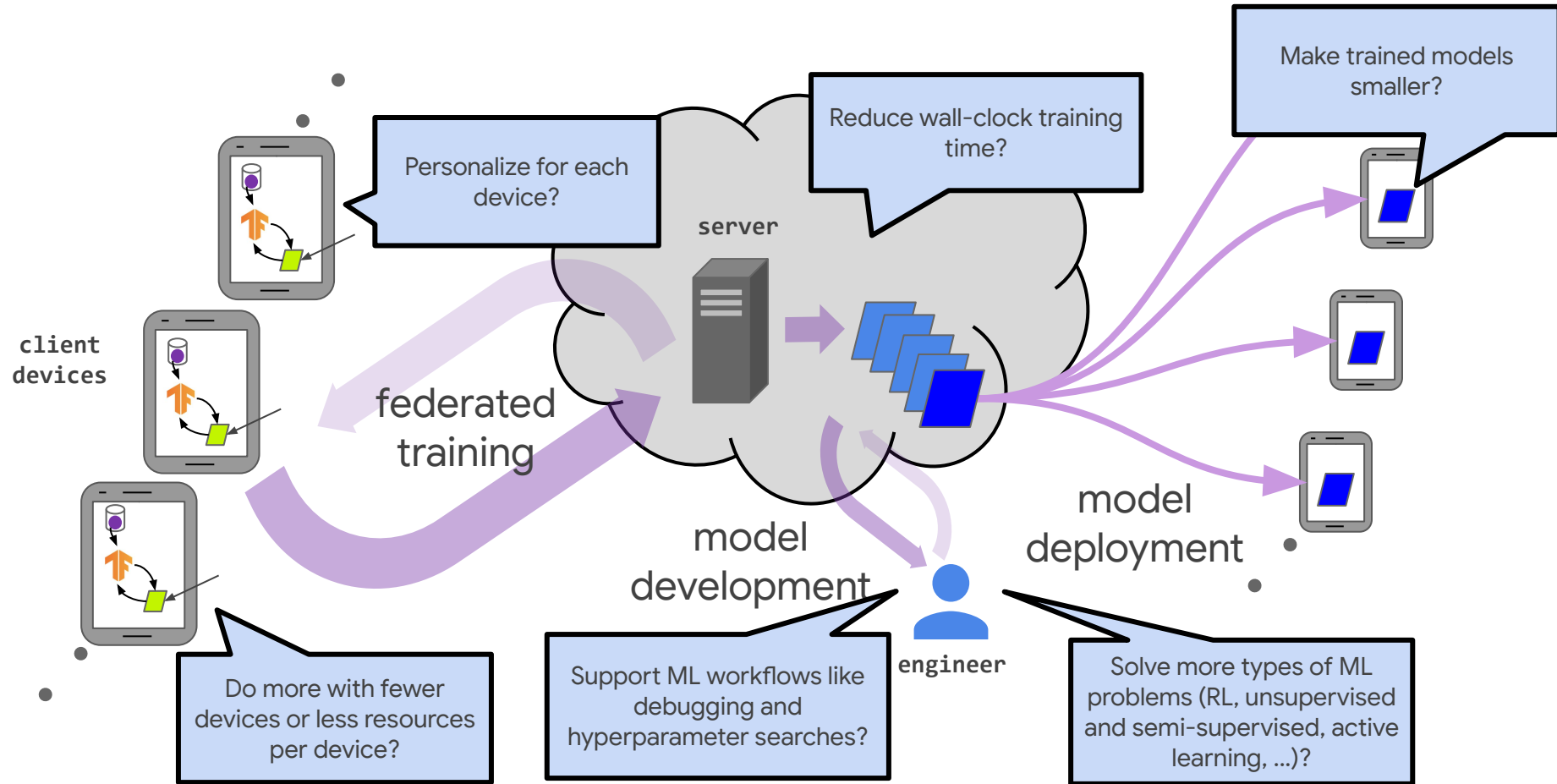
3 Trust via Cryptography



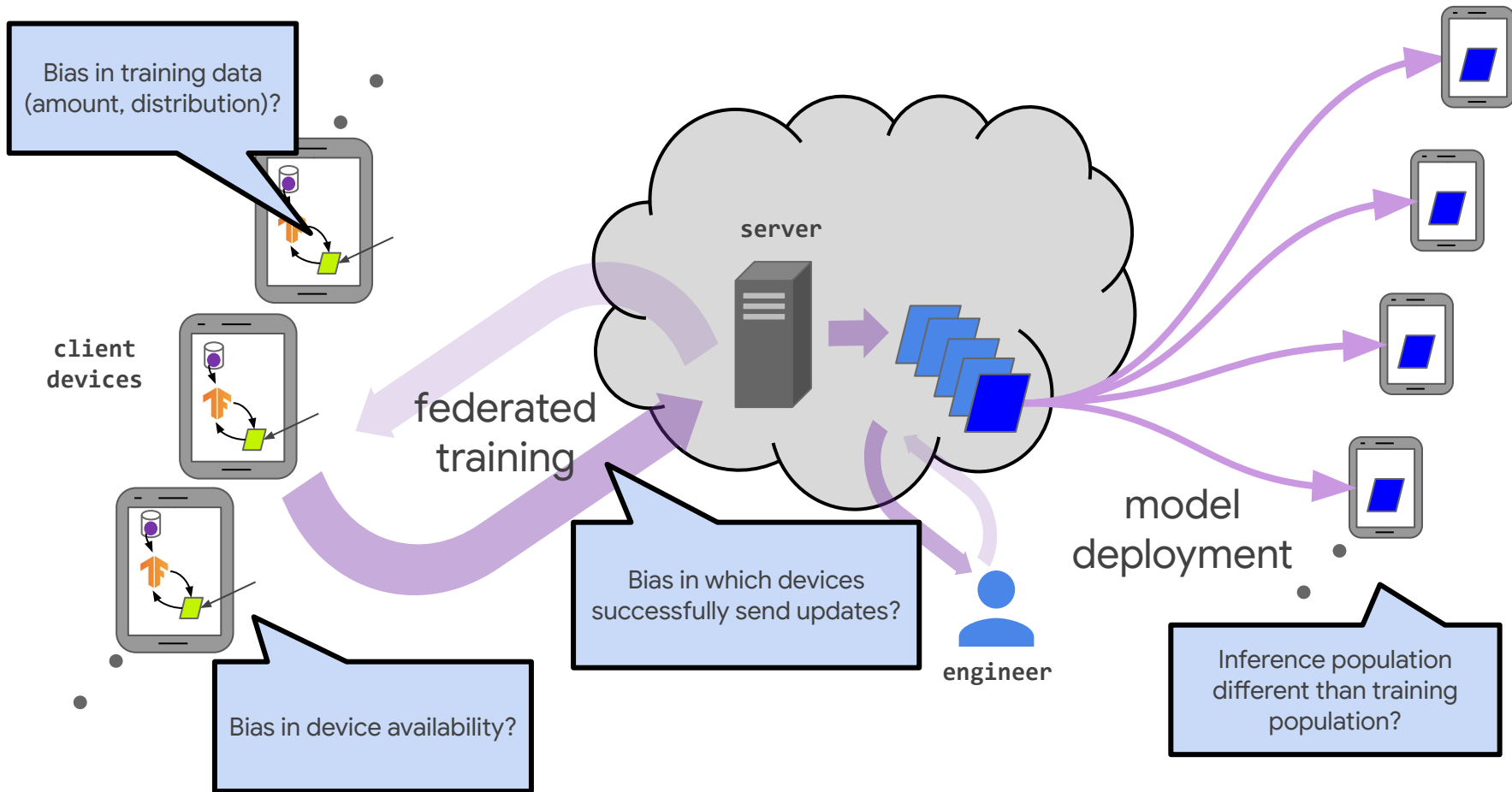
Distributed DP via secure aggregation



Improving efficiency and effectiveness



Ensuring fairness and addressing sources of bias



Robustness to attacks and failures

